

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J16 MLB_IG

LAST_MODIFIED=Fri Mar 22 11:24:26 2013

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Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-0052	PCBA,MLB_IG,DEV,J16	DEVELOPMENT,J16_DEVEL
639-4515	PCBA,MLB_IG,J16	J16,J16_COMMON,CPU:GOOD,SSD:Y,EEEE:FF3T
639-4704	PCBA,MLB_IG,BETTER,J16	J16,J16_COMMON,CPU:BETTER,SSD:Y,EEEE:FGWY
639-4705	PCBA,MLB_IG,CTO,J16	J16,J16_COMMON,CPU:CTO,SSD:Y,EEEE:FGY0

BOM Groups

BOM GROUP	BOM OPTIONS
J16_COMMON	COMMON, ALTERNATE, J16_COMMON1, J16_COMMON2, J16_PROGPARTS
J16_COMMON1	XDP, SPEAKERID, TBTHV:P12V, CPUVCC:3PHASE
J16_COMMON2	VDDQ:P1V35
J16_PROGPARTS	SMC:PROG, BOOTROM:PROG, T29ROM:PROG, CIVROM:PROG, CAMROM:PROG
J16_DEVEL	XDP_CONN, LPCPLUS, DDRVREF_DAC, DEVEL_SENSORS, DEVEL_AUDIO
DEVEL_SENSORS	AP_ISNS:Y, HDD_IVSNS:Y, TEMPSNSDEV
J16_PRODUCTION	AP_ISNS:N, HDD_IVSNS:N

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4515	1	CRW, QEU2, EEU2, C0, 2.7G, 65M, 4+3, 1.15, 4M, BGA	U0500	CRITICAL	CPU: GOOD
337S4516	1	CRW, QEU1, EEU2, C0, 3.0G, 65M, 4+3, 1.13, 4M, BGA	U0500	CRITICAL	CPU: BETTER
337S4517	1	CRW, QEUX, EEU2, C0, 3.2G, 65M, 4+3, 1.3, 6M, BGA	U0500	CRITICAL	CPU: CTO

ASIC Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4483	1	LYNX POINT MOBILE, C1, Q5, QE99, FC8GA995	U1100	CRITICAL	
338S1113	1	IC, TBT, CR-4C, B1, PRQ, C10, 2BB 12X12 FC-CPB	U2800	CRITICAL	
343S0616	1	IC, BCM57766A, CIV+V, A0, 8X8	U3900	CRITICAL	
353S3908	1	IC, LP8561, LED BLKT CTRL, LLP24, B0-F	U8100	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3783	1	IC, EFI, V0839, J16	U5210	CRITICAL	BOOTROM:PROG
335S0867	1	IC, 64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3781	1	IC, SMC, PROGRAMD, V2. 12A30, J16	U5000	CRITICAL	SMC: PROG
338S1159	1	IC, SMC12-A3, 40MHZ/50MIPS, SCPL FW, 1578BA	U5000	CRITICAL	SMC: BLANK
341S3734	1	IC, EEPROM, CR, V16_2, J16	U2890	CRITICAL	T29ROM:PROG
335S0865	1	IC, EEPROM, SERIAL, 256KB, MLP8	U2890	CRITICAL	T29ROM:BLANK
341S3735	1	IC, ENET SPI ROM, NYMONYX, V1.13, D7/D7I	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REV F	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC, CAMERA, FLASH, V7229, J16	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	MLB LABEL, 2D	EEEE_FF3T	CRITICAL	EEEE: FF3T
825-7896	1	MLB LABEL, 2D	EEEE_FGY0	CRITICAL	EEEE: FGY0
825-7896	1	MLB LABEL, 2D	EEEE_FGWY	CRITICAL	EEEE: FGWY

Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
951-0164	1	SCH, MLB_IG, J16	SCH	CRITICAL	J16
820-3588	1	PCBF, MLB_IG, J16	PCB	CRITICAL	J16

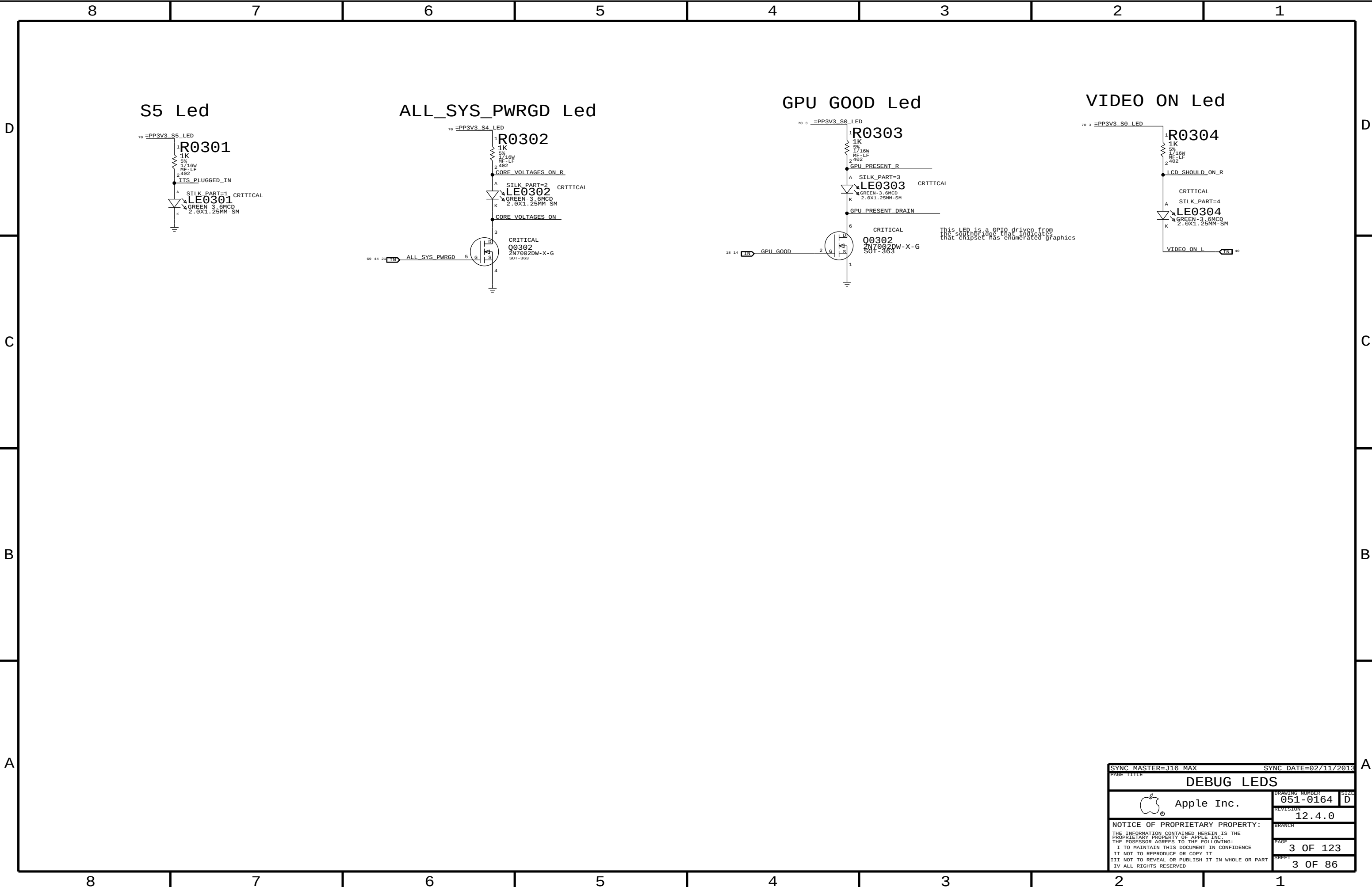
Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB2 diodes
377S0155	377S0104		ALL	USB3 diodes
377S0124	377S0057		ALL	TVS
376S0975	376S1081		ALL	P/Nch dual FET
157S0084	157S0058		Enet	Enet magnetics
155S0578	155S0367		ALL	1200HM EMI BEAD
128S0368	128S0365		ALL	150UF AL POLY
138S0681	138S0638		ALL	Taiyo 10uF 805 alt
197S0479	197S0478		Y4200	12 MHz Cam. Xtal
341S3747	341S3735		U3990	Enet ROM
107S0251	107S0249		ALL	Sense resistor
102S0880	102S0879		ALL	Sense resistor
197S0481	197S0480		ALL	25MHZ Xtal
138S0860	138S0775		ALL	Single-source 1uF 462
138S0859	138S0788		ALL	Single-source 10uF
138S0706	138S0739		ALL	Single-source 1uF 20

R5400, R5520, R5530

R5430

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PAGE TITLE

DEBUG LEDS

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DRAWING NUMBER
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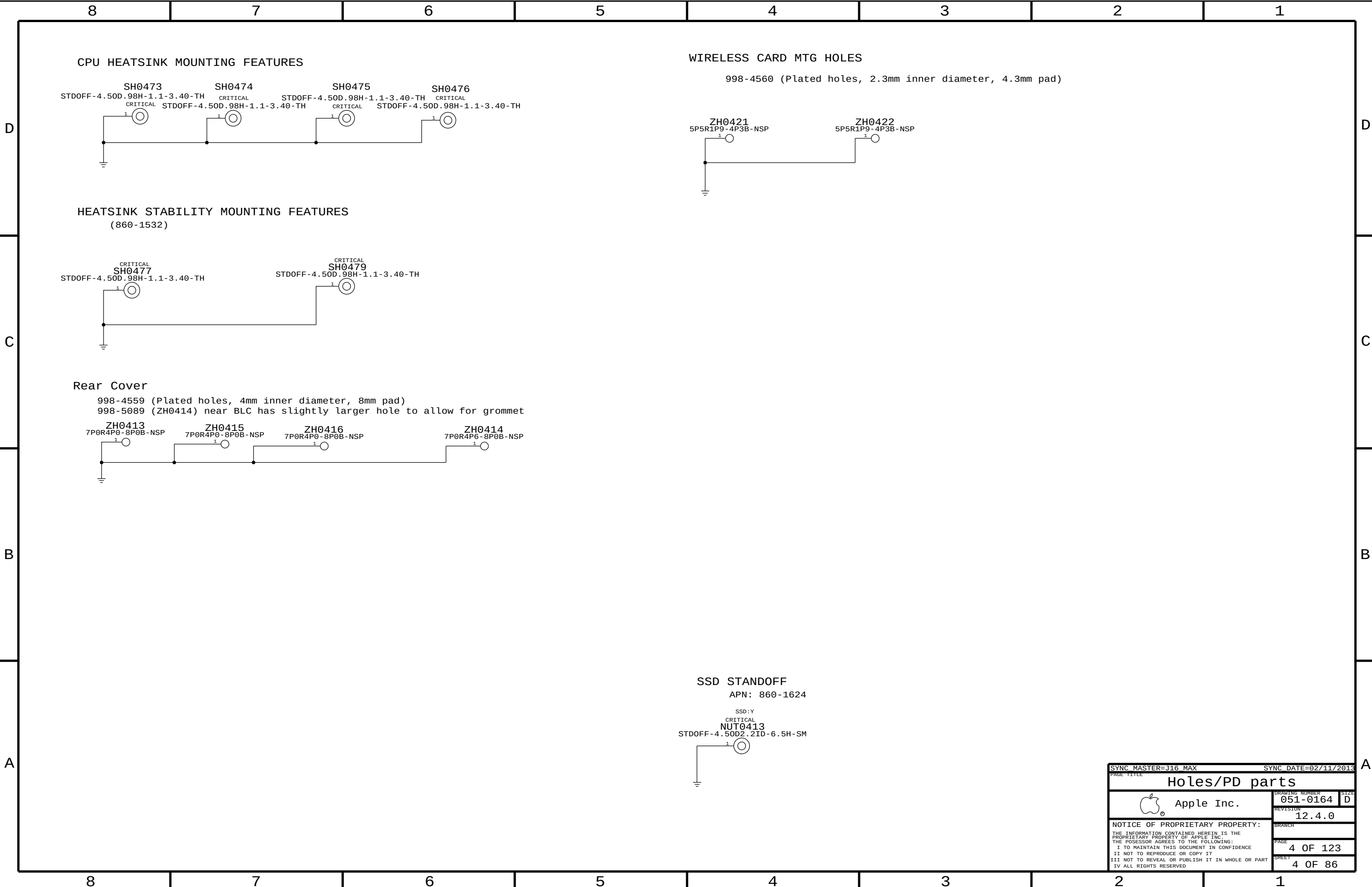
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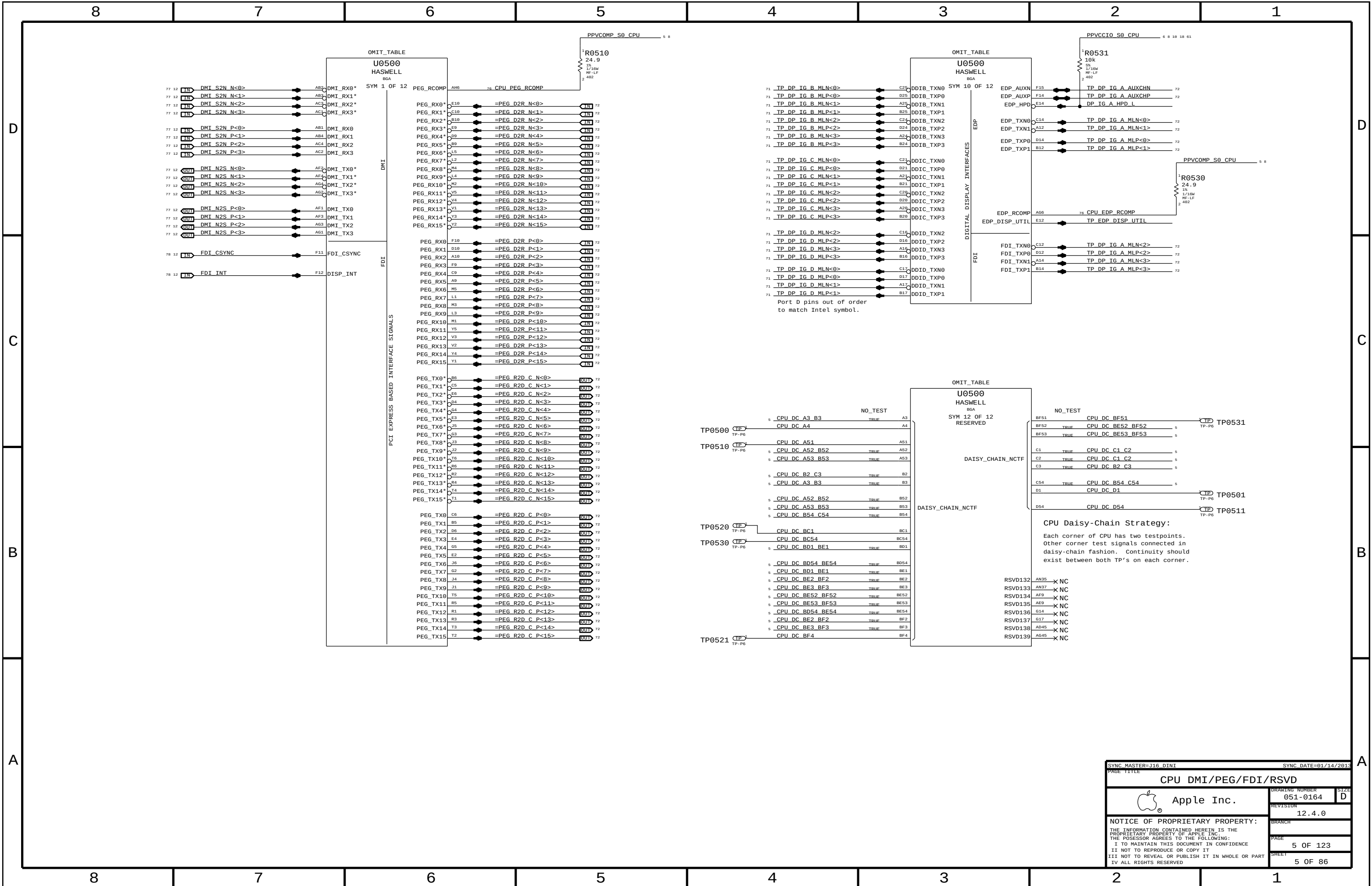
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D

C

B

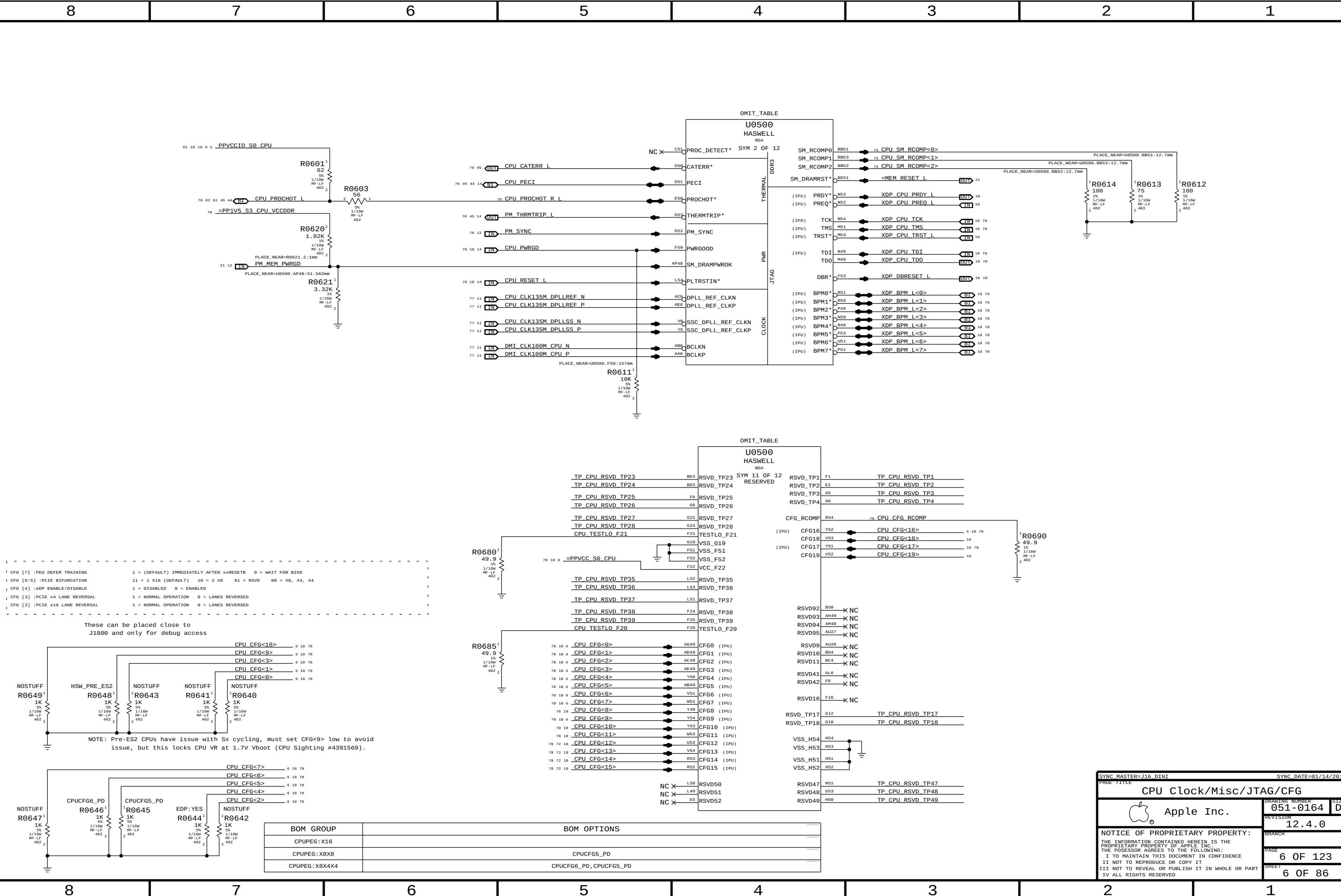
A

D

C

B

A



CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER XRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

These can be placed close to J1809 and only for debug access

NOTE: Pre-ES2 CPUs have issue with Sx cycling, must set CFG<9> low to avoid issue, but this locks CPU VR at 1.7V Vboot (CPU Sighting #4391569).

BOM GROUP	BOM OPTIONS
CPUPEG: X16	
CPUPEG: X8X8	CPUPEG5_PD
CPUPEG: X8X4X4	CPUPEG6_PD, CPUPEG5_PD

SYNC MASTER=J16 DINT

SYNC DATE=01/14/2013

CPU Clock/Misc/JTAG/CFG

Apple Inc.

051-0164

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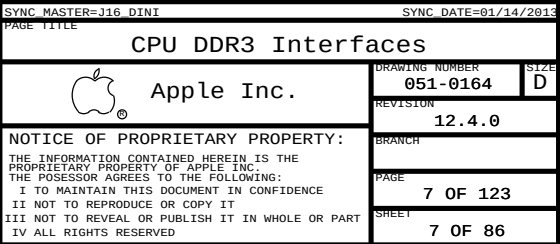
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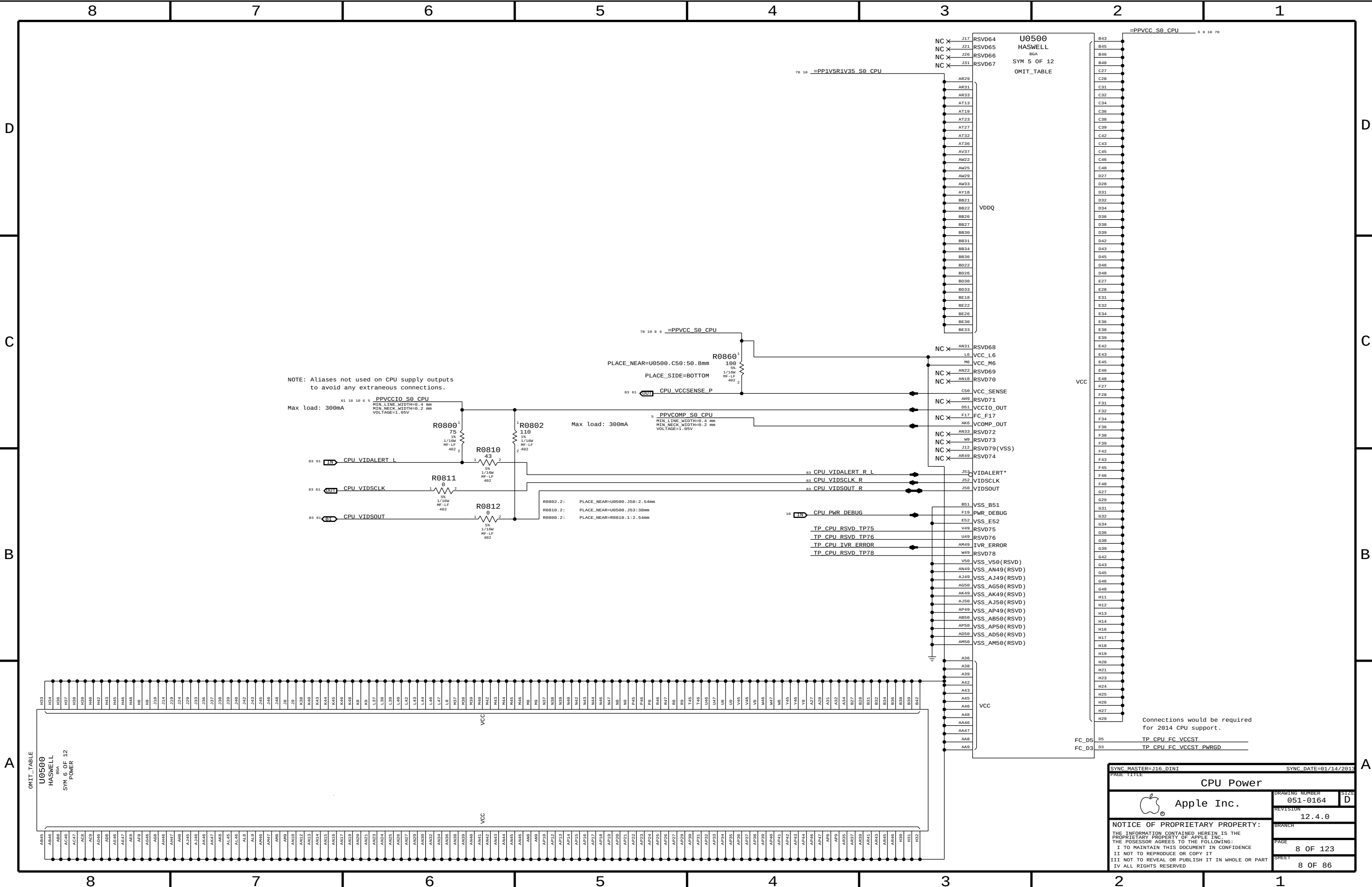
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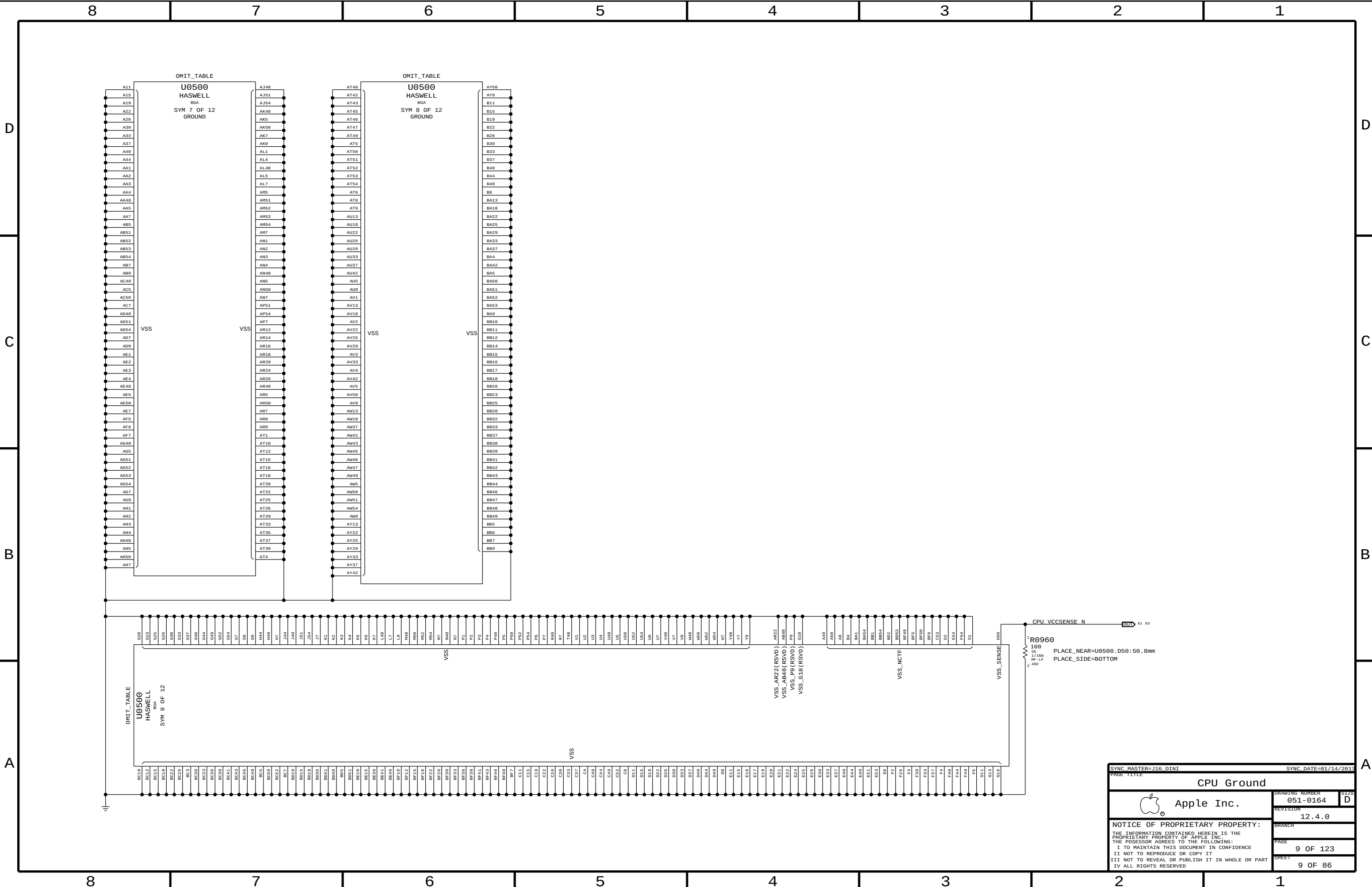
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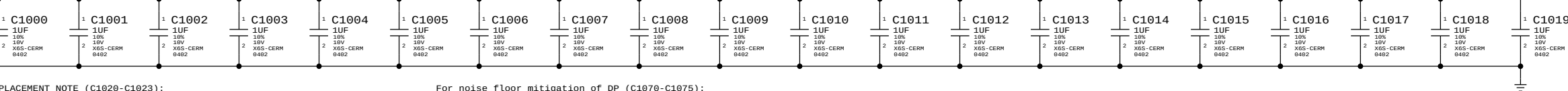


CPU VCORE Decoupling

Intel recommendation: 4x 470uF 4mOhm (3 CPU-side, 1 opposite), 20x 22uF 0805 (10 CPU-side, 10 opposite near edge), 4x 10uF 0603 (2 CPU-side, 2 opposite), 20x 1uF 0402 (under CPU)
Apple Implementation: 9x 210uF 6mOhm, 44x 10uF 0402, 4x 10uF 0402, 20x 1uF 0402

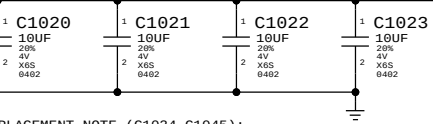
PLACEMENT_NOTE (C1000-C1019):

Place on bottom side of U0500

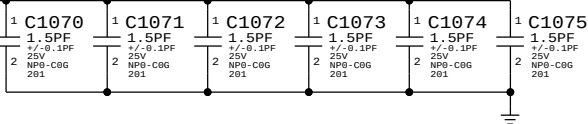


PLACEMENT_NOTE (C1020-C1023):

Place near U0500 on bottom side

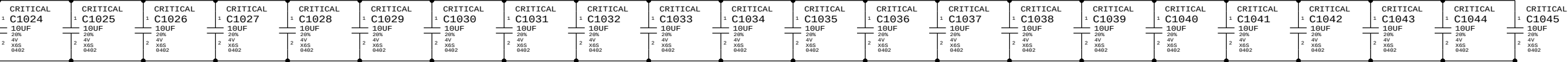


For noise floor mitigation of DP (C1070-C1075):



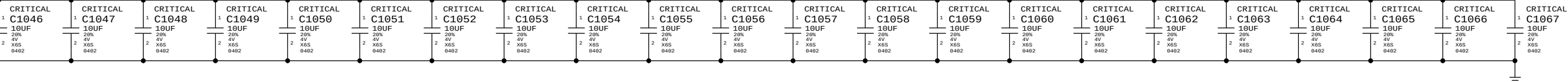
PLACEMENT_NOTE (C1024-C1045):

Place near inductors on bottom side.



PLACEMENT_NOTE (C1046-C1067):

Place near inductors on bottom side.



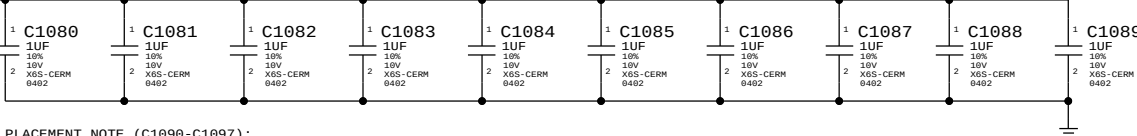
BULK CAPS ON REGULATOR PAGE

CPU VDDQ Decoupling

Intel recommendation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402
Apple Implementation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402

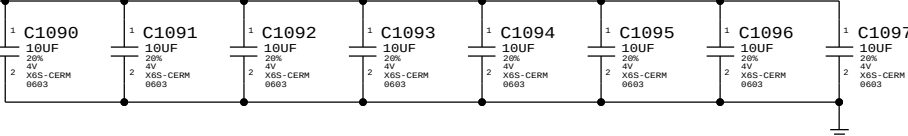
PLACEMENT_NOTE (C1080-C1089):

Place on bottom side of U0500



PLACEMENT_NOTE (C1090-C1097):

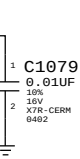
Place near U0500 on bottom side



BULK CAPS ON REGULATOR PAGE

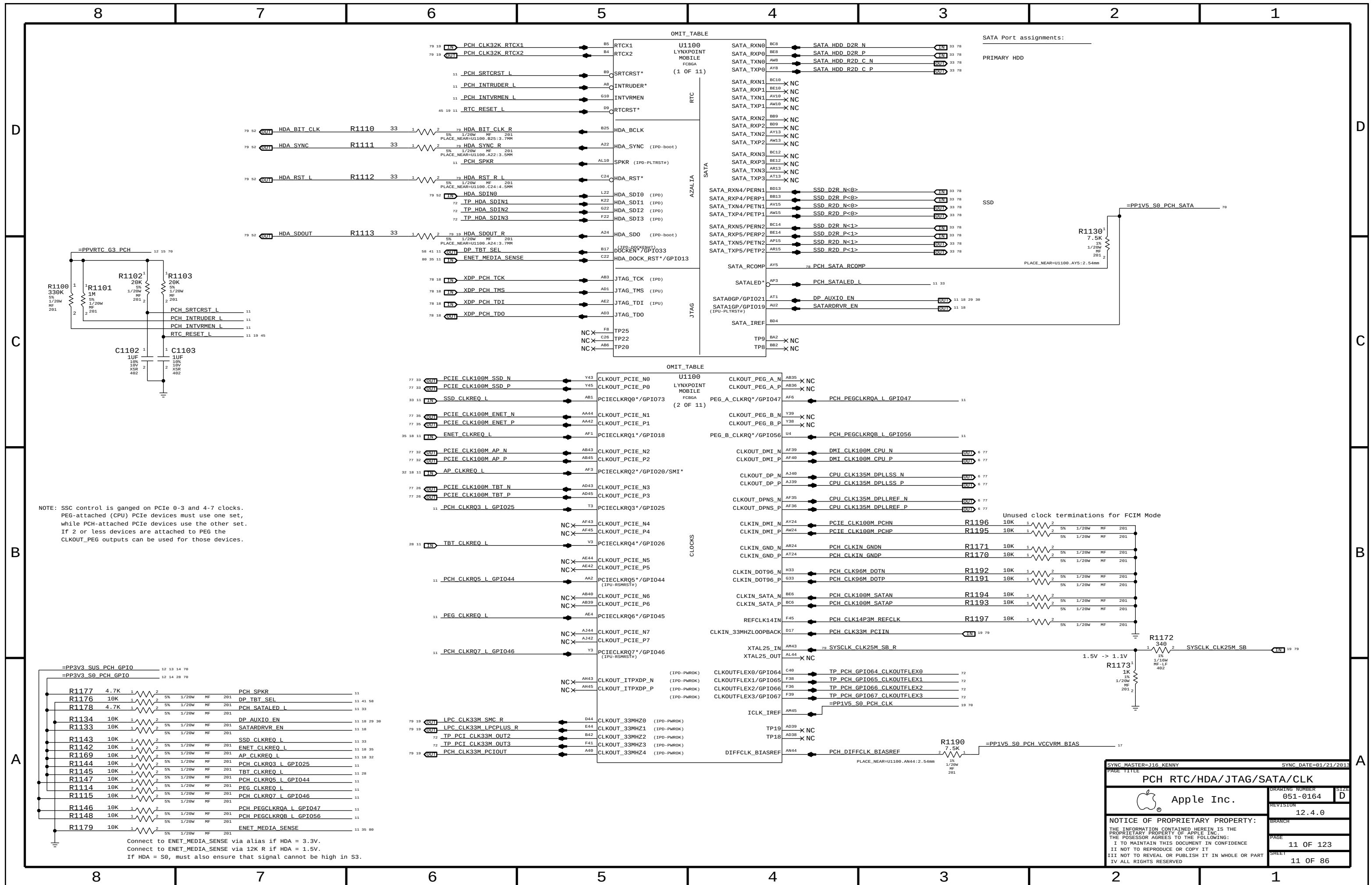
CPU VCCIO Decoupling

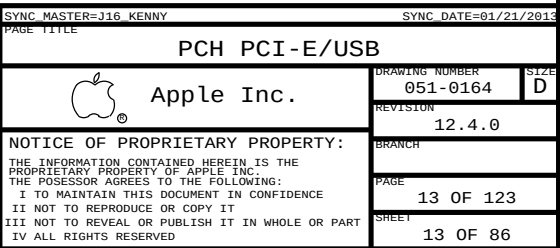
Intel recommendation: 2x 0.01uF 0402 (1 near CPU, 1 near SVID pull-ups)
Apple Implementation: 2x 0.01uF 0402 (second cap is on CPU VR page)

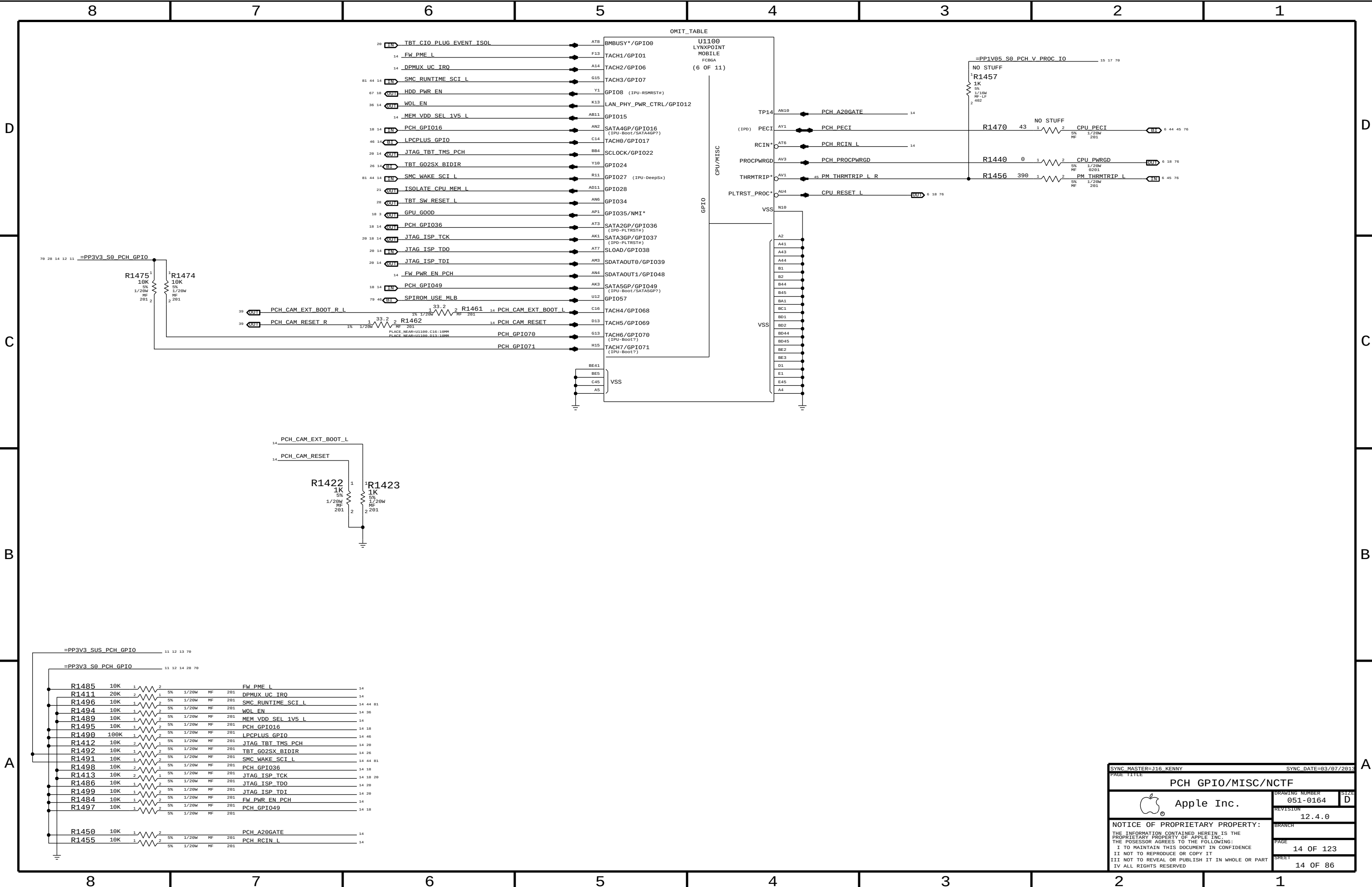


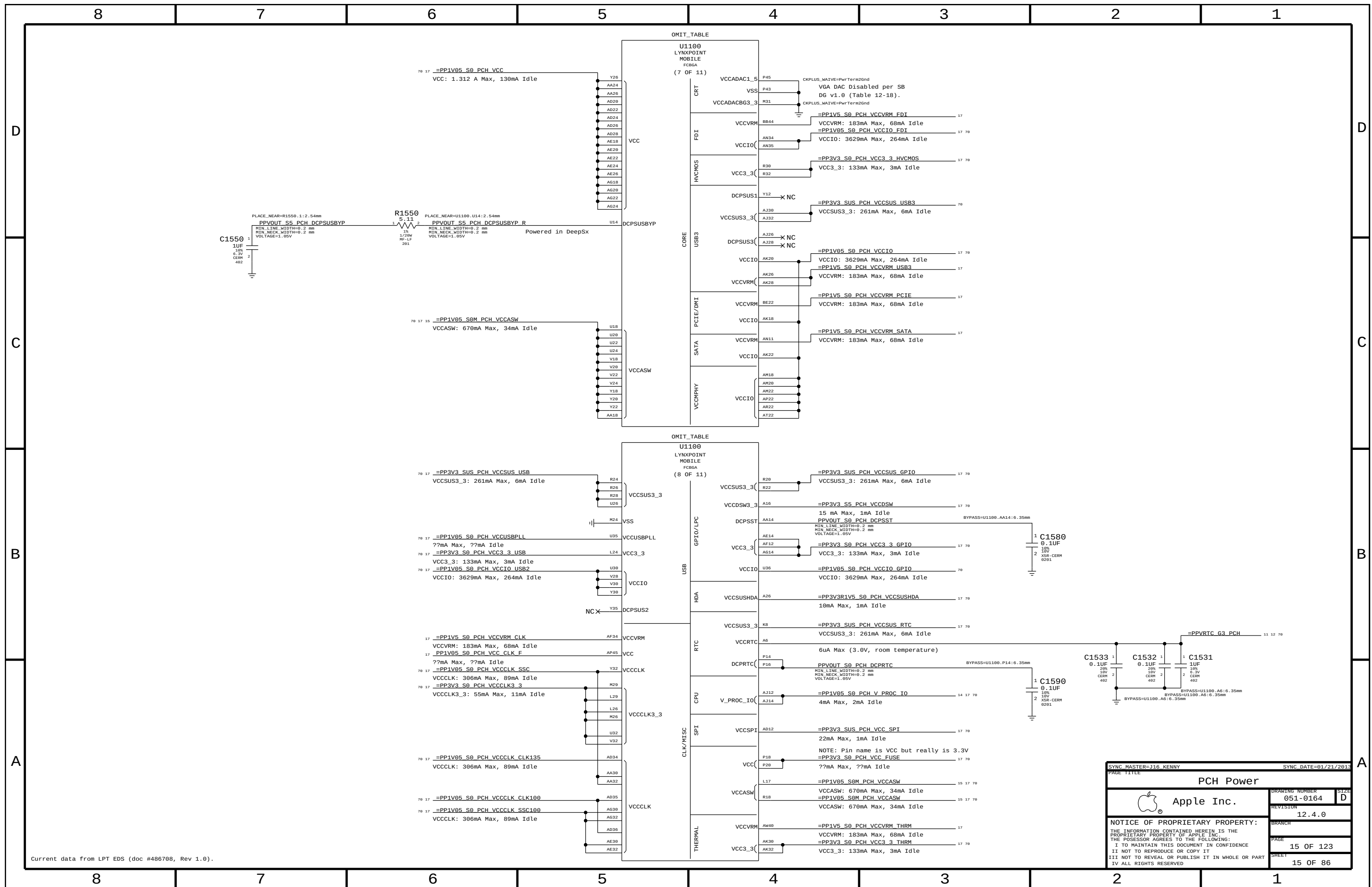
NOTE: Intel decoupling recommendations from Shark Bay Mobile Platform Power Delivery Design Guide (doc #487822, Rev 0.8 dated January 2012), Section 5.

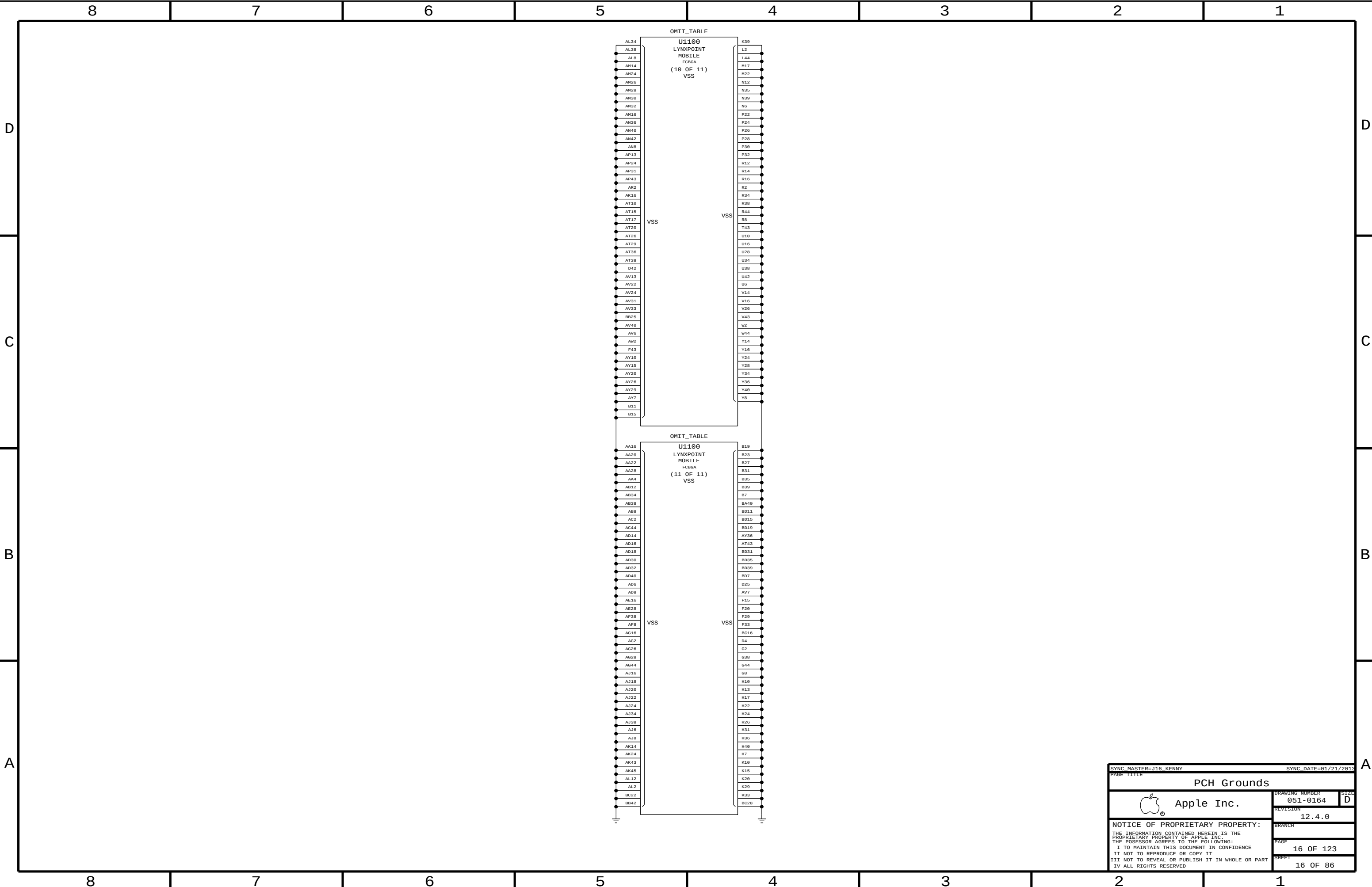
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SYNC MASTER=J16 KENNY

SYNC DATE=01/21/2013

PAGE TITLE

PCH Grounds

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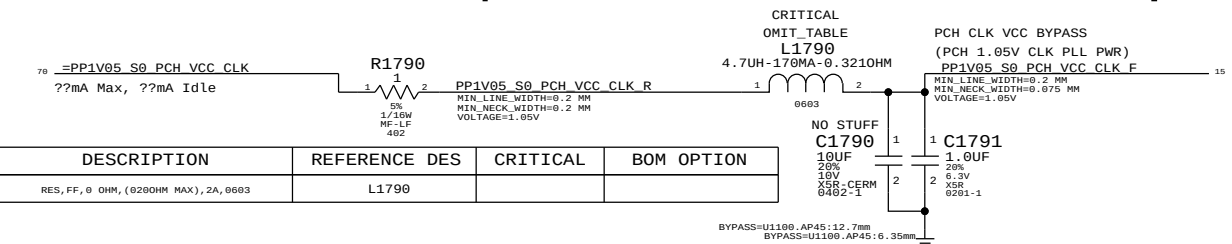
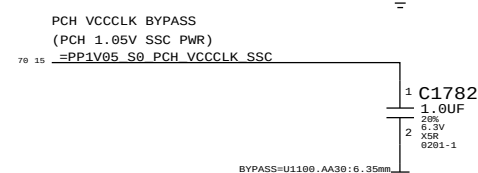
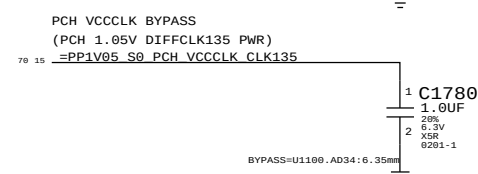
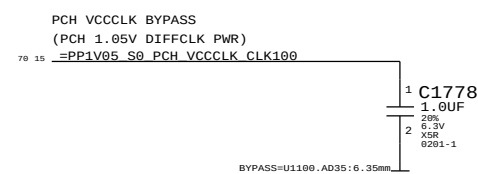
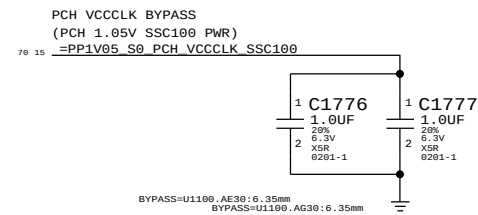
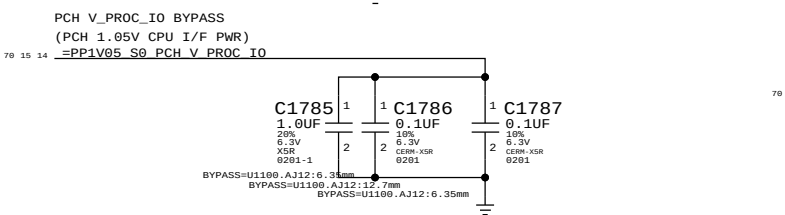
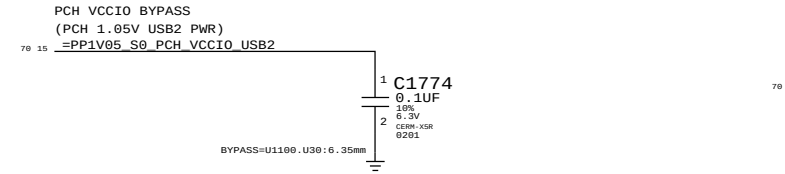
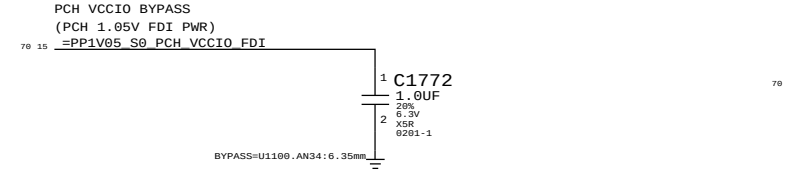
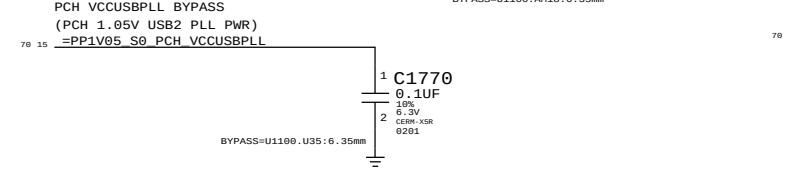
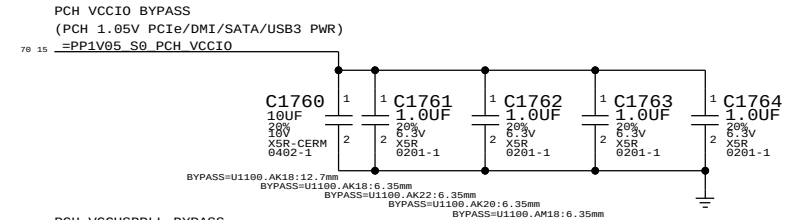
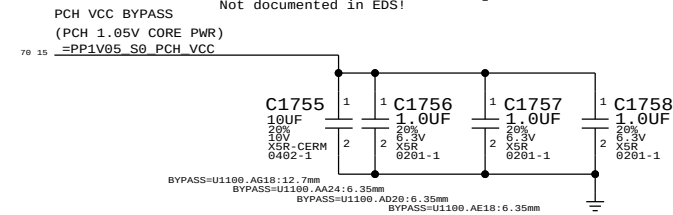
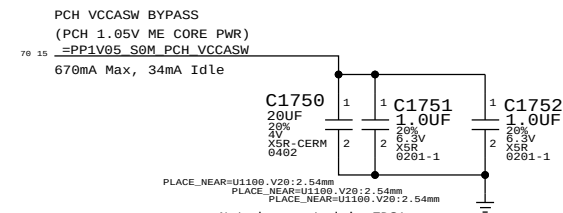
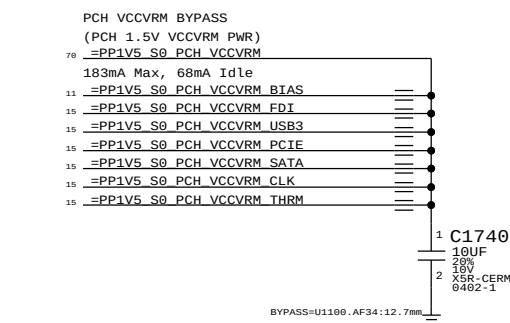
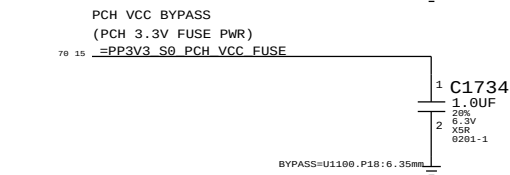
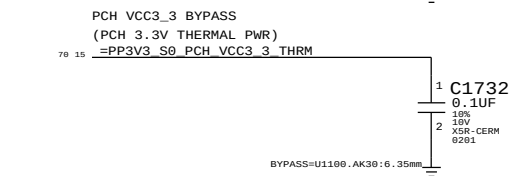
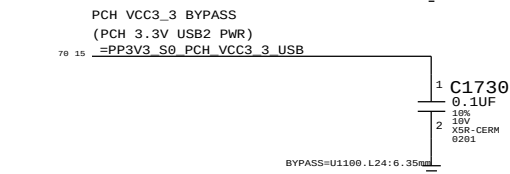
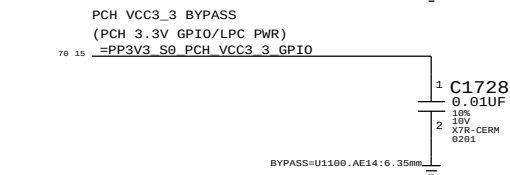
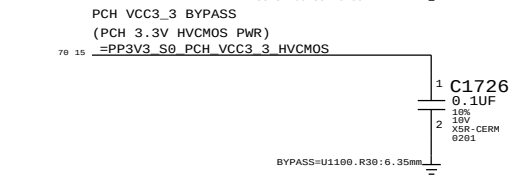
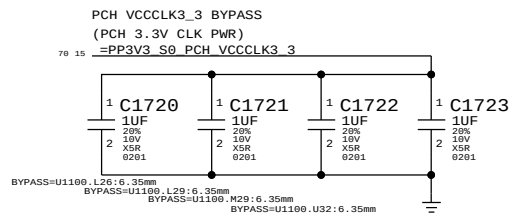
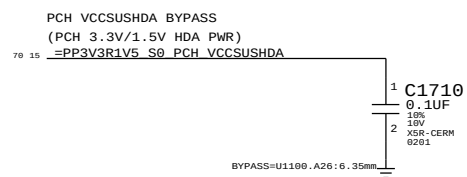
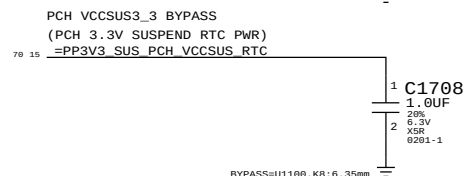
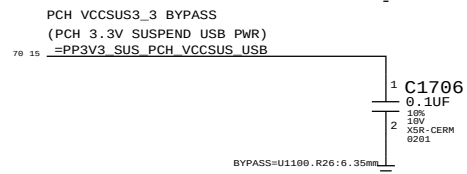
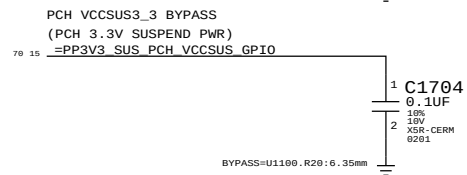
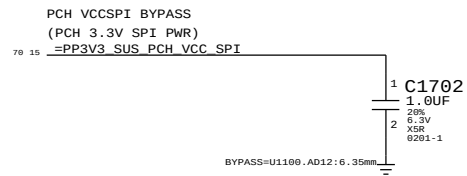
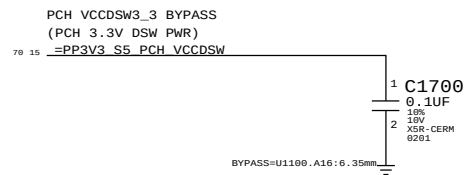
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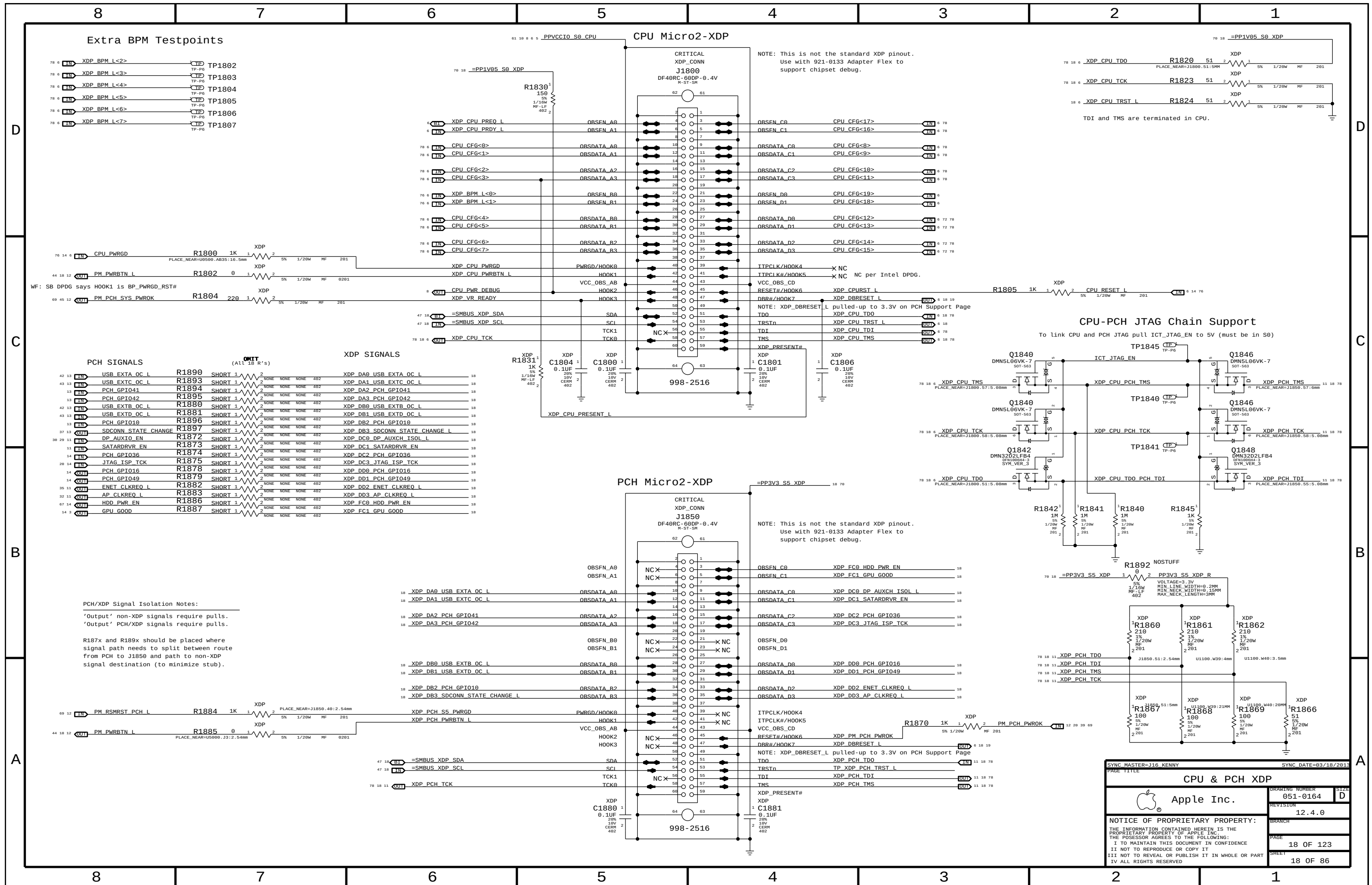
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SHEET

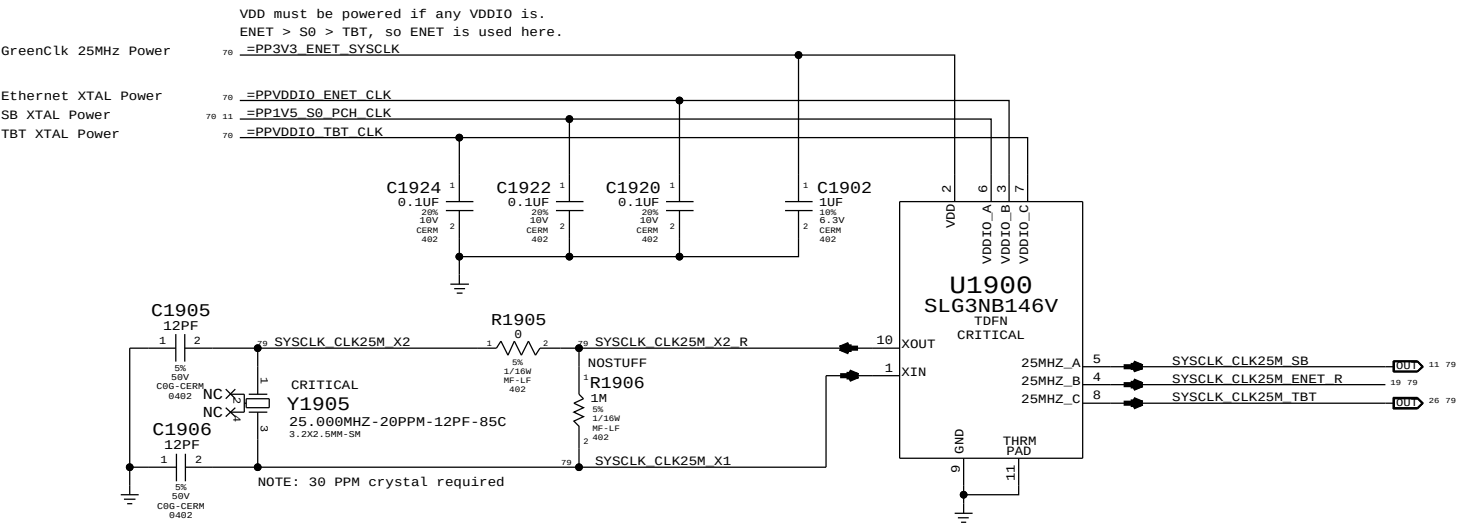
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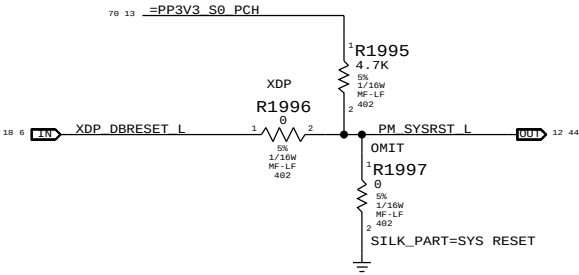
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	1	RES, FF, 0 OHM, (020OHM MAX), 2A, 0603	L1790		



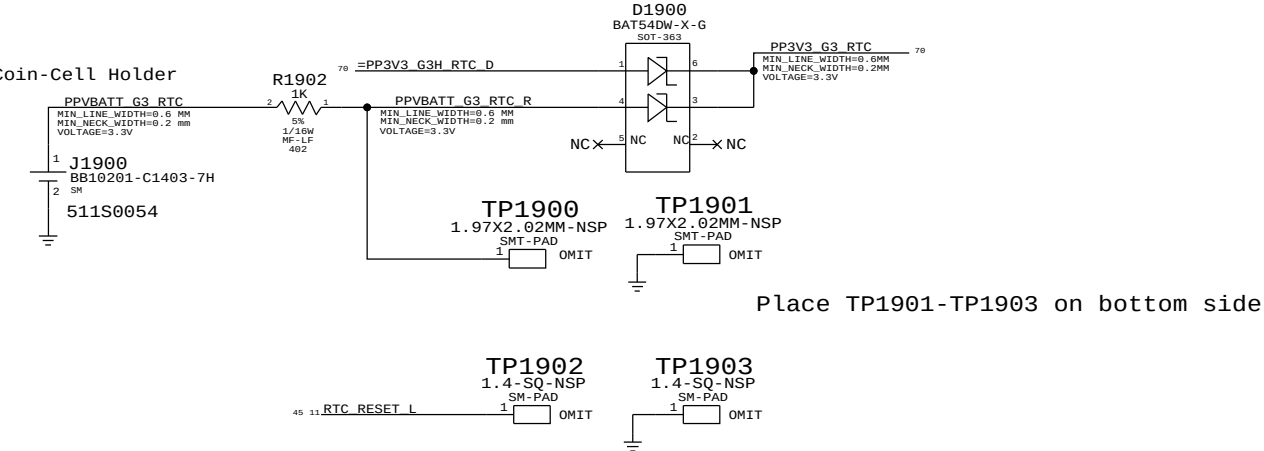
System 25MHz Clock Generator



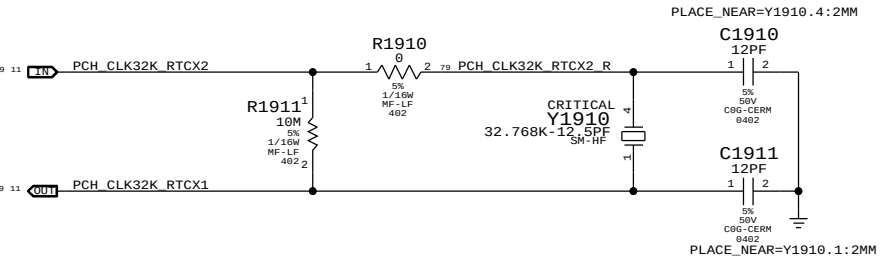
PCH Reset Button



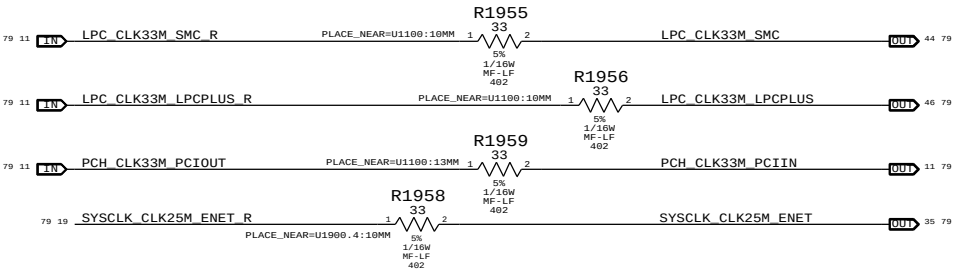
RTC Power Sources



PCH RTC Crystal

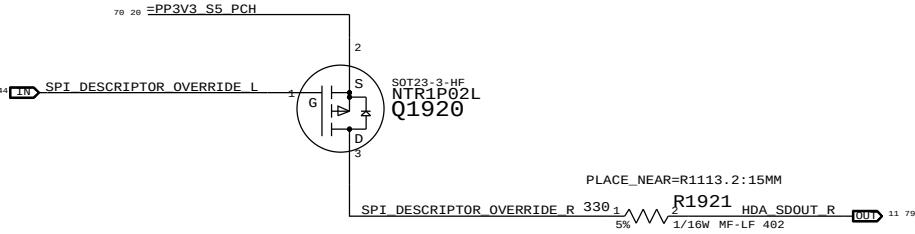


Clock series termination

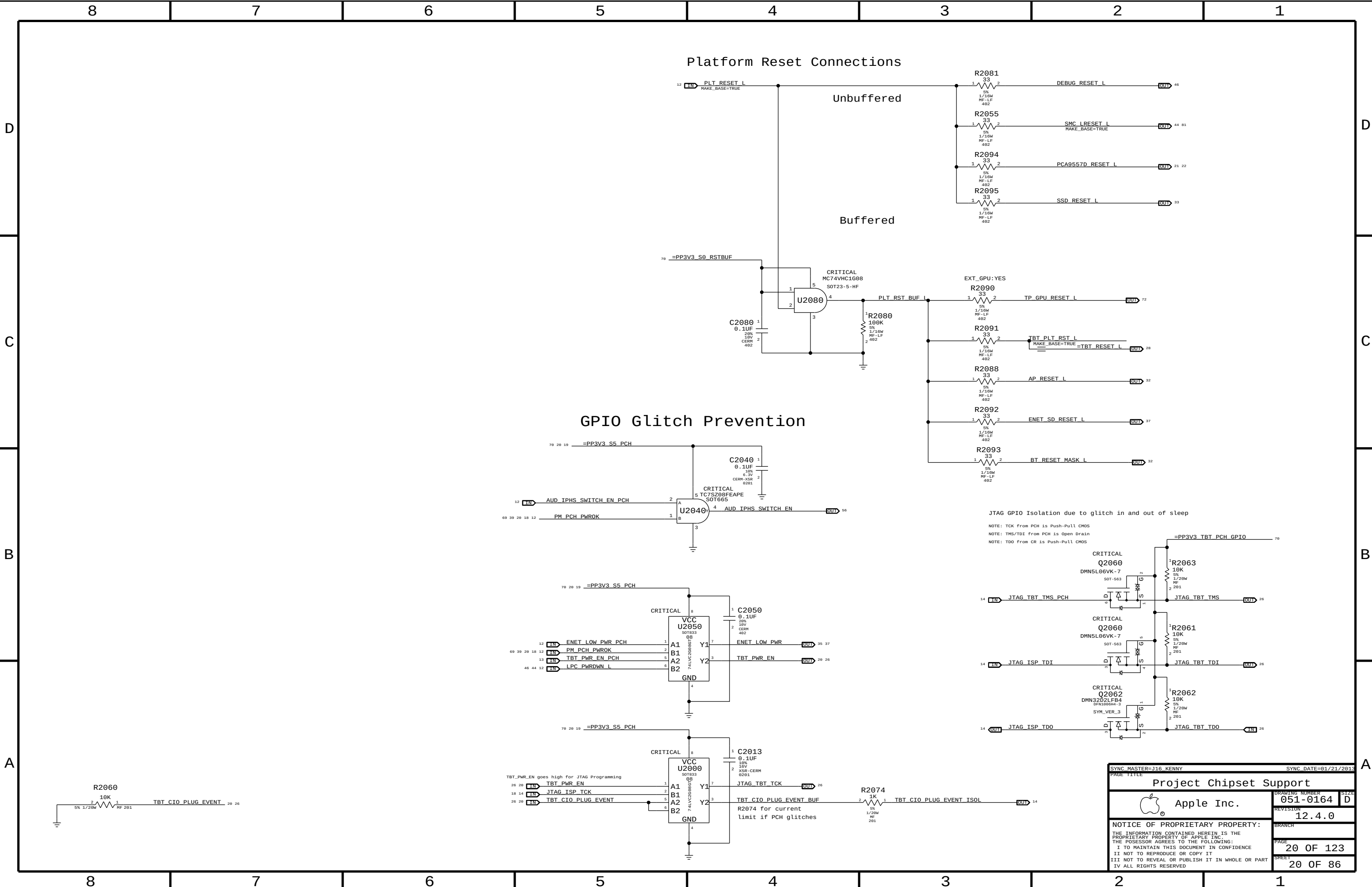


PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting.



SYNC MASTER=J16 KENNY		SYNC DATE=01/21/2013	
PAGE TITLE			
Chipset Support		DRAWING NUMBER	SIZE
Apple Inc.		051-0164	D
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SYNC MASTER=J16 KENNY

SYNC DATE=01/21/2013

PROJECT TITLE

Project Chipset Support

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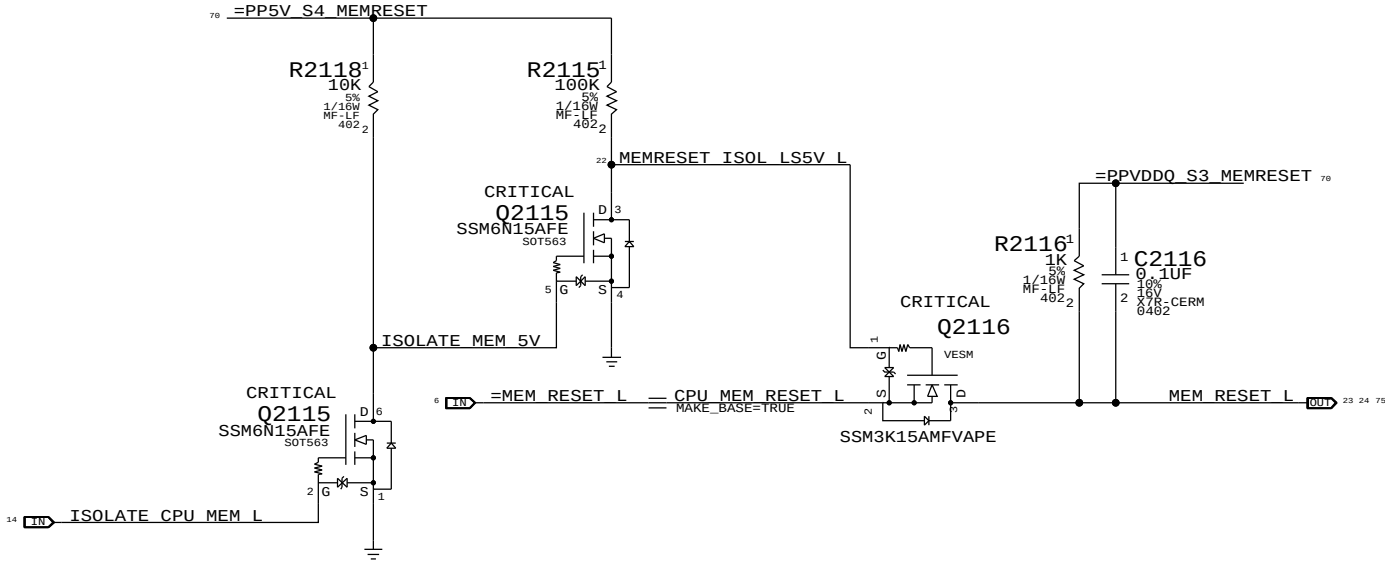
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

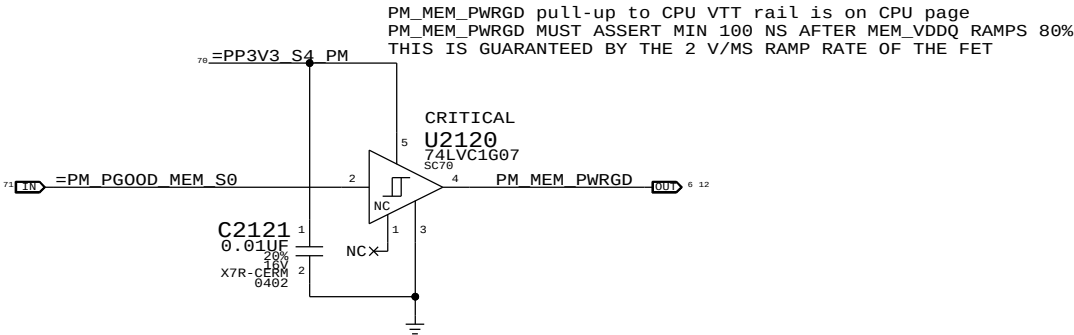
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

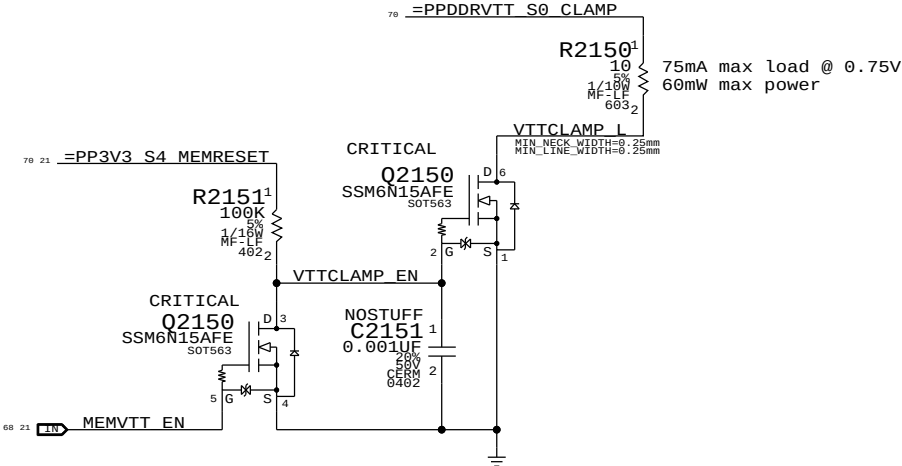
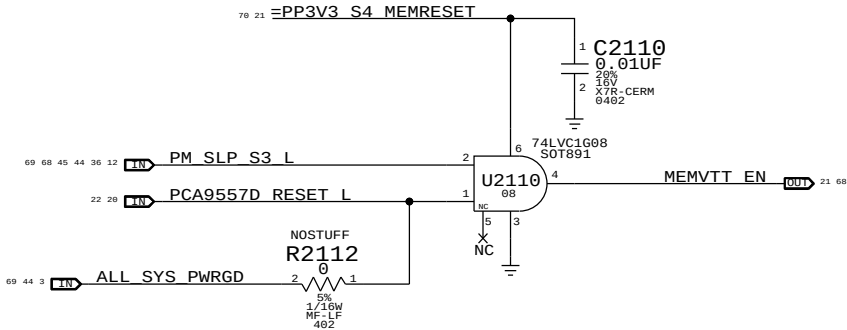


MEM S0 "PGOOD" FOR CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1
1	1	0	1	1	1	1
2	0	0	1	1	1	0
3	0	0	0	X	1	0
4	0	0	1	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=J16 NICK

SYNC DATE=12/11/2012

CPU Memory S3 Support

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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

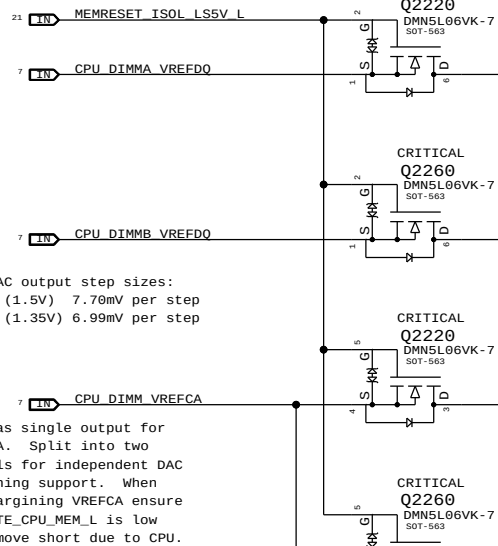
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

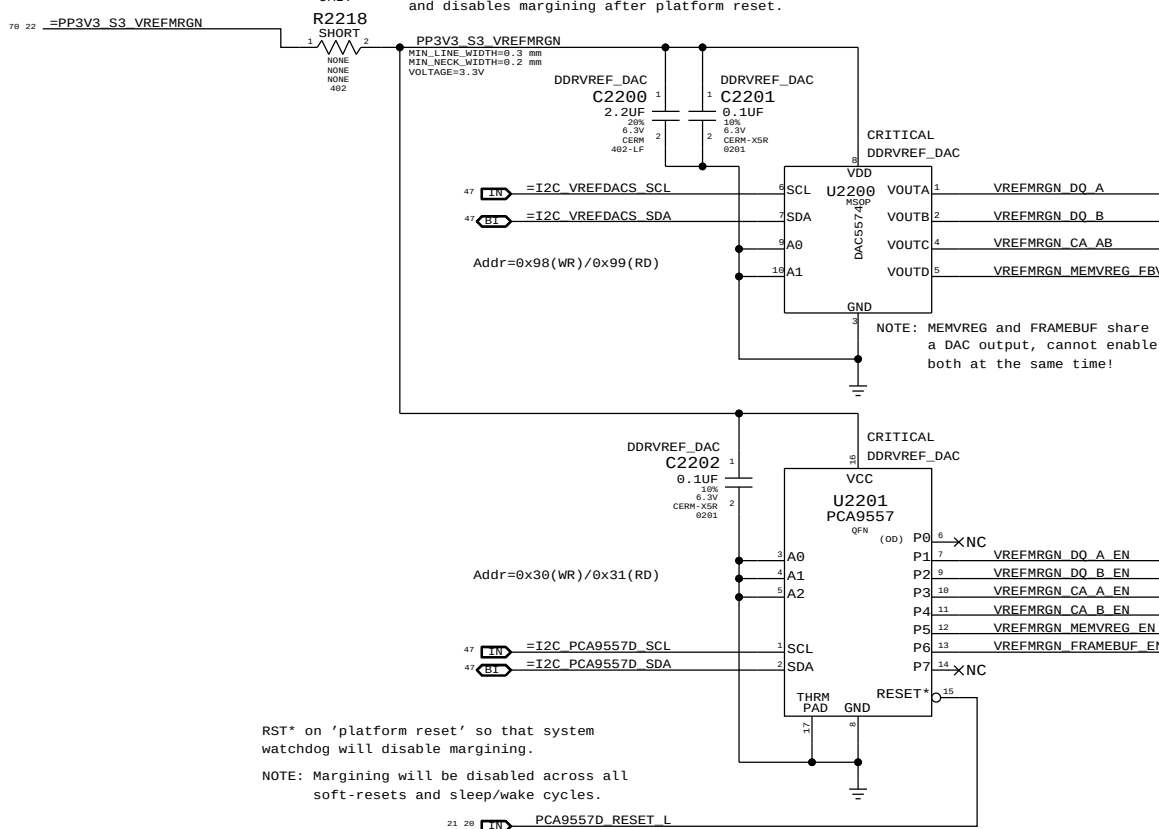
CPU-Based Margining

FETs for CPU isolation during S3



DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.



VRef Dividers

Always used, regardless of margining option.

=PPDDR_S3_MEMVREF

R2221

R2222

R2223

R2224

R2225

R2226

R2227

R2228

R2229

R2230

R2231

R2232

R2233

R2234

R2235

R2236

R2237

R2238

R2239

R2240

R2241

R2242

R2243

R2244

R2245

R2246

R2247

R2248

R2249

R2250

R2251

R2252

R2253

R2254

R2255

R2256

R2257

R2258

R2259

R2260

R2261

R2262

R2263

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R2302

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R2304

R2305

R2306

R2307

R2308

R2309

R2310

R2311

R2312

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R2314

R2315

R2316

R2317

R2318

R2319

R2320

R2321

R2322

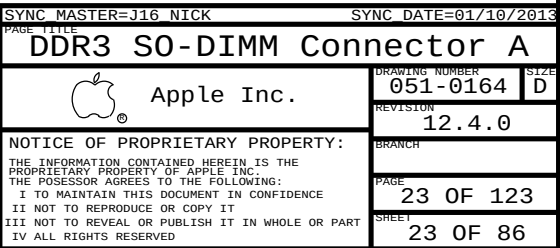
R2323

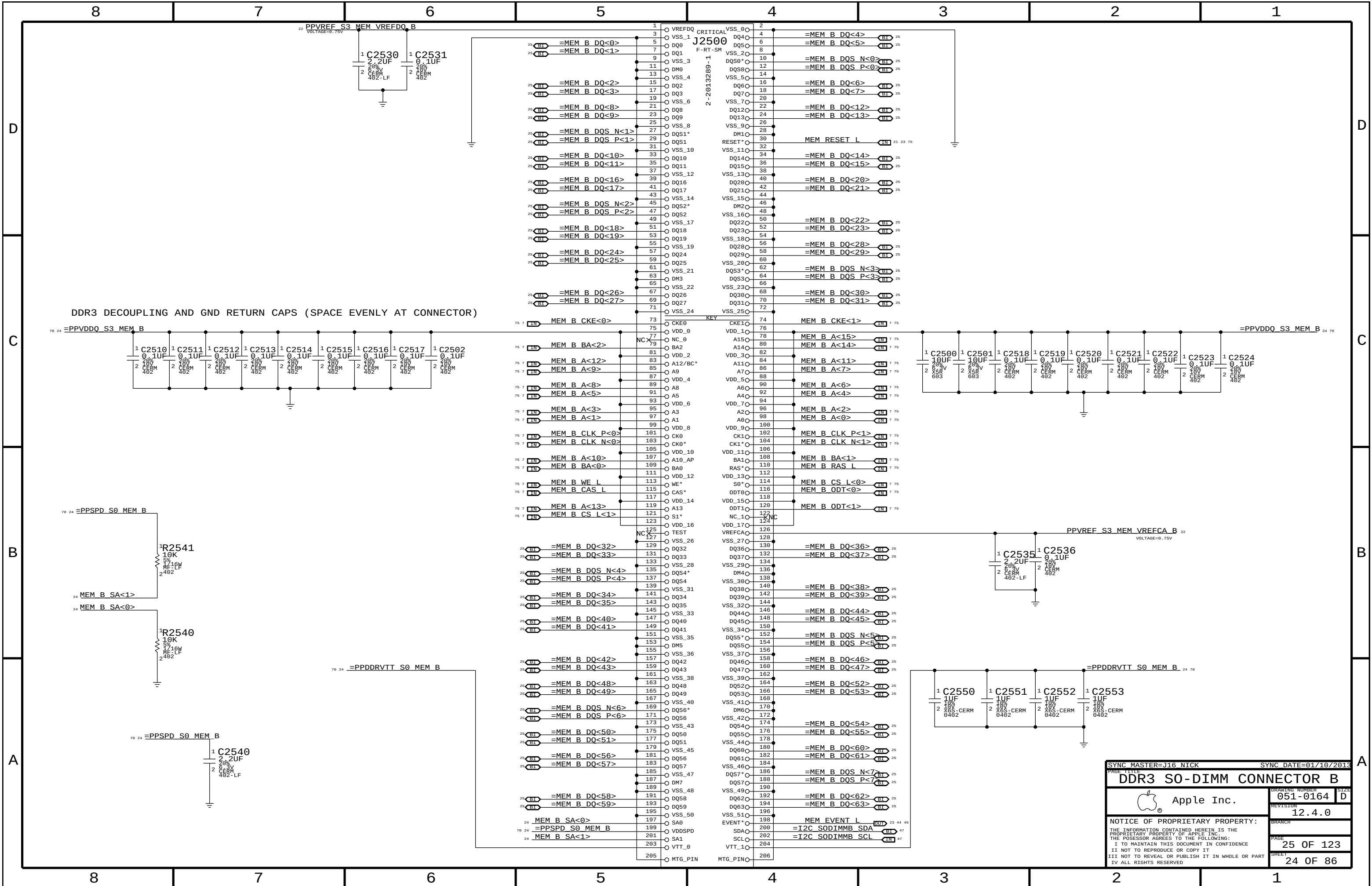
R2324

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	DDR3 (1.5V)		DDR3L (1.35V)		
Nominal value	0.750V (DAC: 0x3A = 0.747mV)		0.675V (DAC: 0x34 = 0.670mV)		1.500V (DAC: 0x74 = 1.495V)
Margined target:	0.300V - 1.200V (+/- 450mV)		0.275V - 1.075V (+/- 400mV)		1.200V - 1.800V (+/- 300mV)
DAC range:	0.000V - 1.500V (0x00 - 0x75)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 3.004V (0x00 - 0xE9)
Margined range:	0.299V - 1.206V (+/- 453mV)		0.269V - 1.083V (+/- 406mV)		1.199V - 1.801V (+/- 301mV)
VRef current:	+901uA - 911uA (- = sourced)		+811uA - 816uA (- = sourced)		+36uA - 36uA (- = sourced)
DAC step size:	7.68mV / step @ output		7.67mV / step @ output		2.575mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC MASTER=J16 NICK		SYNC DATE=01/10/2013	
PAGE TITLE		DDR3 VREF MARGINING	
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8		7		6		5		4		3		2		1	
THERE ARE NO PIN SWAPS															
D	75 7	MEM A DQS N<0>				=MEM A DQS N<0>	23	75 7	MEM B DQS N<0>					=MEM B DQS N<0>	24
	75 7	MEM A DQS P<0>	MAKE_BASE=TRUE			=MEM A DQS P<0>	23	75 7	MEM B DQS P<0>	MAKE_BASE=TRUE				=MEM B DQS P<0>	24
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	75 7	MEM A DQ<1>	MAKE_BASE=TRUE			=MEM A DQ<1>	23	75 7	MEM B DQ<1>	MAKE_BASE=TRUE				=MEM B DQ<1>	24
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	75 7	MEM A DQ<26>	MAKE_BASE=TRUE			=MEM A DQ<26>	23	75 7	MEM B DQ<26>	MAKE_BASE=TRUE				=MEM B DQ<26>	24
	75 7	MEM A DQ<25>	MAKE_BASE=TRUE			=MEM A DQ<25>	23	75 7	MEM B DQ<25>	MAKE_BASE=TRUE				=MEM B DQ<25>	24
	75 7	MEM A DQ<24>	MAKE_BASE=TRUE			=MEM A DQ<24>	23	75 7	MEM B DQ<24>	MAKE_BASE=TRUE				=MEM B DQ<24>	24
SYNC_MASTER=J16_NICK SYNC_DATE=01/10/2013 DDR3 ALIASES AND BITSWAPS  Apple Inc. DRAWING NUMBER 051-0164 SIZE D REVISION 12.4.0 NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED PAGE 27 OF 123 SHEET 25 OF 86															
8	7	6	5	4	3	2	1								

SYNC MASTER=J16 NICK

SYNC DATE=01/10/2013

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

 Apple Inc.

DRAWING NUMBER
051-0164

SIZE
D

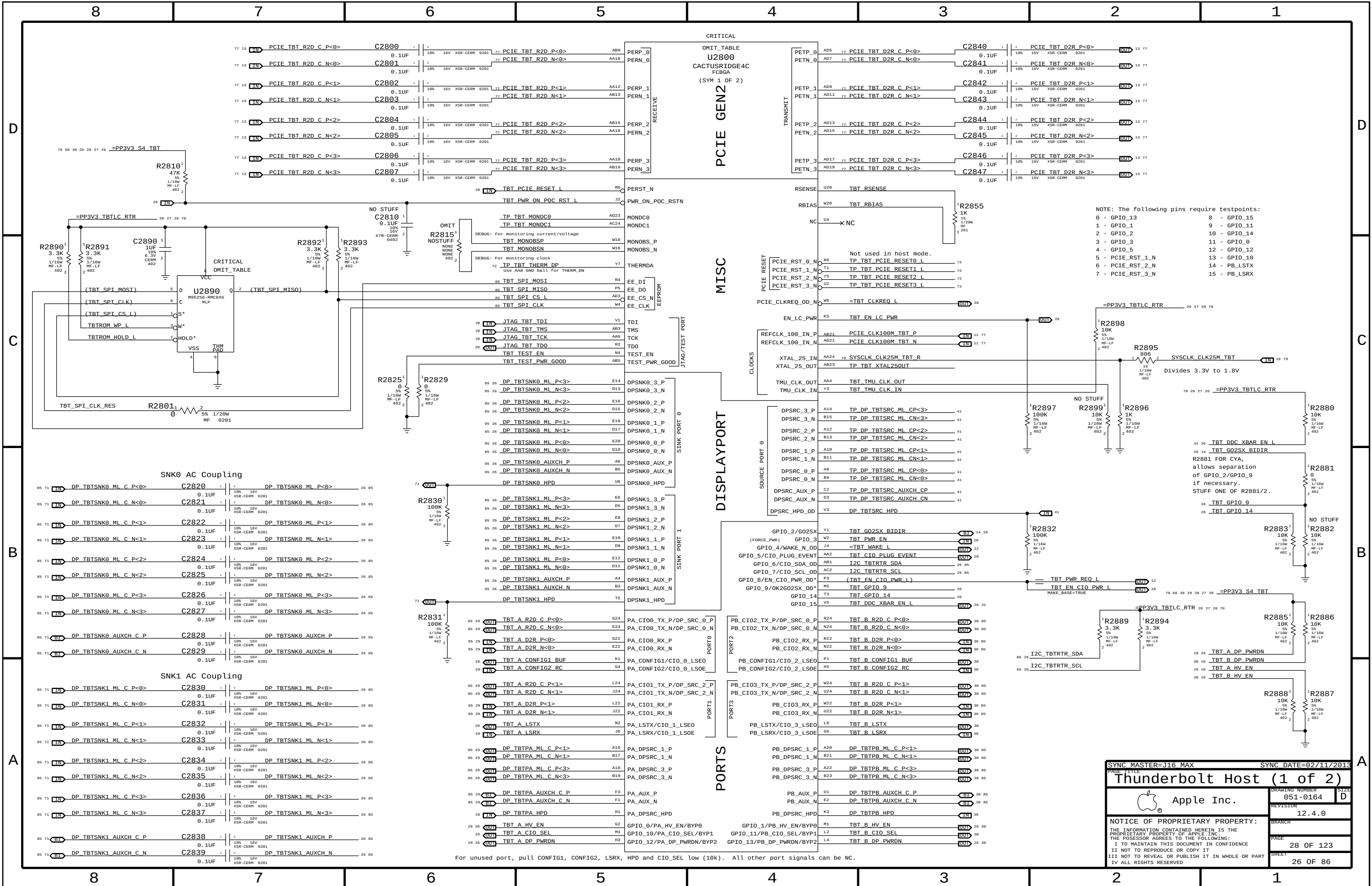
REVISION
12.4.0

BRANCH

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PAGE
27 OF 123

SHEET
25 OF 86



NOTE: The following pins require testpoints:

0 - GPIO_13	8 - GPIO_15
1 - GPIO_1	9 - GPIO_11
2 - GPIO_2	10 - GPIO_14
3 - GPIO_3	11 - GPIO_0
4 - GPIO_5	12 - GPIO_12
5 - PCIE_RST_1_N	13 - GPIO_10
6 - PCIE_RST_2_N	14 - PB_LSTX
7 - PCIE_RST_3_N	15 - PB_LSRX

SYNC MASTER=J16 MAX

SYNC DATE=02/11/2013

Thunderbolt Host (1 of 2)

Apple Inc.

051-0164

12.4.0

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PAGE

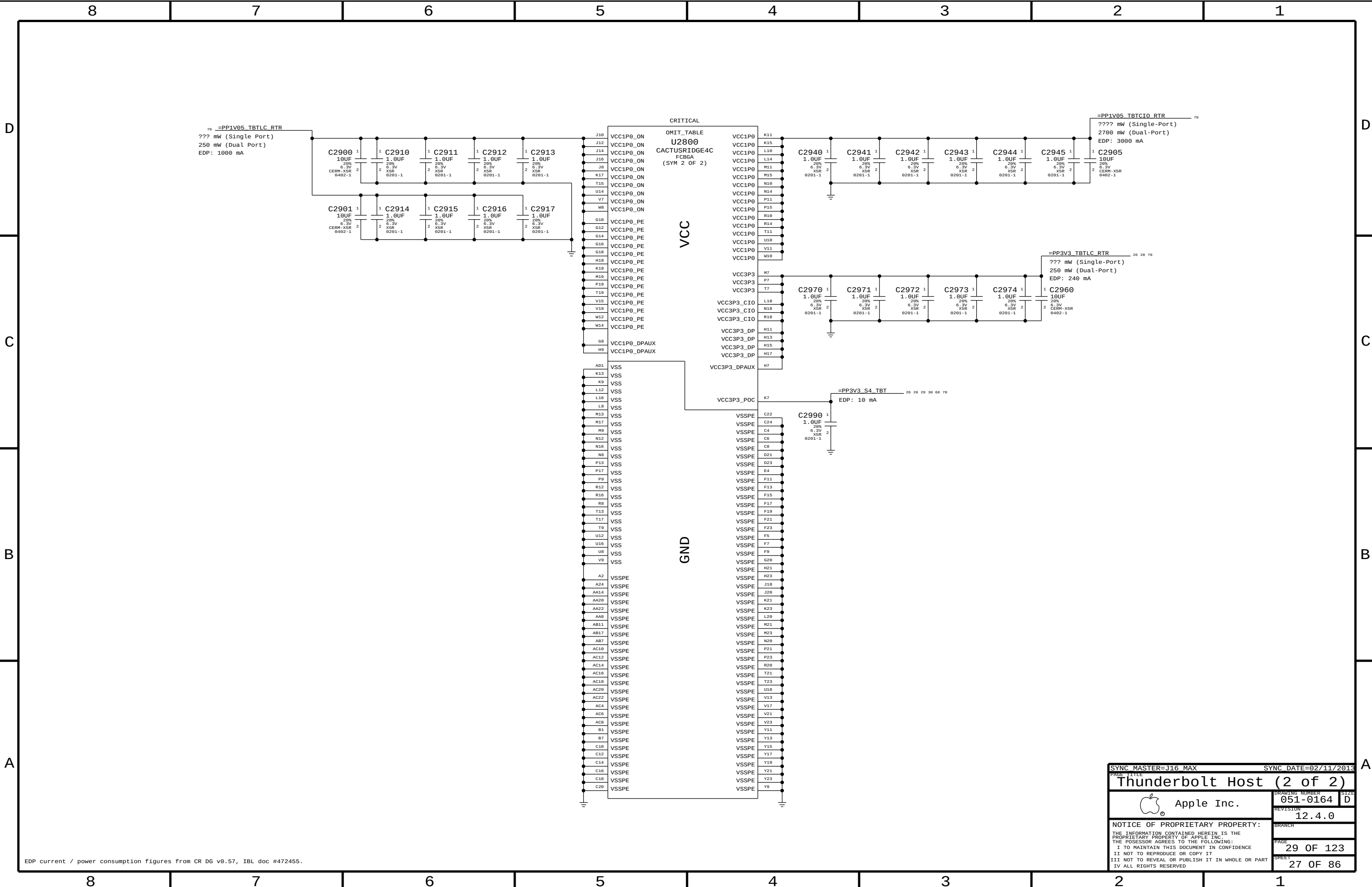
SHEET

SIZE

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26 OF 86



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC MASTER=J16 MAX		SYNC DATE=02/11/2013	
PAGE TITLE		Thunderbolt Host (2 of 2)	
Apple Inc.		DRAWING NUMBER	051-0164
		REVISION	12.4.0
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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

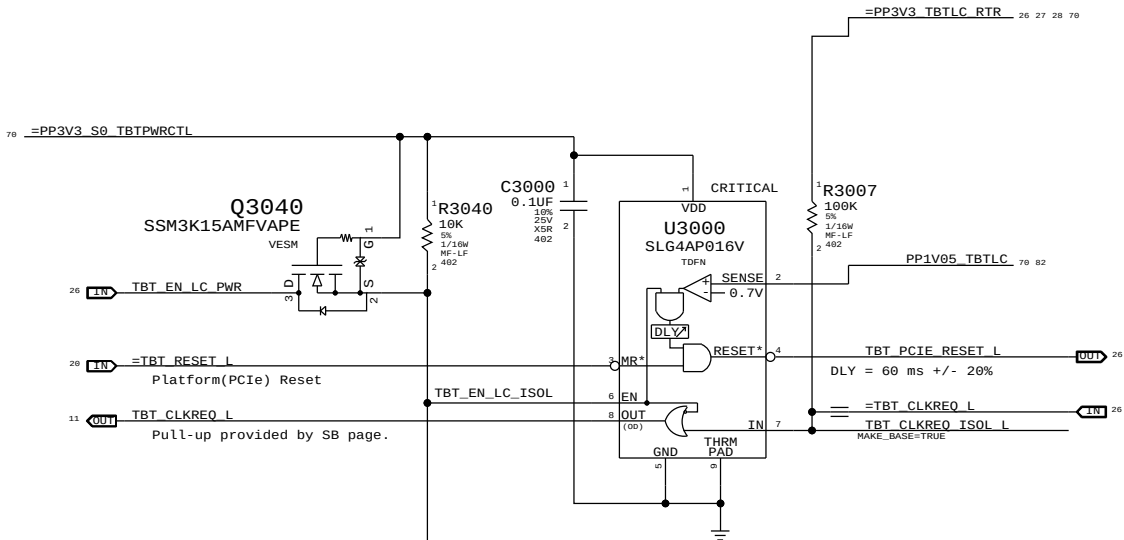
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

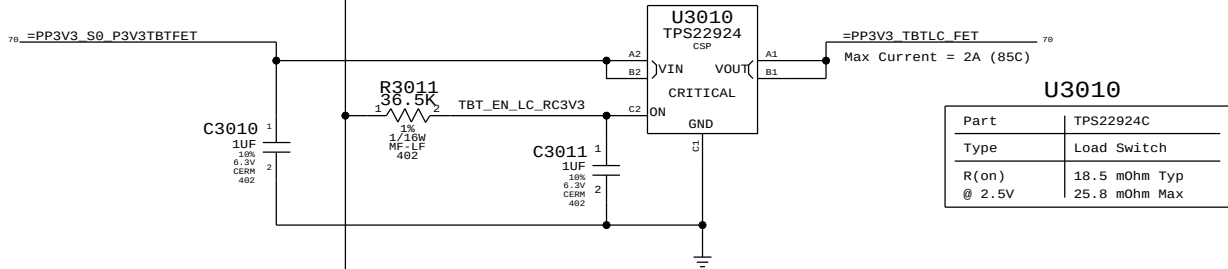
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

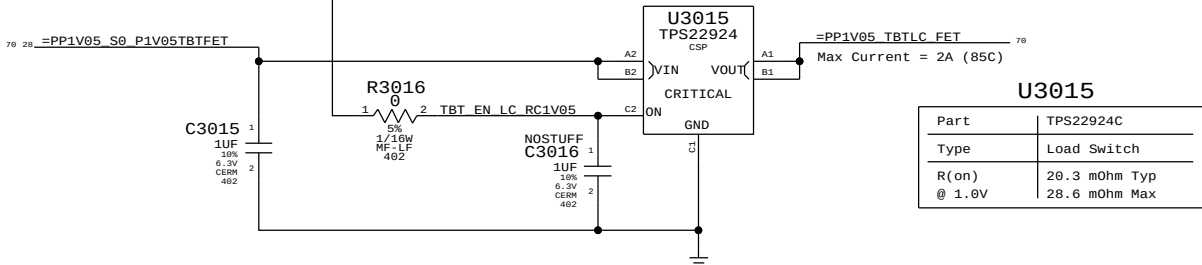
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

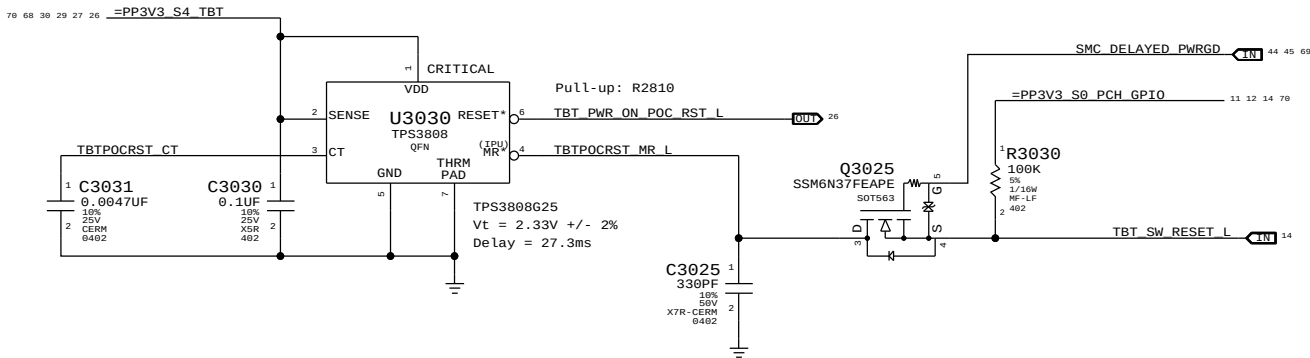


1.05V TBT "LC" Switch

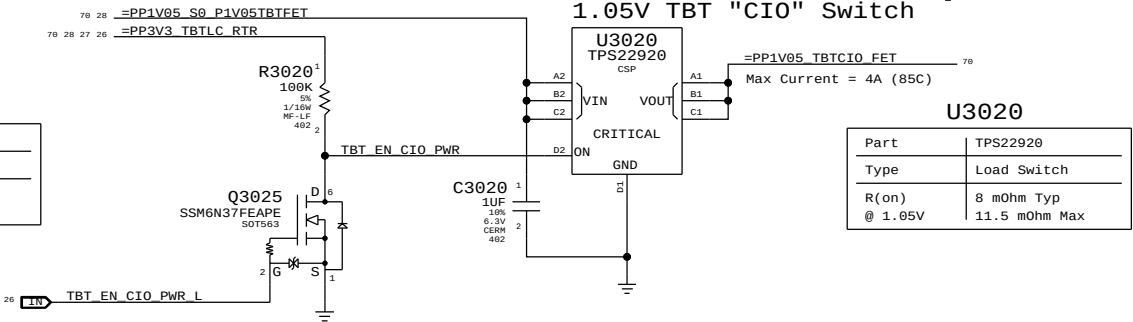


TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



SYNC MASTER=J16 MAX		SYNC DATE=02/11/2013	
PAGE TITLE		Thunderbolt Power Support	
Apple Inc.		DRAWING NUMBER	051-0164
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0596	311S0593		ALL	TI Alternate
128S0398	128S0220		ALL	3.3V INPUT CAP

D

B

A

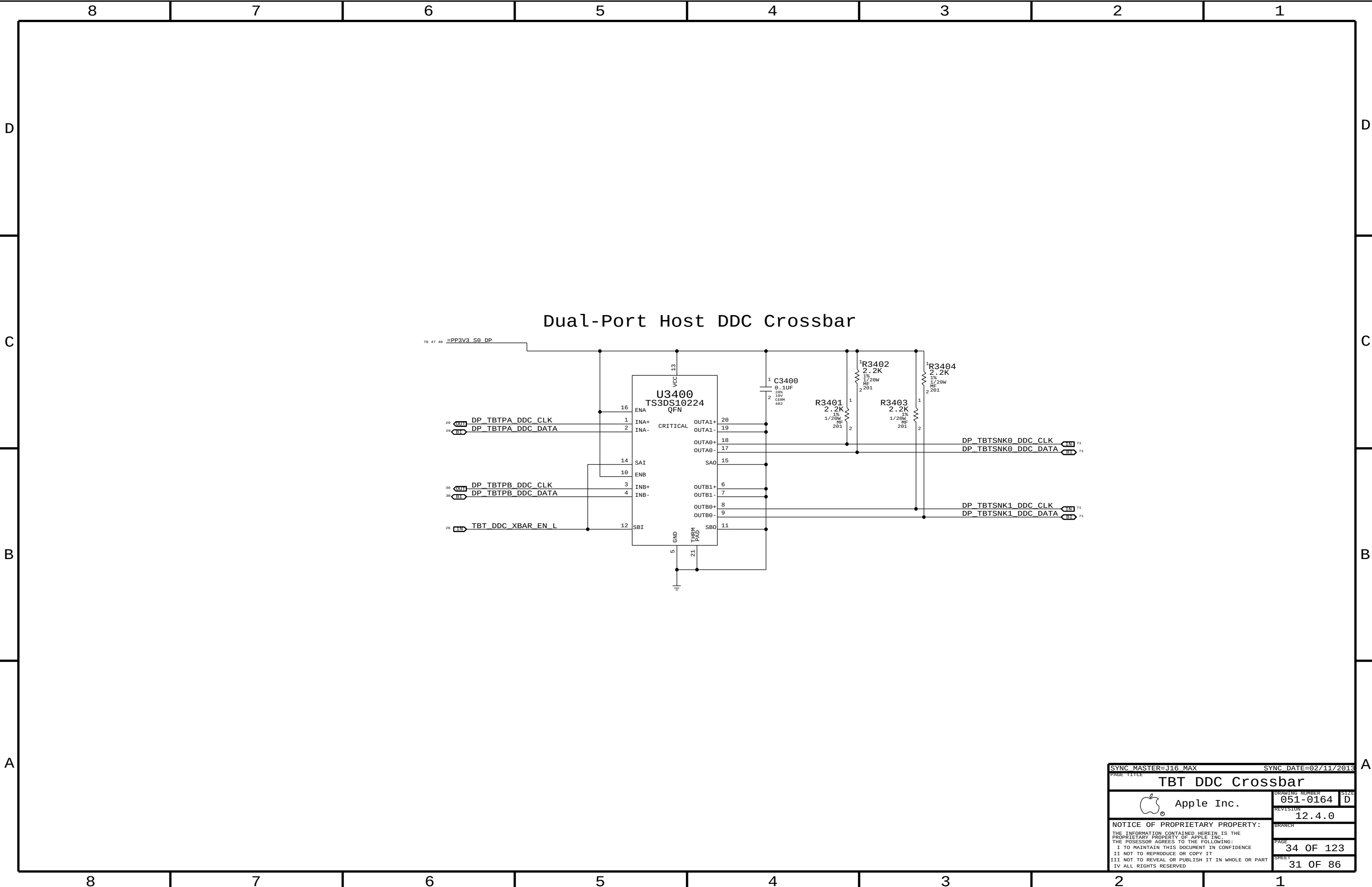


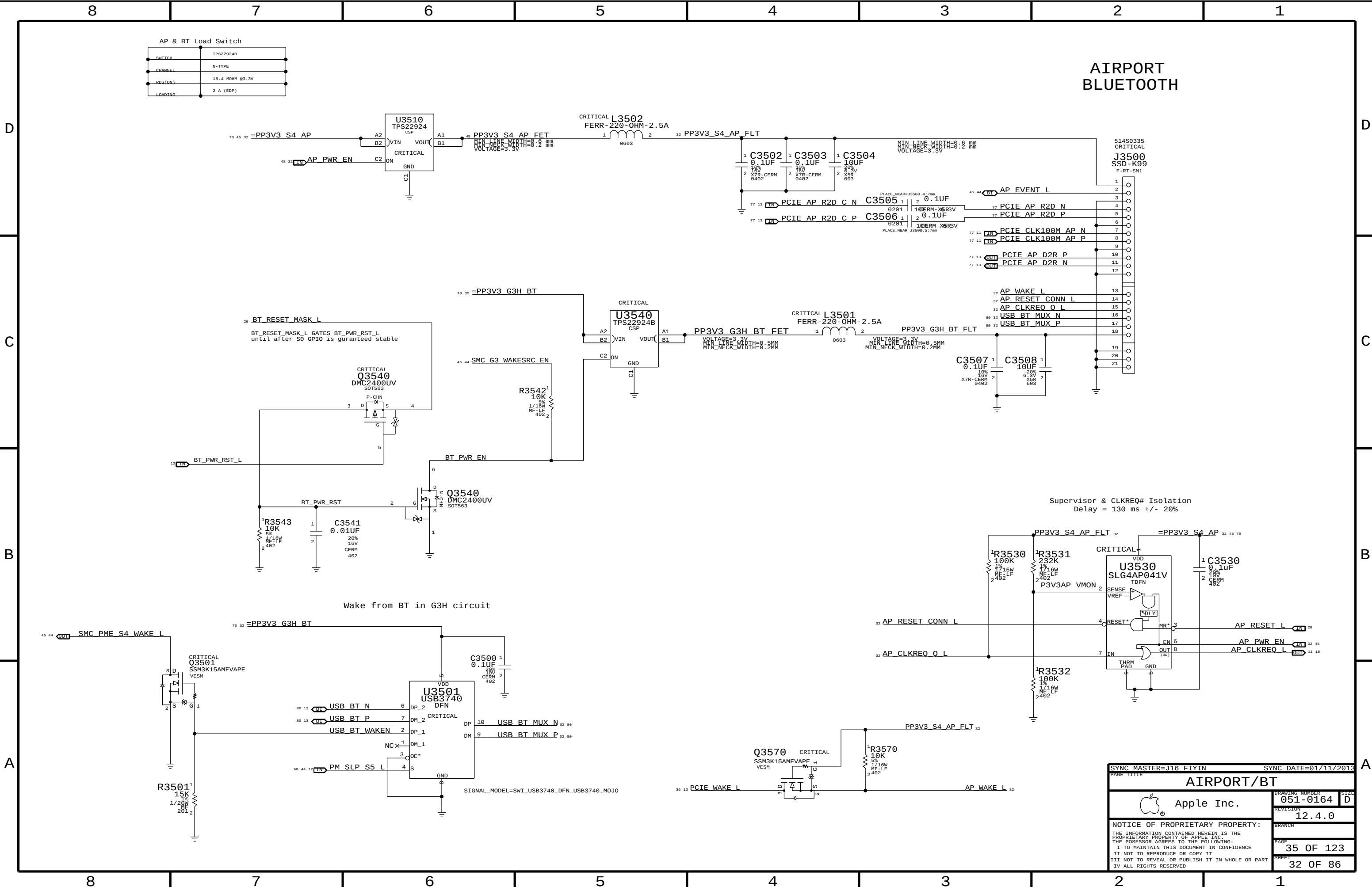
A

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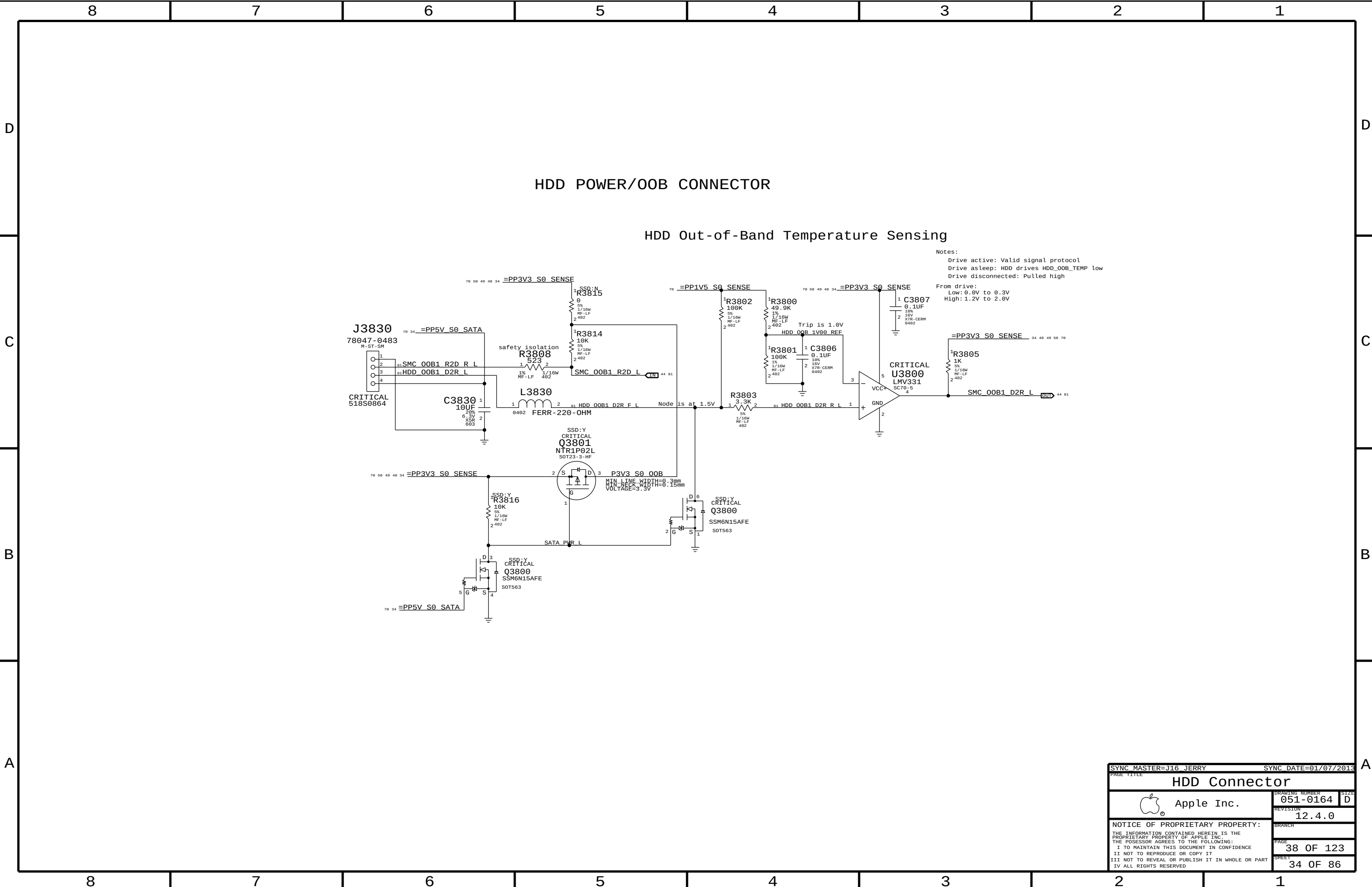
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PAGE TITLE			
Thunderbolt Connector B			
 Apple Inc.	DRAWING NUMBER	SIZE	
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		33 OF 123	
		SHEET	
		30 OF 86	

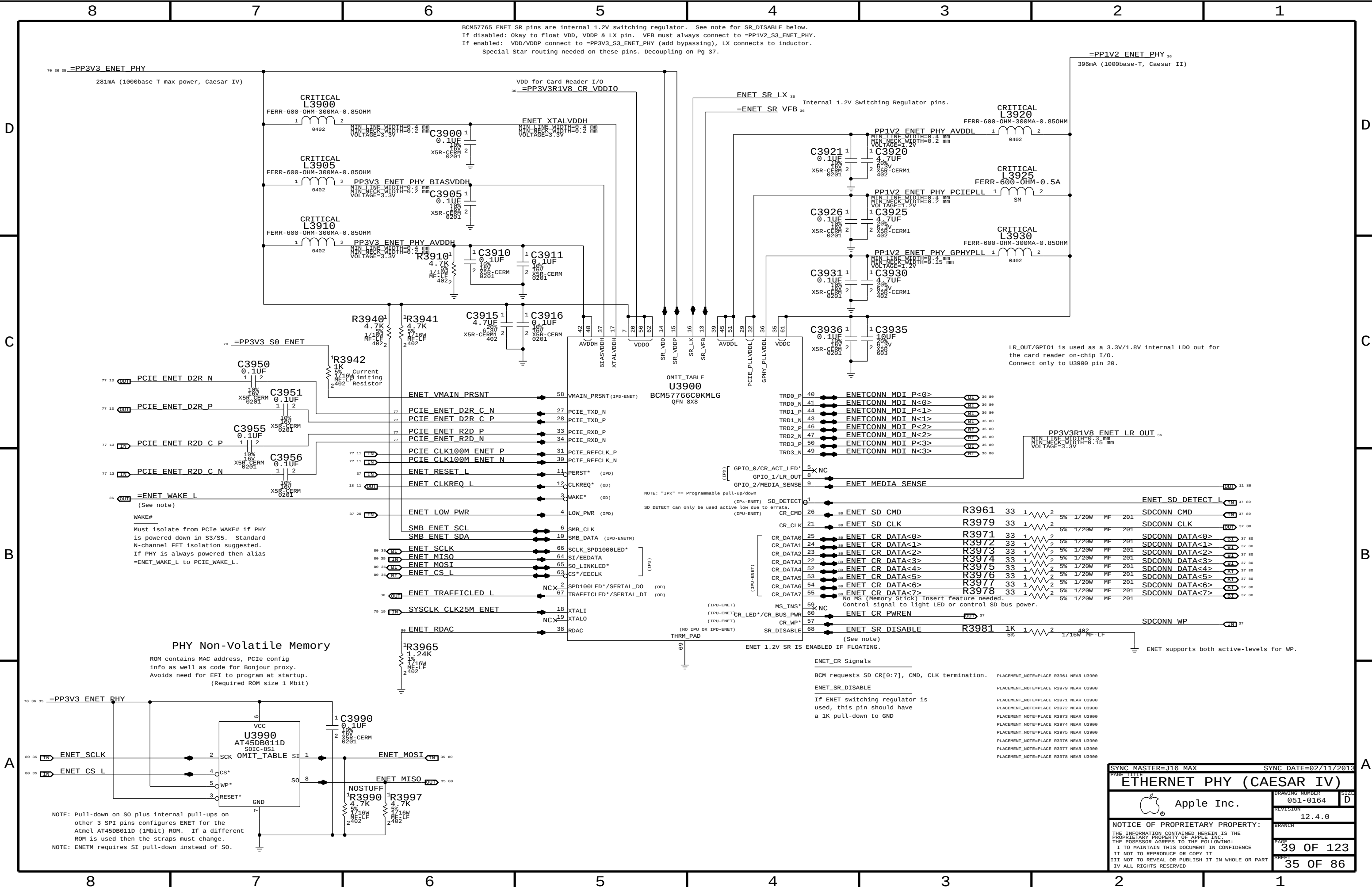




AIRPORT BLUETOOTH

SYNC MASTER=J16 FIYIN		SYNC DATE=01/11/2013	
PAGE TITLE		AIRPORT/BT	
Apple Inc.		DRAWING NUMBER	051-0164
		REVISION	12.4.0
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		PAGE	35 OF 123
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BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.

=PP1V2 ENET PHY 36
396mA (1000base-T, Caesar II)

=PP3V3 ENET PHY
281mA (1000base-T max power, Caesar IV)

ENET SR LX 36
=ENET_SR_VFB 36
Internal 1.2V Switching Regulator pins.

CRITICAL L3900
FERR-600-0HM-300MA-0.850HM

CRITICAL L3905
FERR-600-0HM-300MA-0.850HM

CRITICAL L3910
FERR-600-0HM-300MA-0.850HM

ENET_XTALVDDH
MIN LINE WIDTH=8.2 mm
MIN NECK WIDTH=8.2 mm
VOLTAGE=3.3V

C3900
0.1uF
X5R-CERM
0201

C3905
0.1uF
X5R-CERM
0201

C3910
0.1uF
X5R-CERM
0201

C3911
0.1uF
X5R-CERM
0201

C3915
4.7uF
X5R-CERM
0201

C3916
0.1uF
X5R-CERM
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C3917
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X5R-CERM
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C3918
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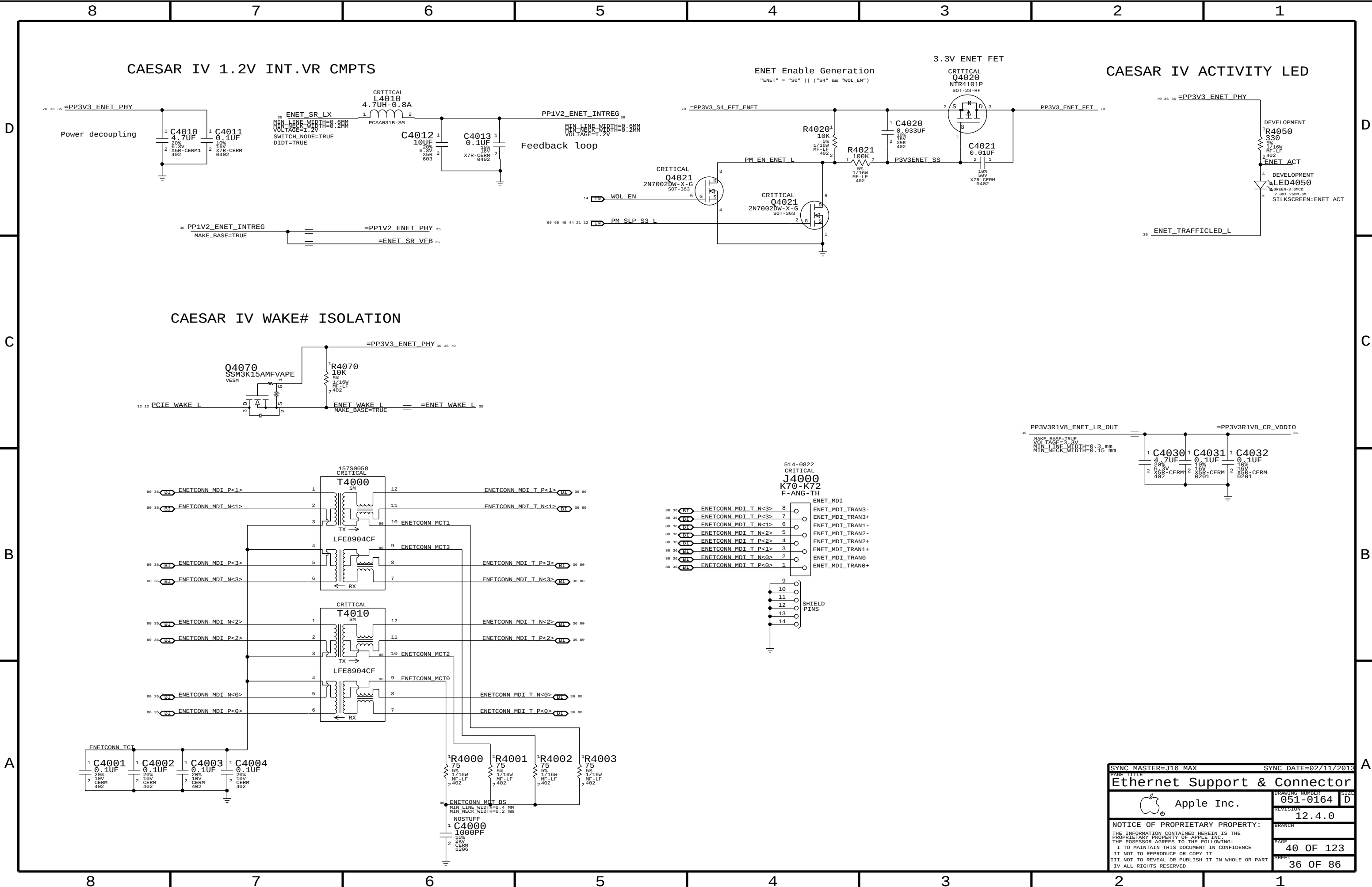
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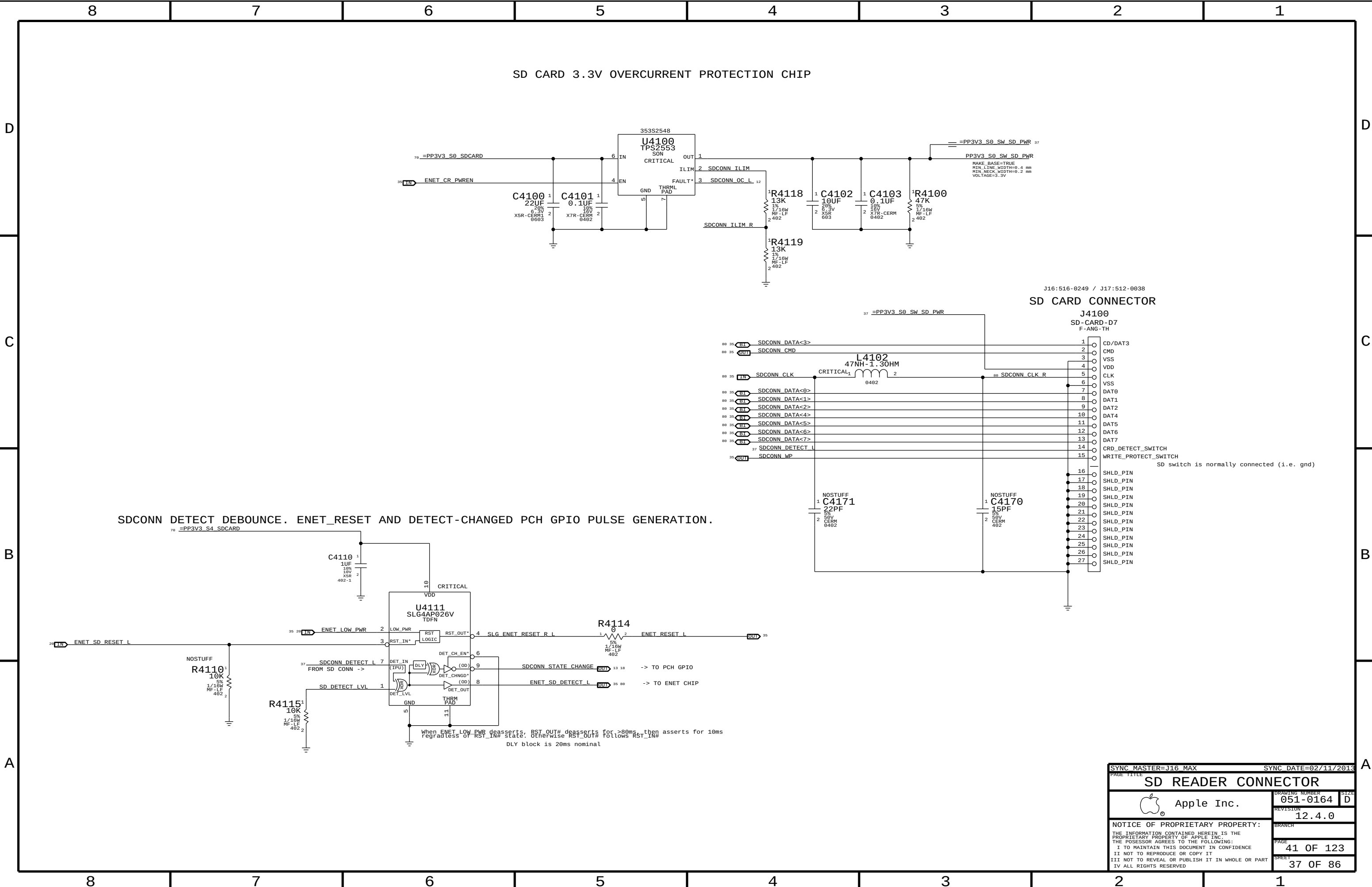
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0201

C4098
0.1uF
X5R-CERM
0201

C4099
0.1uF
X5R-CERM
0201

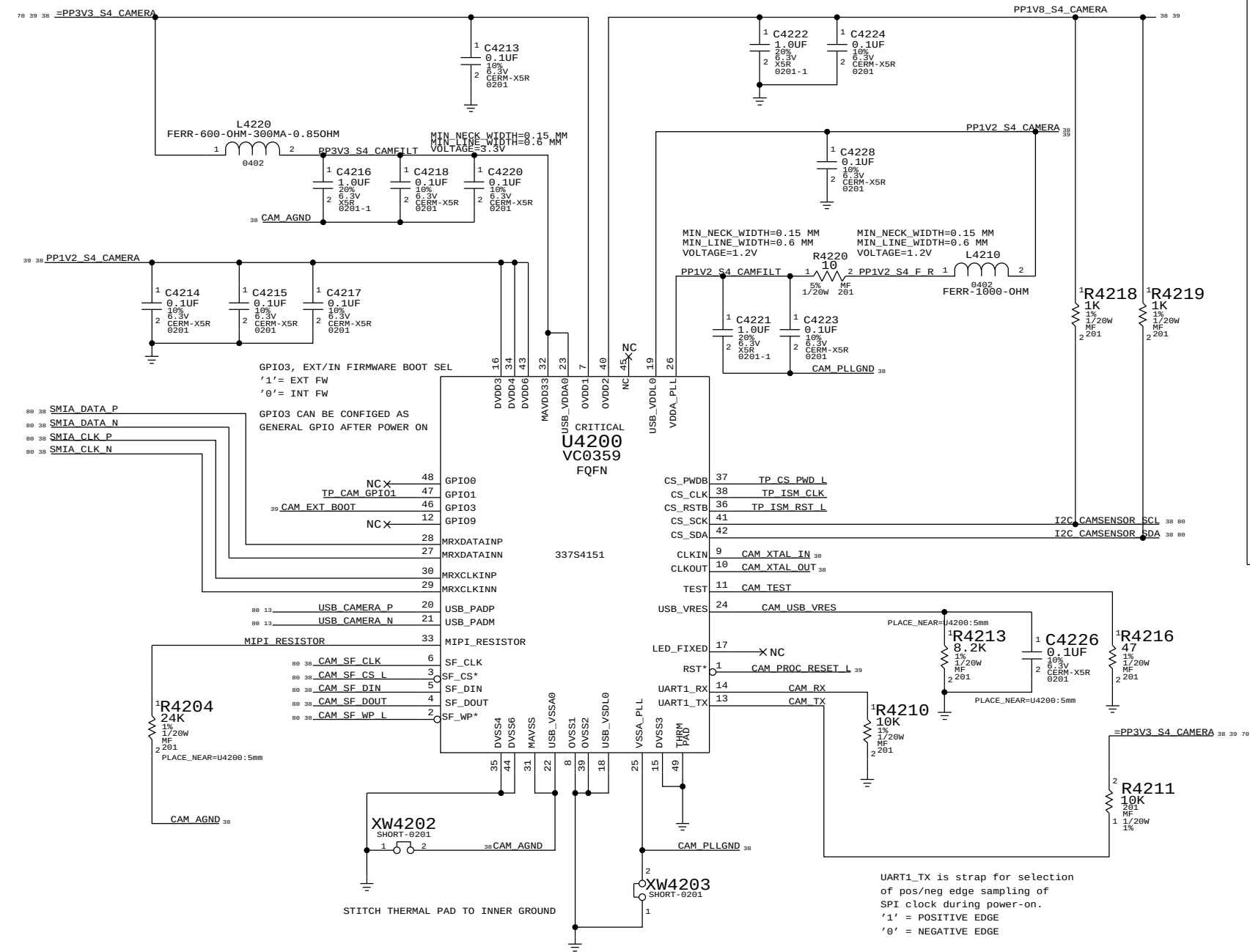




SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.

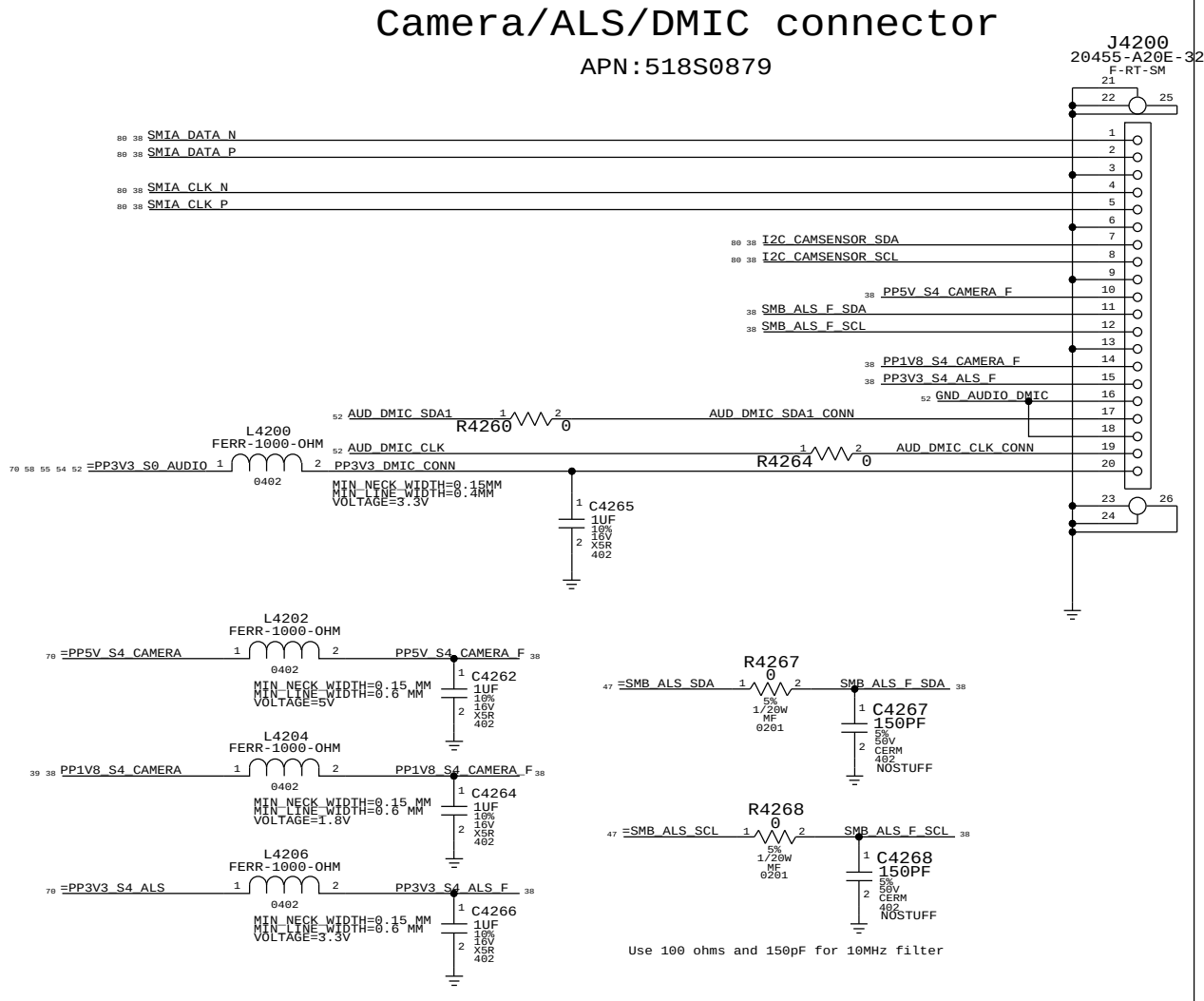
PAGE TITLE		PAGE TITLE	
SD READER CONNECTOR		SD READER CONNECTOR	
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USB CAMERA CONTROLLER

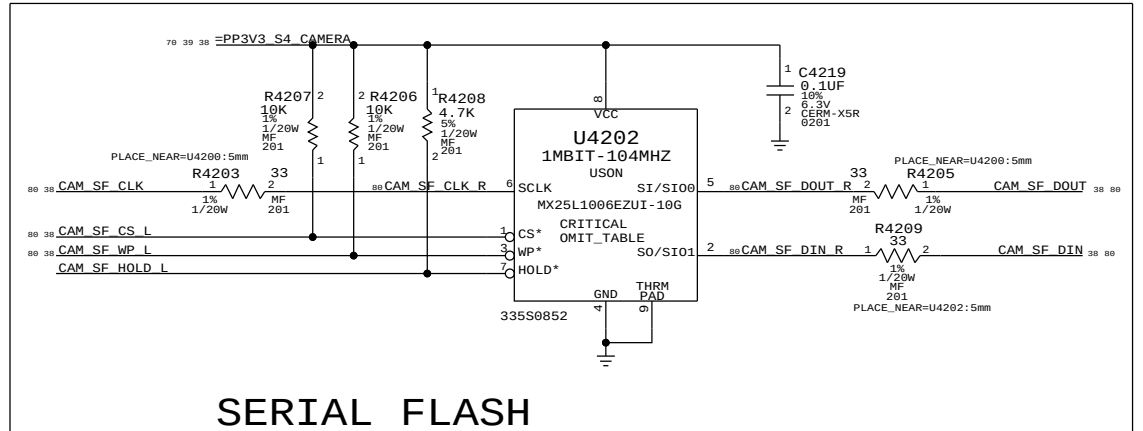


Camera/ALS/DMIC connector

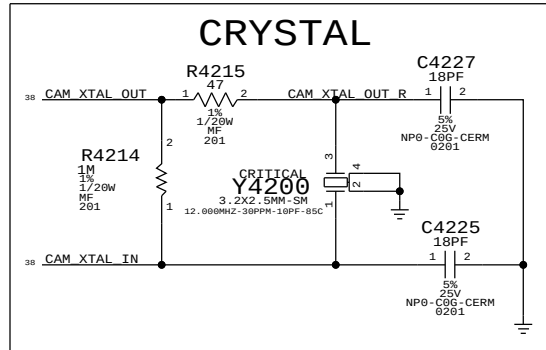
APN:518S0879



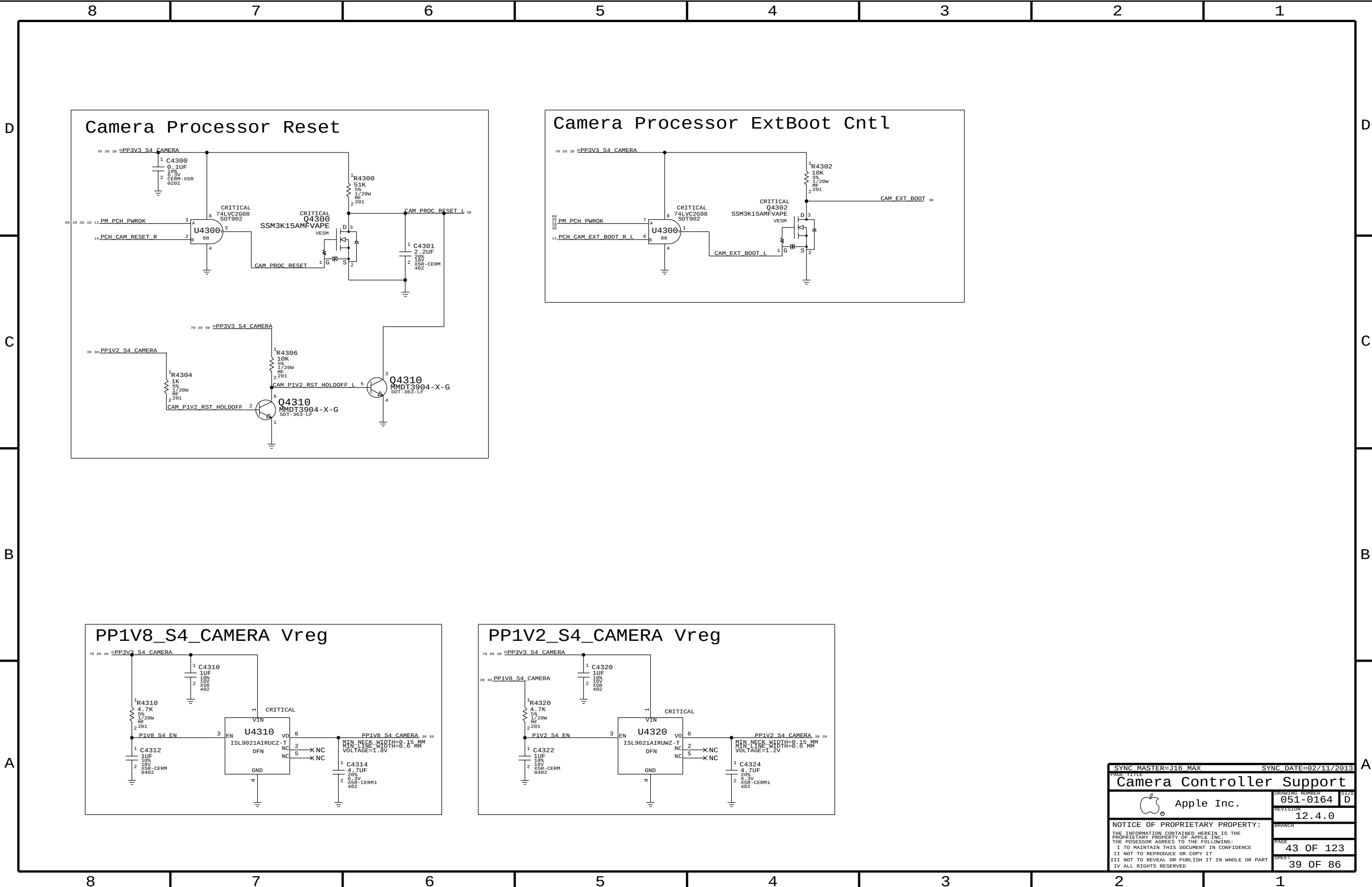
SERIAL FLASH



CRYSTAL



SYNC MASTER=J16 MAX		SYNC DATE=02/11/2013	
PAGE TITLE		Camera Controller	
Apple Inc.		DRAWING NUMBER	051-0164
REVISION		12.4.0	SIZE D
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Internal DP Connector

518S0829
CRITICAL
J4400
20525-130E-01
F-RT-SM

Backlight Control

Delay applies only on a L->H transition on VIDEO_ON. This guarantees video is valid before the backlight is enabled.
On a H->L transition, output follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video

Internal DP Connector Pinout:

Pin	Signal
1	PP12V_S0_LCD
2	PP12V_LCD_F
3	PP12V_LCD
4	PP12V_LCD
5	PP12V_LCD
6	PP12V_LCD
7	PP12V_LCD
8	PP12V_LCD
9	PP12V_LCD
10	PP12V_LCD
11	PP12V_LCD
12	PP12V_LCD
13	SMB_DP_TCON_SDA
14	SMB_DP_TCON_SCL
15	SMB_DP_TCON_SLA_SDA
16	SMB_DP_TCON_SLA_SCL
17	DP_INT_SPDIF_AUDIO
18	DP_INTPNL_HPD
19	DP_INTPNL_AUX_N
20	DP_INTPNL_AUX_P
21	DP_INTPNL_AUX_P
22	DP_INTPNL_ML_P<0>
23	DP_INTPNL_ML_N<0>
24	DP_INTPNL_ML_P<1>
25	DP_INTPNL_ML_N<1>
26	DP_INTPNL_ML_P<1>
27	DP_INTPNL_ML_N<1>
28	VIDEO_ON
29	BKLT_VSYNC
30	BKLT_VSYNC
31	BKLT_VSYNC
32	BKLT_VSYNC
33	BKLT_VSYNC
34	BKLT_VSYNC
35	BKLT_VSYNC
36	BKLT_VSYNC
37	BKLT_VSYNC
38	BKLT_VSYNC
39	BKLT_VSYNC
40	BKLT_VSYNC
41	BKLT_VSYNC
42	BKLT_VSYNC
43	BKLT_VSYNC
44	BKLT_VSYNC
45	BKLT_VSYNC
46	BKLT_VSYNC
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48	BKLT_VSYNC
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50	BKLT_VSYNC
51	BKLT_VSYNC
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54	BKLT_VSYNC
55	BKLT_VSYNC
56	BKLT_VSYNC
57	BKLT_VSYNC
58	BKLT_VSYNC
59	BKLT_VSYNC
60	BKLT_VSYNC
61	BKLT_VSYNC
62	BKLT_VSYNC
63	BKLT_VSYNC
64	BKLT_VSYNC
65	BKLT_VSYNC
66	BKLT_VSYNC

Backlight Control Circuitry:

- U4450:** 74AUP2G14GM, SOT886, 74AUP2G14GM
- D4450:** BAT54XV2T1, SOT886, BAT54XV2T1
- R4450:** 30.1K, 1/20W, MF, 201
- R4451:** 100K, 1/20W, MF, 201
- C4450:** 0.1uF, 6.3V, CERM-X5R, 0201
- C4451:** 10uF, 20%, 6.3V, CERM-X5R, 0402-1

Internal DP Connector Pinout Table:

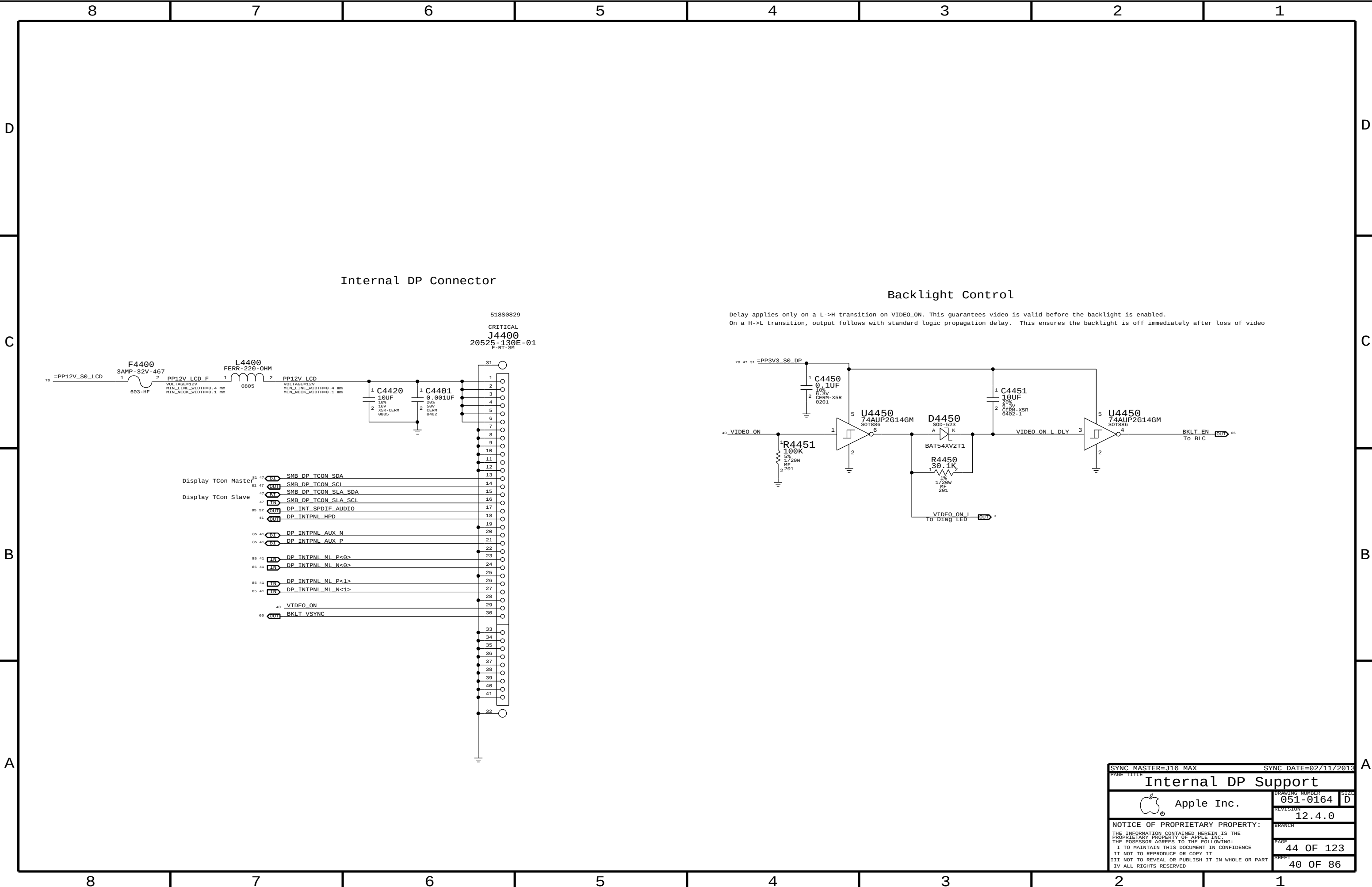
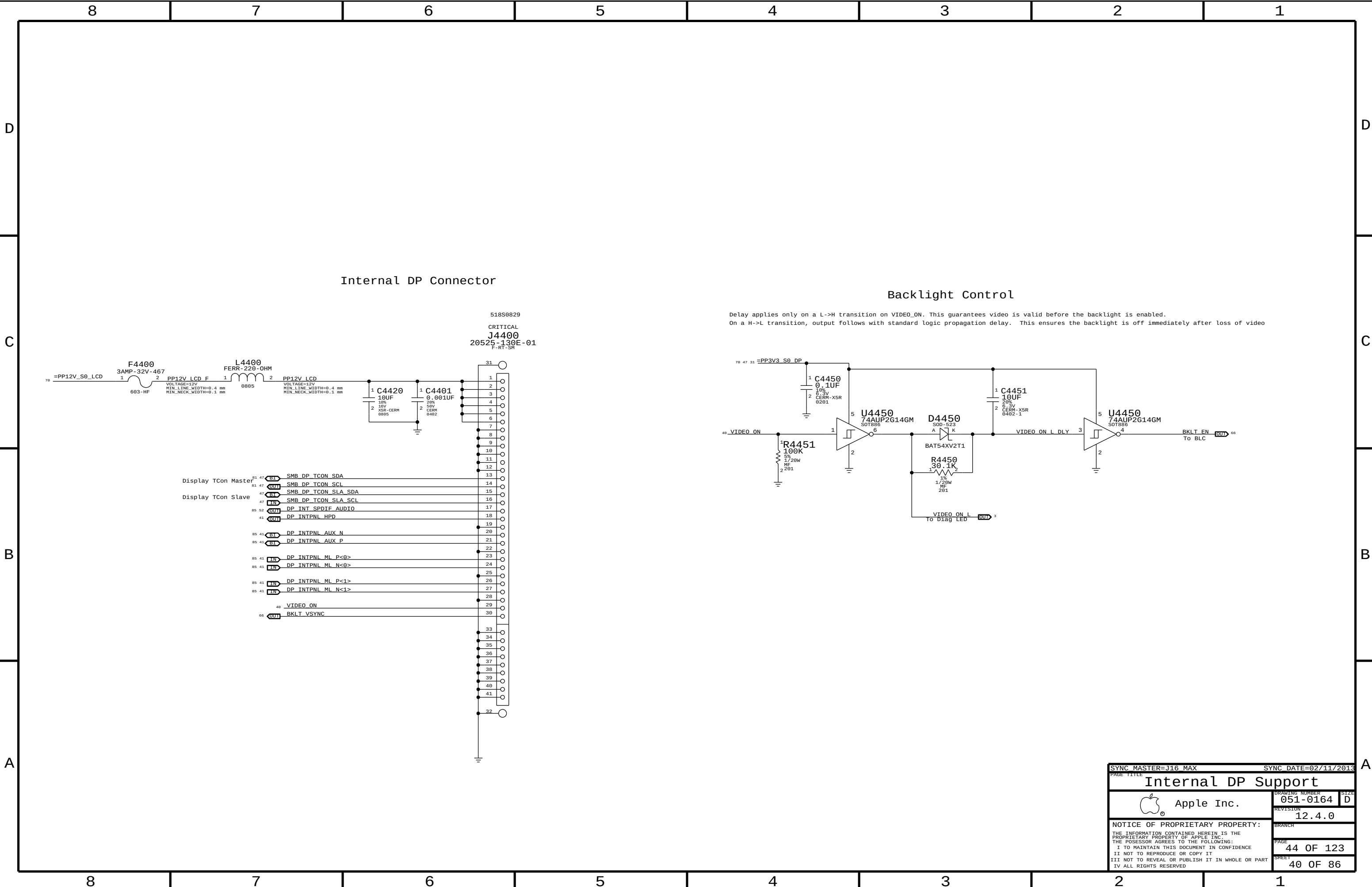
Pin	Signal
1	PP12V_S0_LCD
2	PP12V_LCD_F
3	PP12V_LCD
4	PP12V_LCD
5	PP12V_LCD
6	PP12V_LCD
7	PP12V_LCD
8	PP12V_LCD
9	PP12V_LCD
10	PP12V_LCD
11	PP12V_LCD
12	PP12V_LCD
13	SMB_DP_TCON_SDA
14	SMB_DP_TCON_SCL
15	SMB_DP_TCON_SLA_SDA
16	SMB_DP_TCON_SLA_SCL
17	DP_INT_SPDIF_AUDIO
18	DP_INTPNL_HPD
19	DP_INTPNL_AUX_N
20	DP_INTPNL_AUX_P
21	DP_INTPNL_AUX_P
22	DP_INTPNL_ML_P<0>
23	DP_INTPNL_ML_N<0>
24	DP_INTPNL_ML_P<1>
25	DP_INTPNL_ML_N<1>
26	DP_INTPNL_ML_P<1>
27	DP_INTPNL_ML_N<1>
28	VIDEO_ON
29	BKLT_VSYNC
30	BKLT_VSYNC
31	BKLT_VSYNC
32	BKLT_VSYNC
33	BKLT_VSYNC
34	BKLT_VSYNC
35	BKLT_VSYNC
36	BKLT_VSYNC
37	BKLT_VSYNC
38	BKLT_VSYNC
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40	BKLT_VSYNC
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66	BKLT_VSYNC

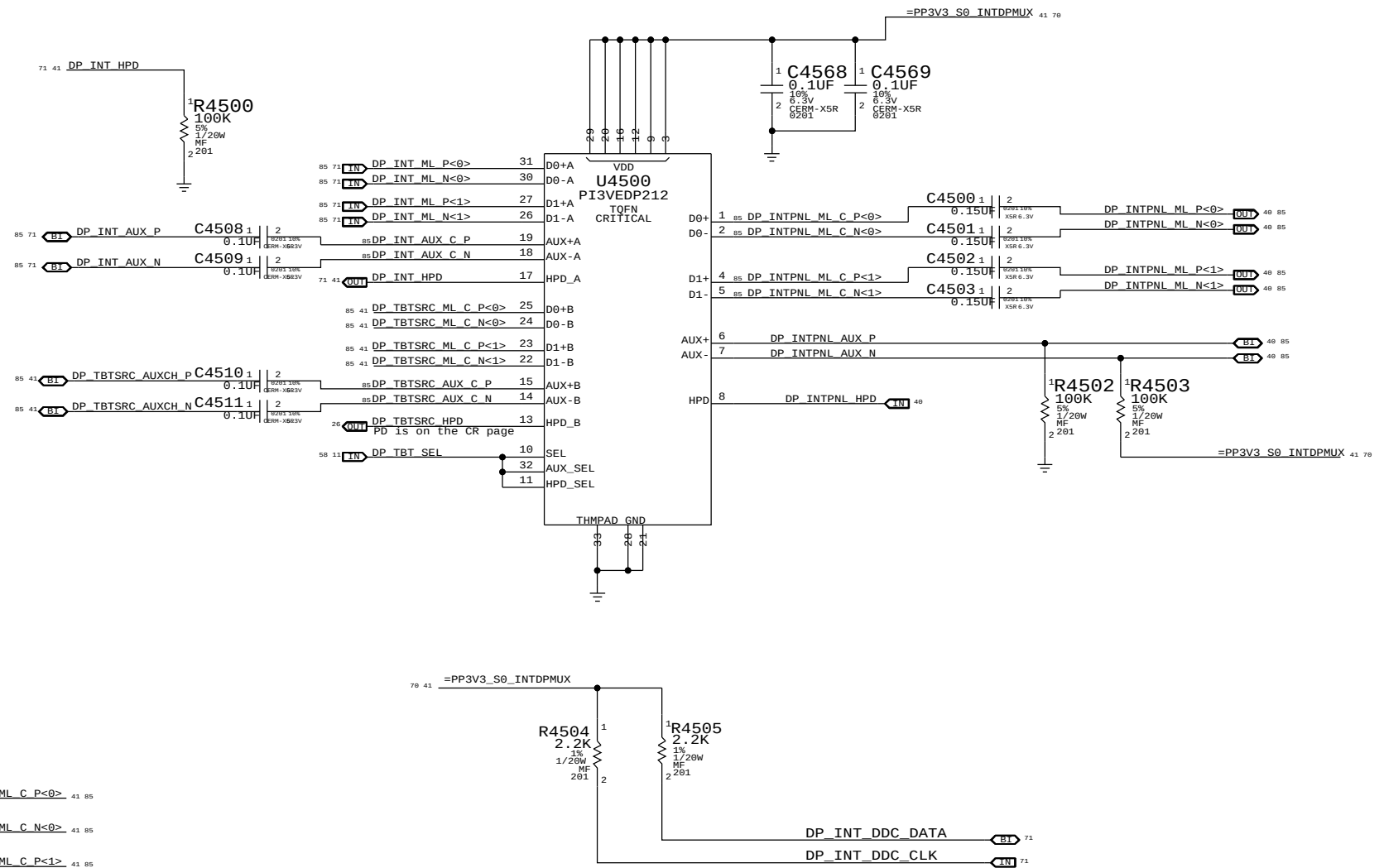
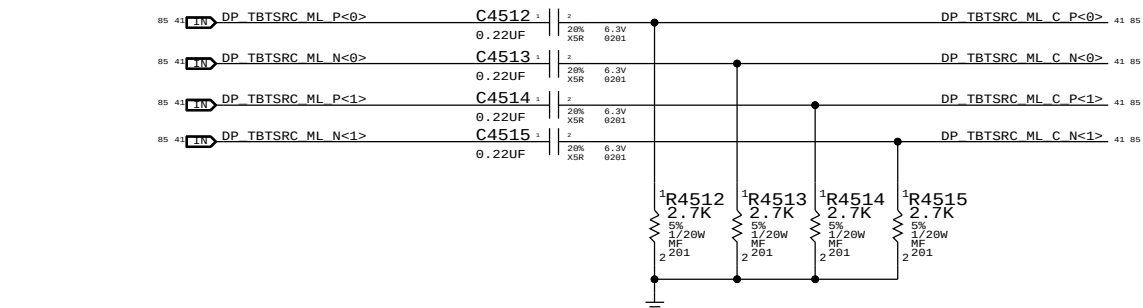
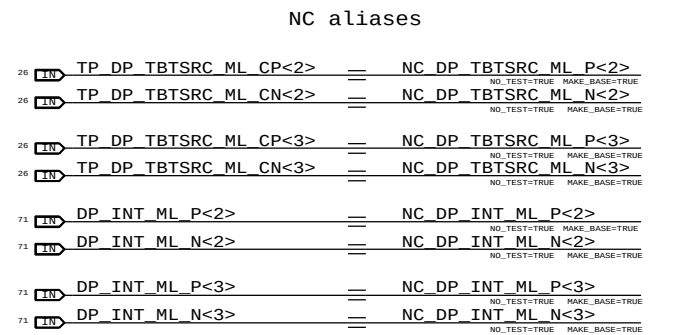
Backlight Control Circuitry:

- U4450:** 74AUP2G14GM, SOT886, 74AUP2G14GM
- D4450:** BAT54XV2T1, SOT886, BAT54XV2T1
- R4450:** 30.1K, 1/20W, MF, 201
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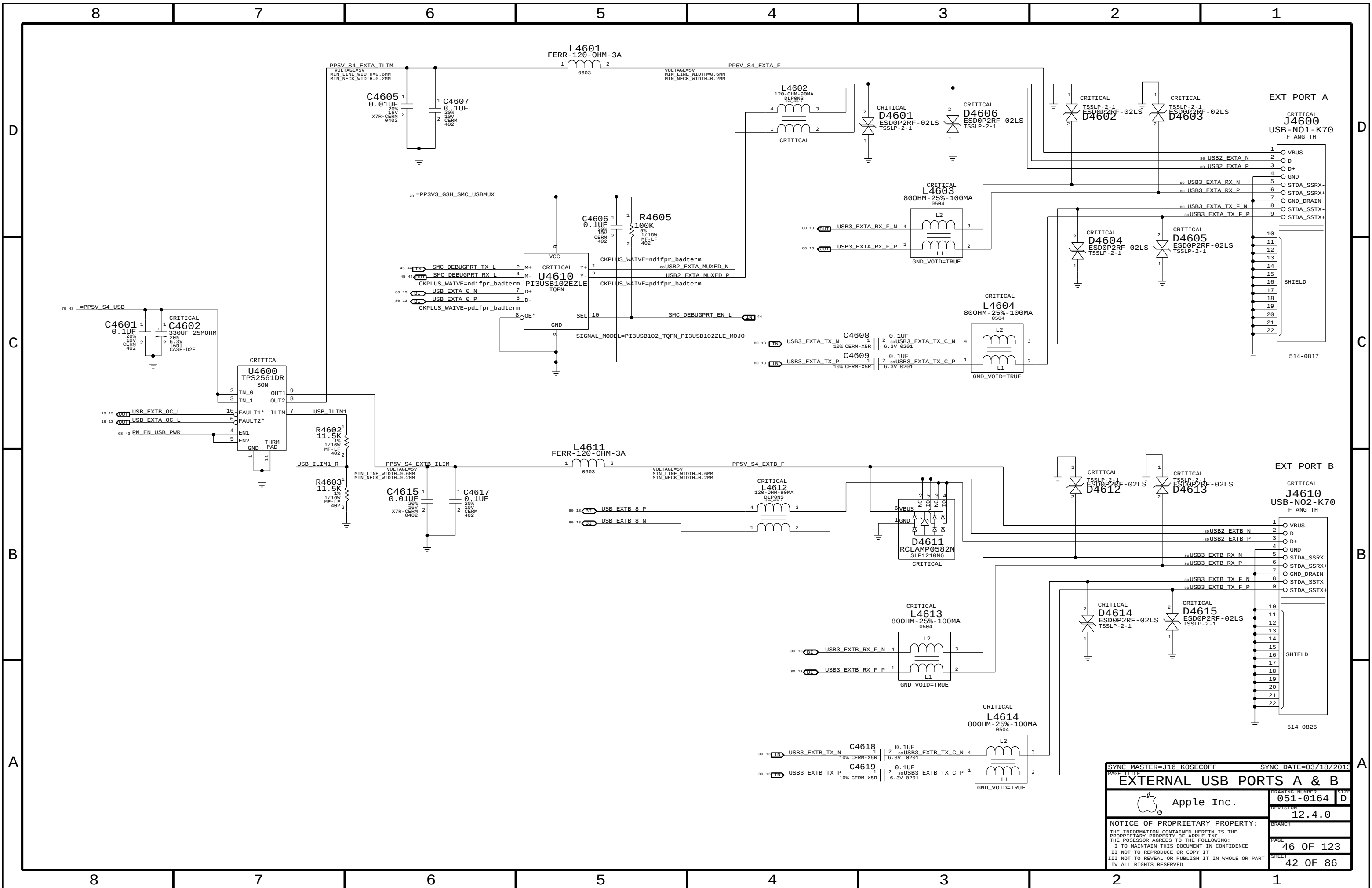
Internal DP Connector Pinout Table:

Pin	Signal
1	PP12V_S0_LCD
2	PP12V_LCD_F
3	PP12V_LCD
4	PP12V_LCD
5	PP12V_LCD
6	PP12V_LCD

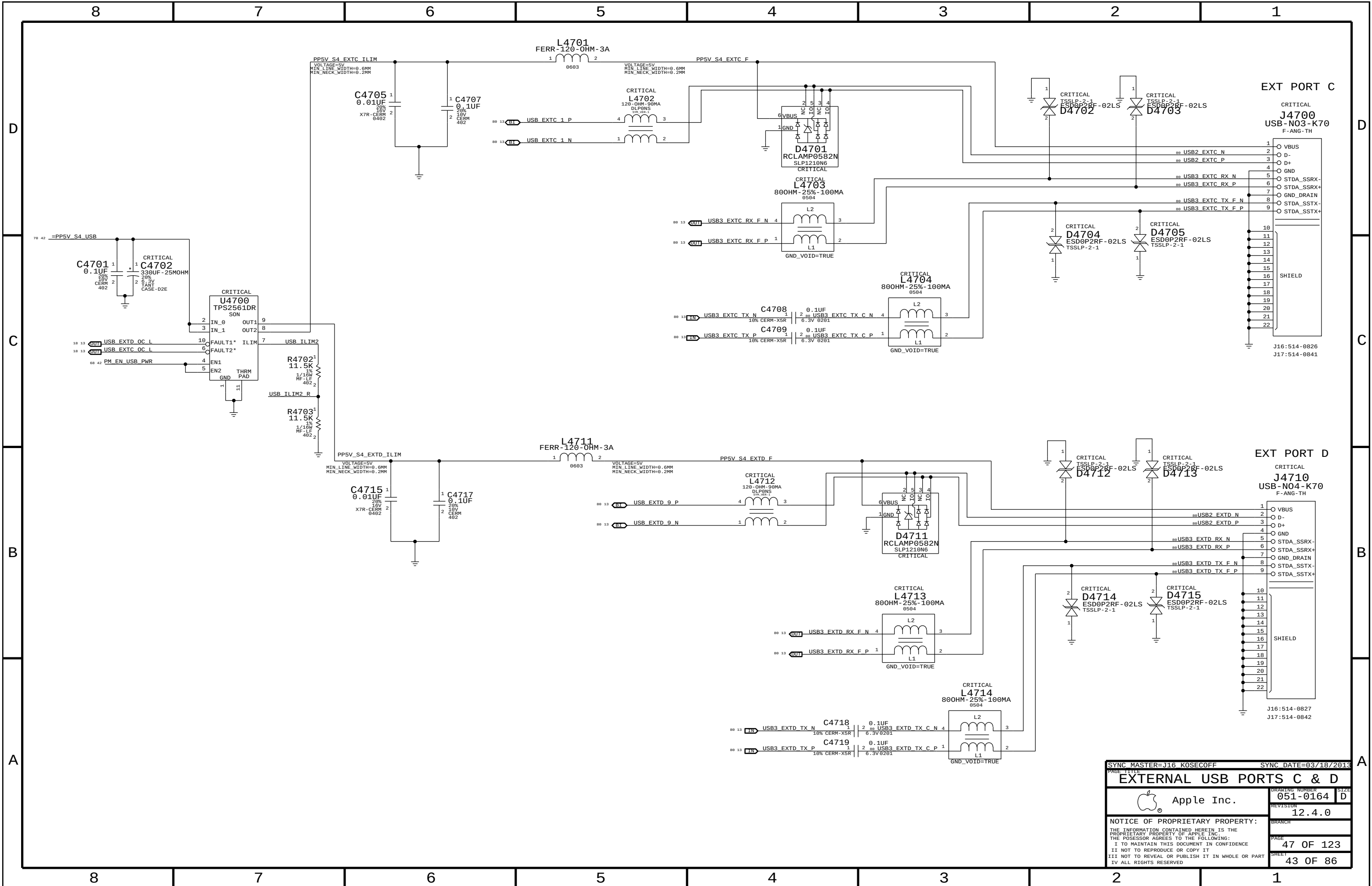
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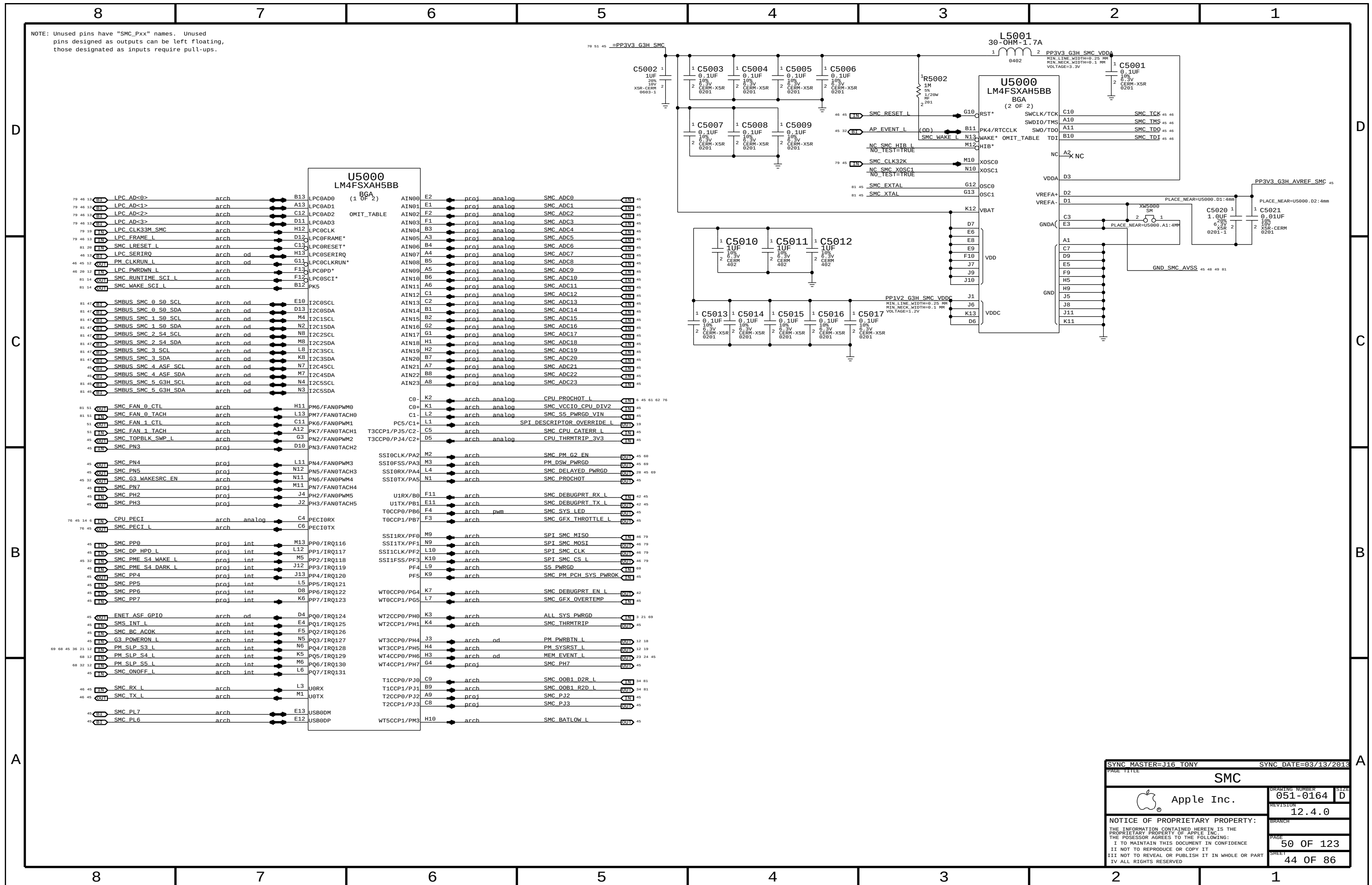
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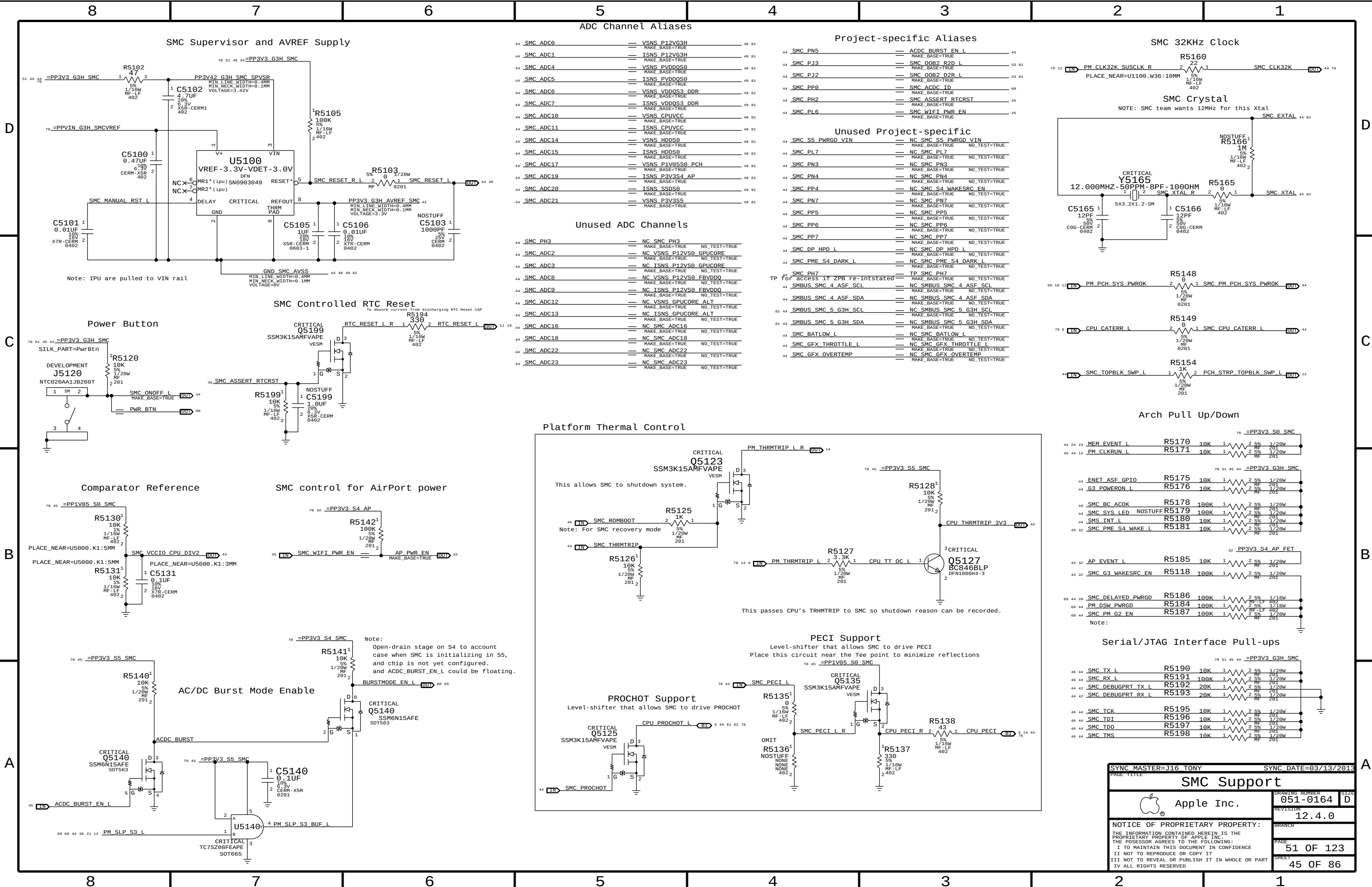


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SMC Supervisor and AVREF Supply

ADC Channel Aliases

Project-specific Aliases

SMC 32KHz Clock

SMC Crystal

Unused Project-specific

Unused ADC Channels

Arch Pull Up/Down

Power Button

SMC Controlled RTC Reset

Comparator Reference

SMC control for AirPort power

Platform Thermal Control

PECI Support

PROCHOT Support

Serial/JTAG Interface Pull-ups

SYNC MASTER=J16 TONY

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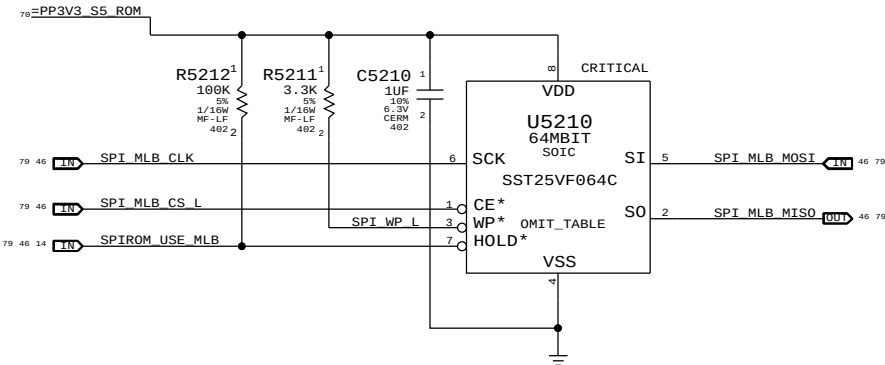
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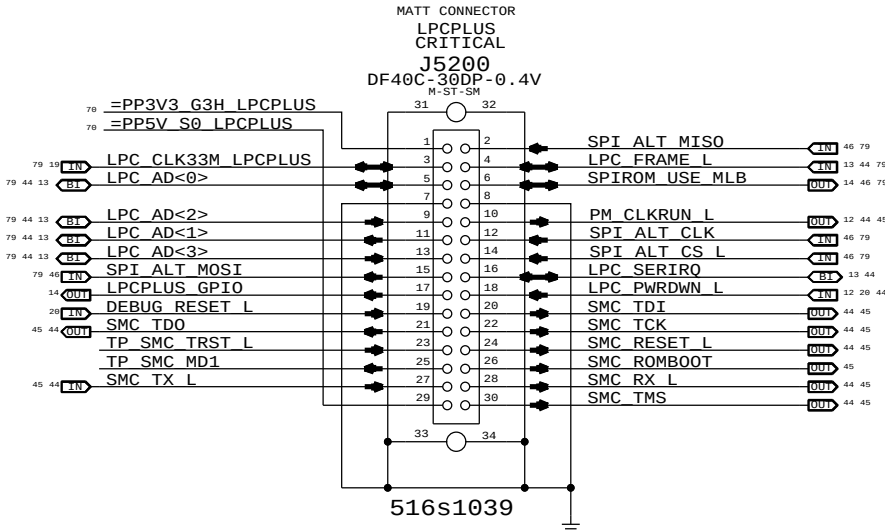
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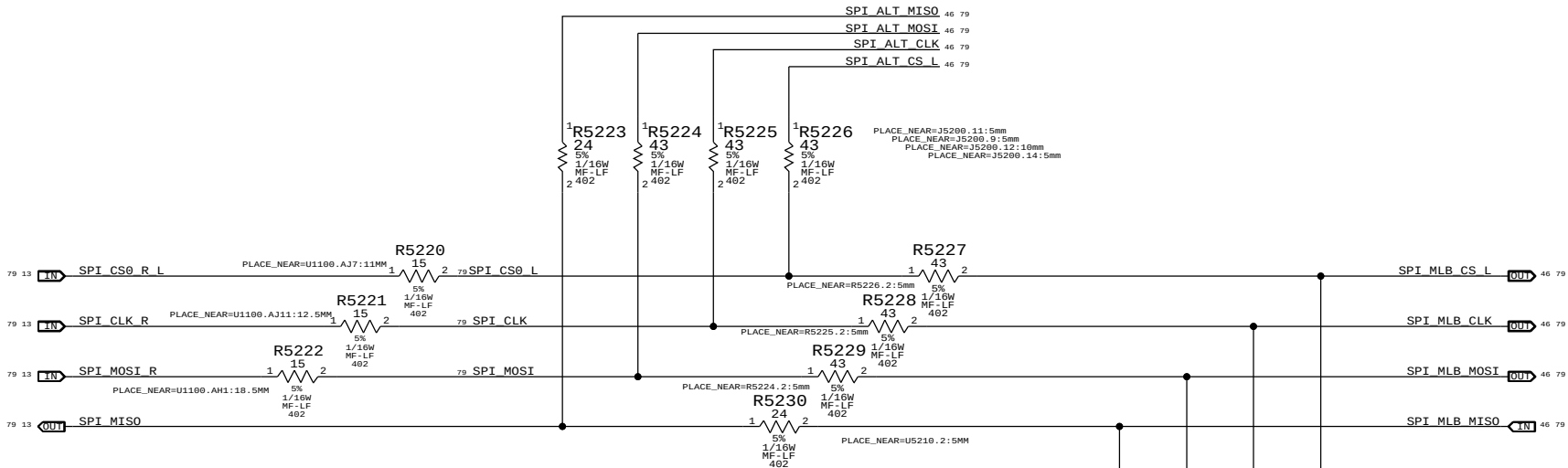
SPI BootROM



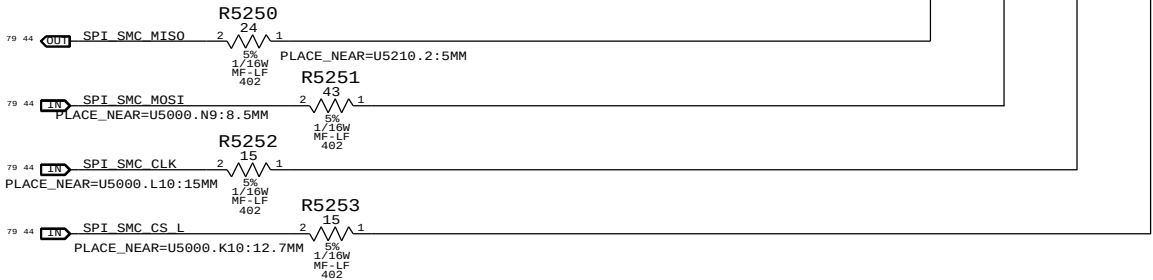
LPC+SPI Connector



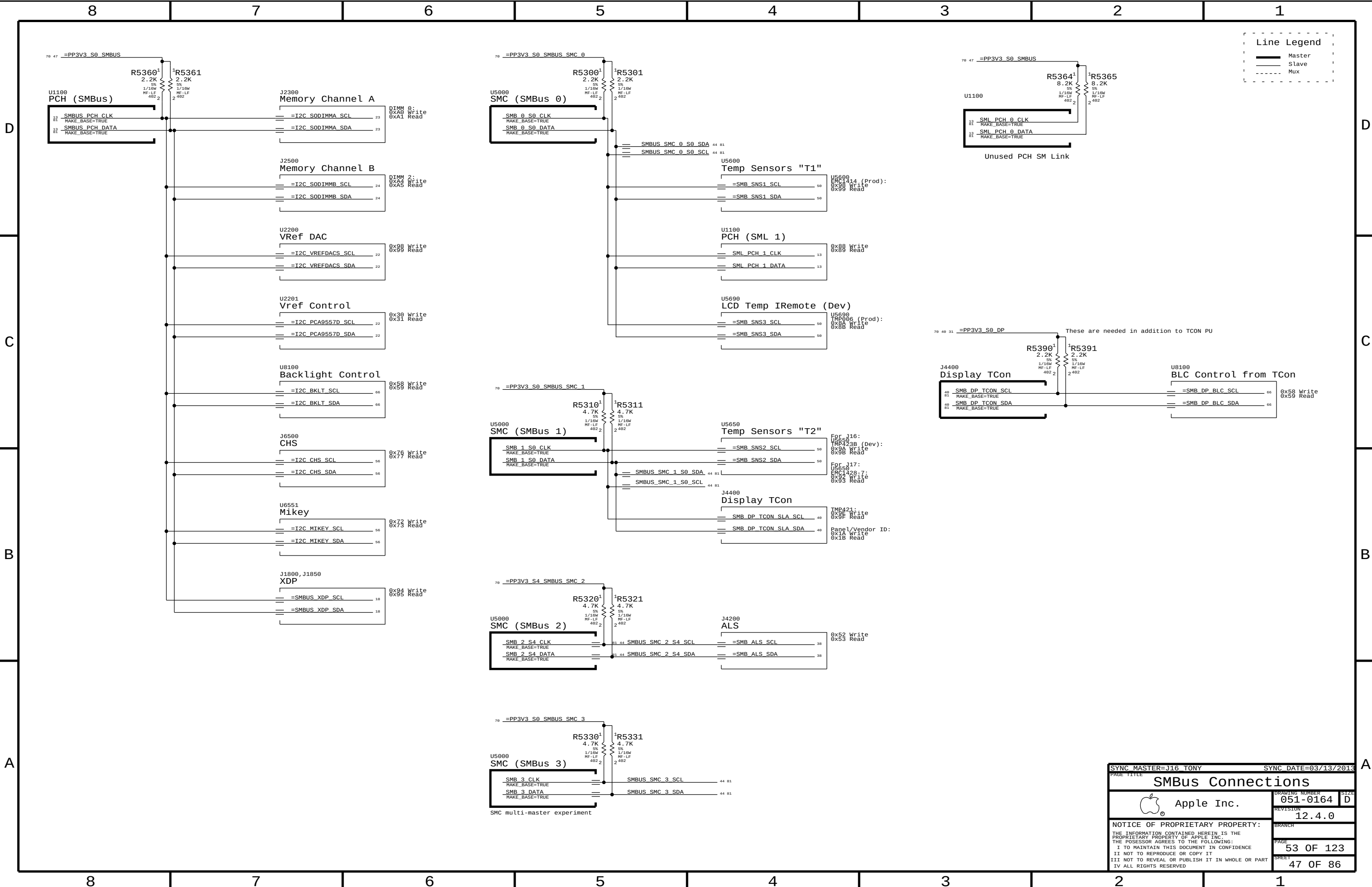
SPI Series Termination

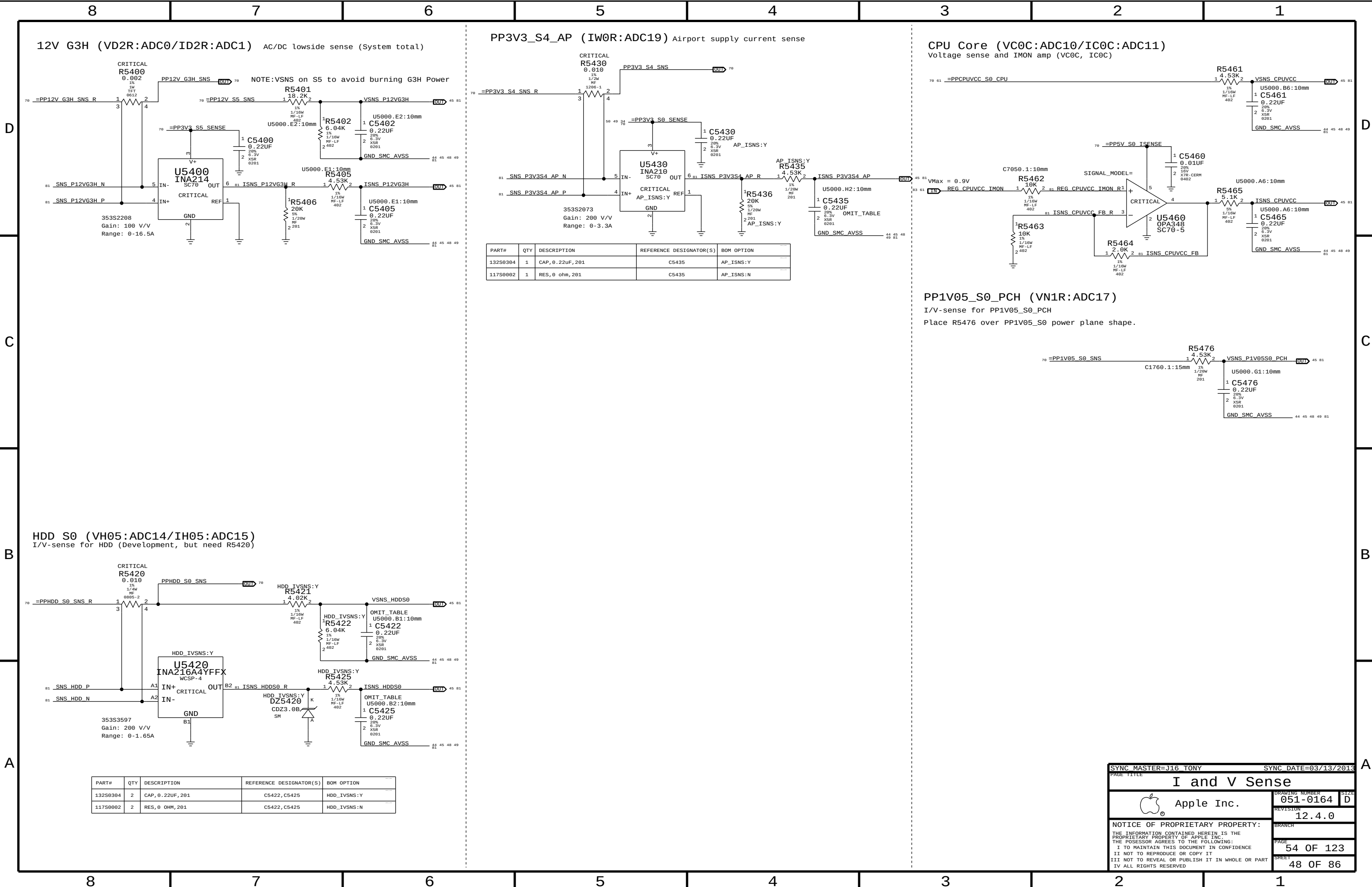


SMC SPI Support

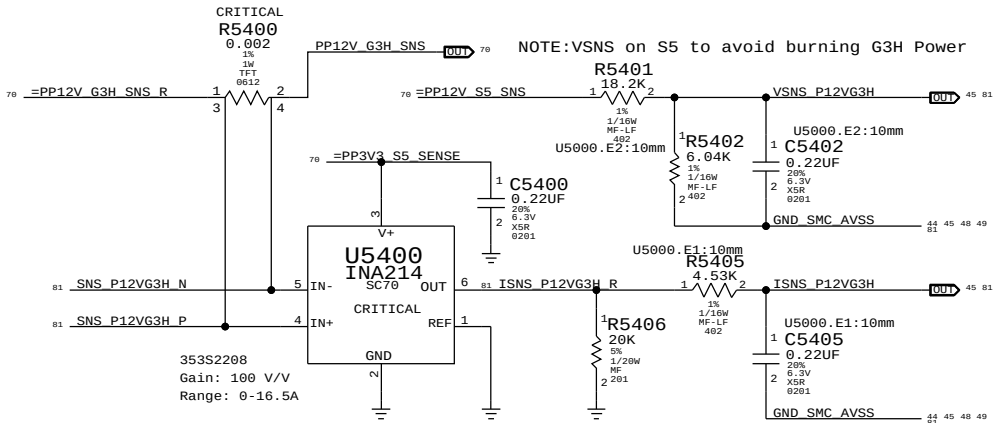


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SPI and Debug Connector		051-0164	
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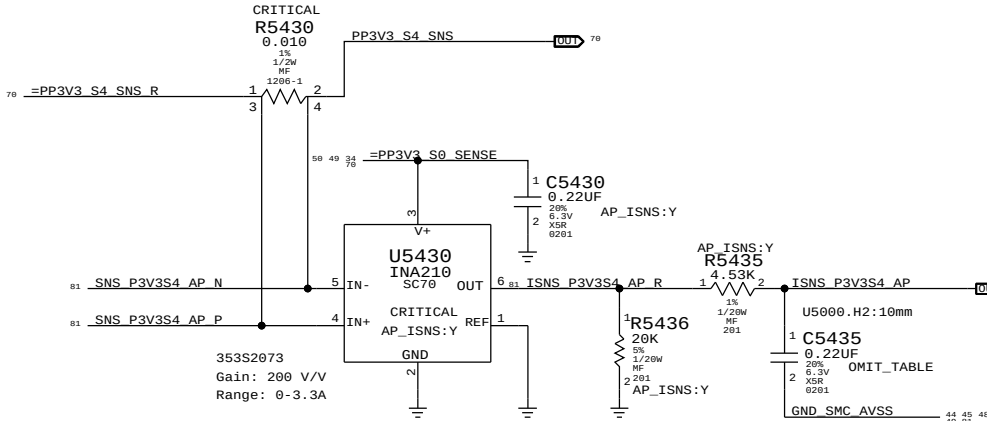




12V G3H (VD2R:ADC0/ID2R:ADC1) AC/DC lowside sense (System total)

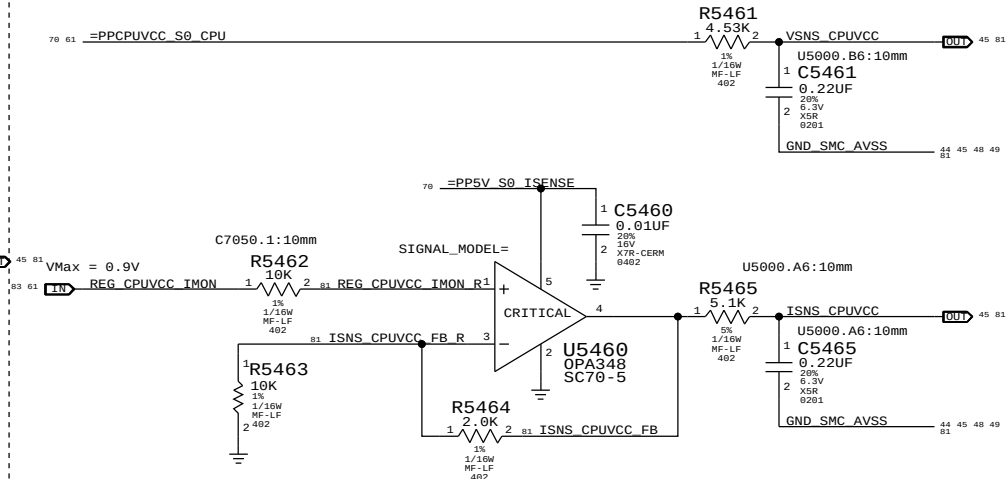


PP3V3_S4_AP (IW0R:ADC19) Airport supply current sense



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	1	CAP, 0.22uF, 201	C5435	AP_ISNS:Y
117S0002	1	RES, 0 ohm, 201	C5435	AP_ISNS:N

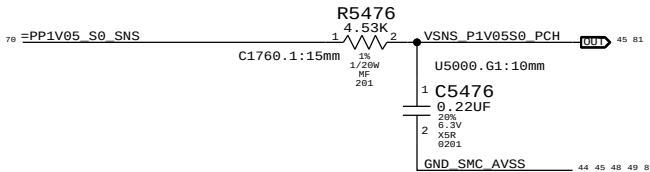
CPU Core (VC0C:ADC10/IC0C:ADC11) Voltage sense and IMON amp (VC0C, IC0C)



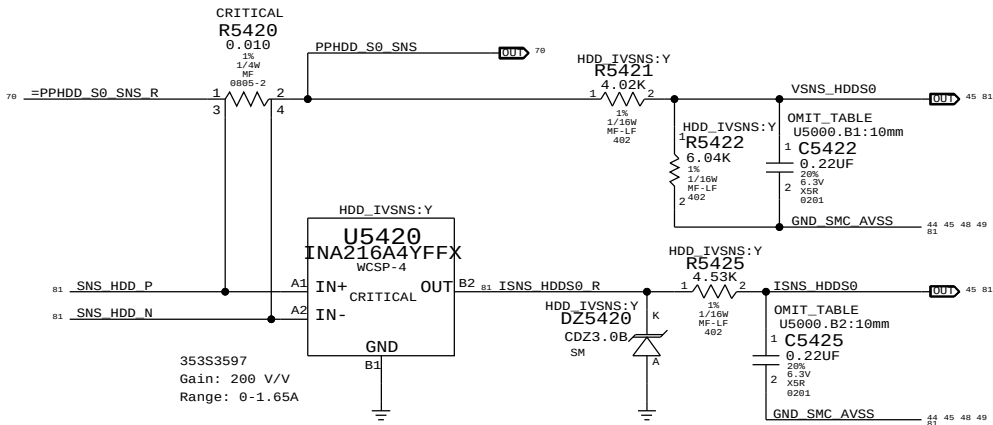
PP1V05_S0_PCH (VN1R:ADC17)

I/V-sense for PP1V05_S0_PCH

Place R5476 over PP1V05_S0 power plane shape.



HDD S0 (VH05:ADC14/IH05:ADC15) I/V-sense for HDD (Development, but need R5420)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP, 0.22uF, 201	C5422, C5425	HDD_IVSNS:Y
117S0002	2	RES, 0 OHM, 201	C5422, C5425	HDD_IVSNS:N

SYNC MASTER=J16 TONY		SYNC DATE=03/13/2013	
PAGE TITLE			
I and V Sense			
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		REVISION	12.4.0
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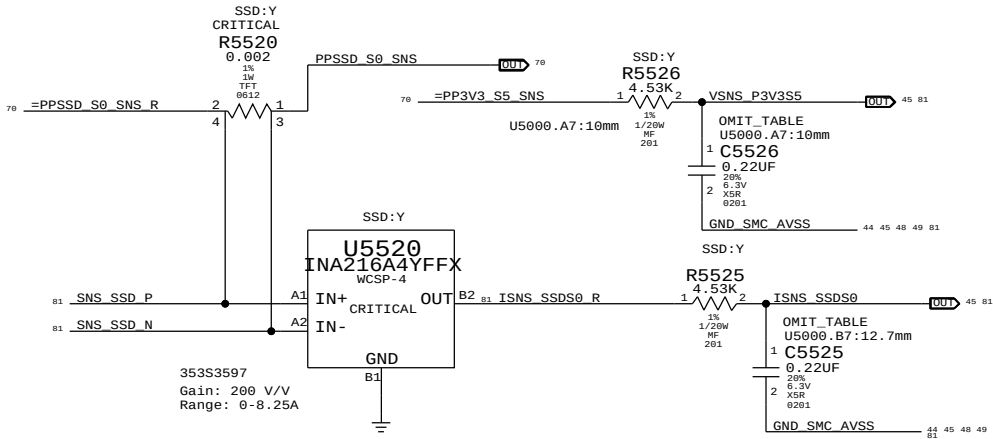
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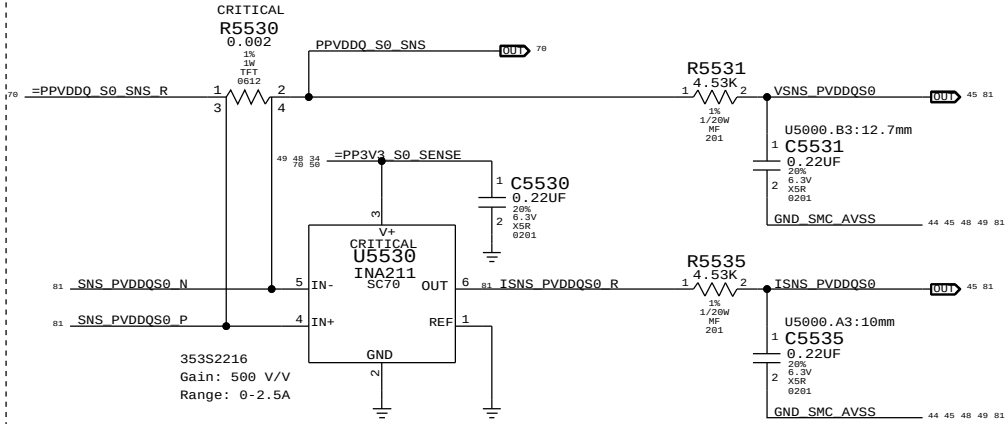
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SSD S0 (IH1R:ADC20/VR3R:ADC21) I-sense for SSD / V-sense for PP3V3_S5)

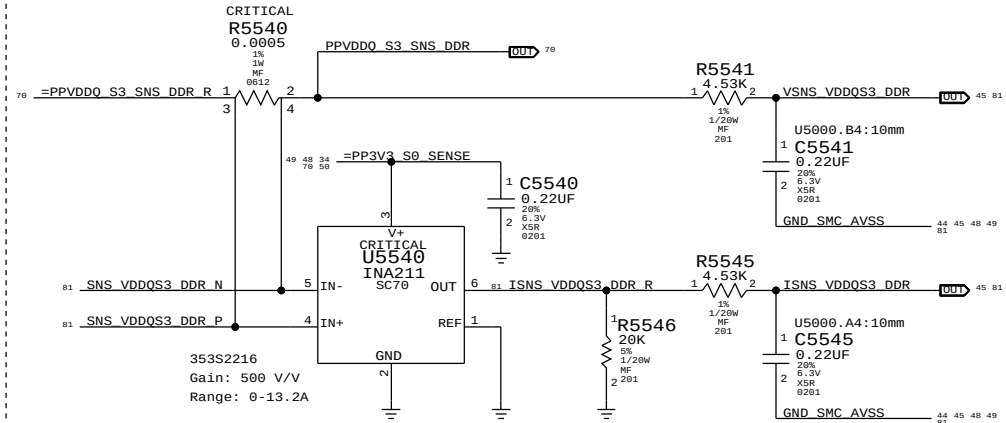


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP, 0.22UF, 201	C5525, C5526	SSD:Y
117S0002	2	RES, 0 OHM, 201	C5525, C5526	SSD:N

PPVDDQ_S0 (VC0M:ADC4/IC0M:ADC5)
lowside sense for CPU mem rail (2.5A max draw, 3.3A max sense capability)



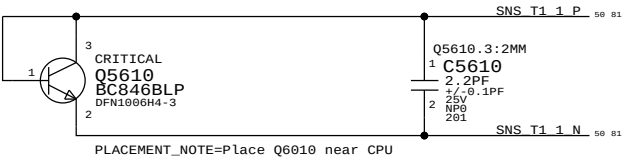
VDDQ S3 (VM0R:ADC6/IM0R:ADC7)
VDDQ lowside sense for S0-DIMM modules



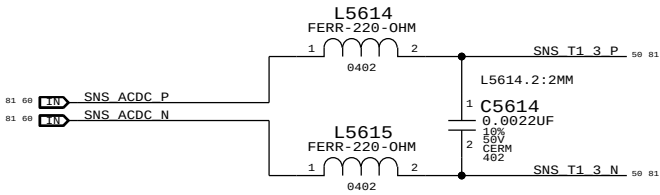
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PAGE TITLE			
I and V Sense(Continued)			
 Apple Inc.		DRAWING NUMBER	SIZE
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		SHEET	
		49 OF 86	

Temperature Sensor T1

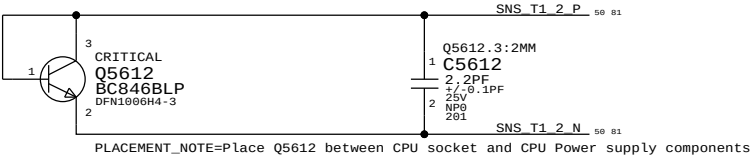
MLB Proximity



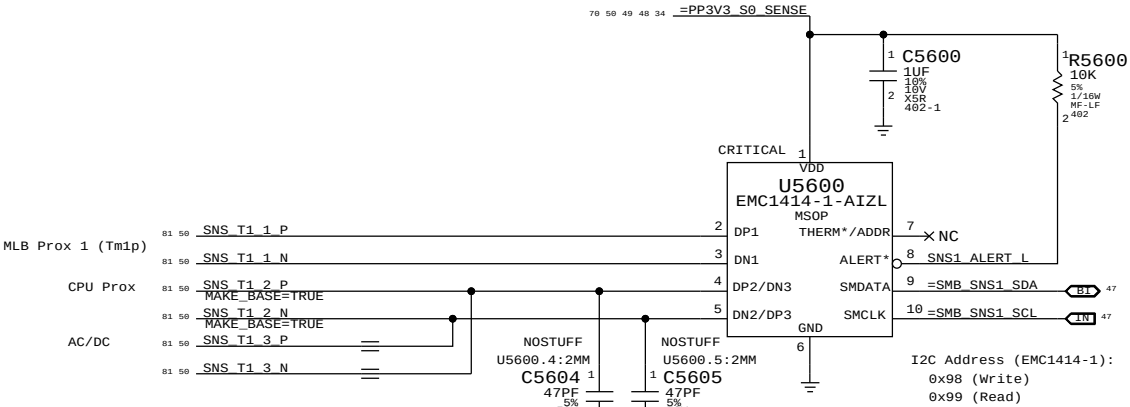
AC/DC Diode on supply



CPU Proximity



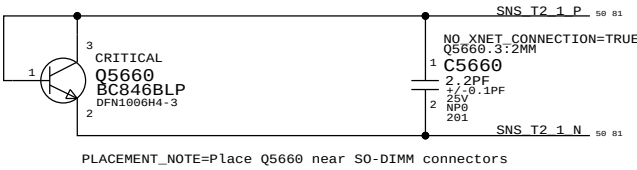
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode



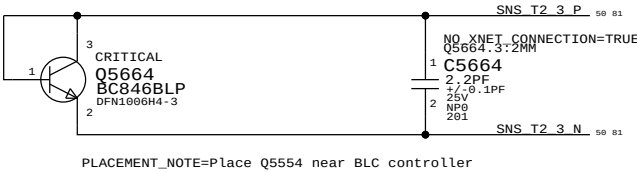
Note:
Internal sensor of the EMC 1414 will be used as the ambient sensor. Place U5600 at the coolest location on the MLB.

Temperature Sensor T2

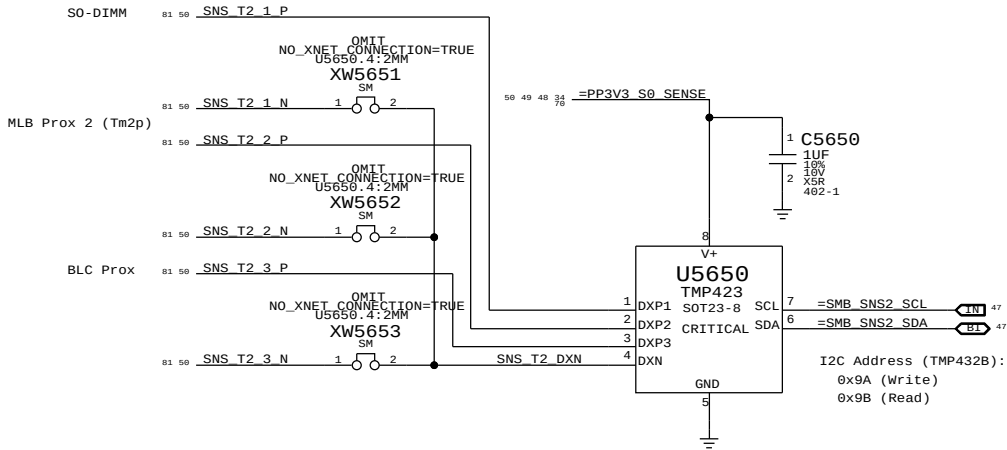
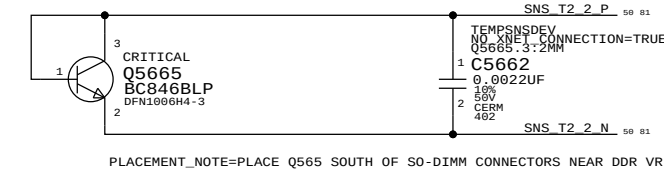
S0-DIMM Proximity



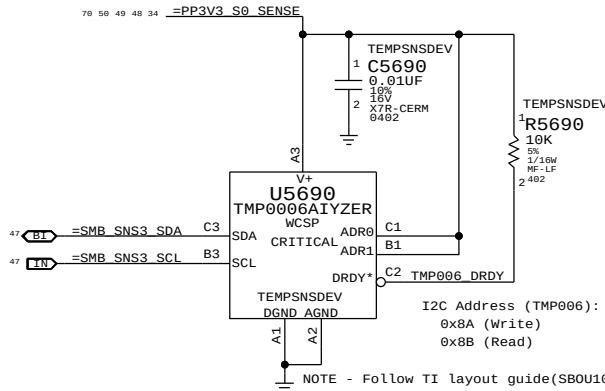
BLC Proximity



MLB Proximity



Temperature Sensor T3: LCD Remote Sensor(Dev Only)



This PD part is a rubber bumper to protect TMP006 Added to board BOM (DEV only) to clean up PD BOM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
875-6433	1	BUMPER, U5690, D7	BUMPER_U5690	TEMPSNSDEV

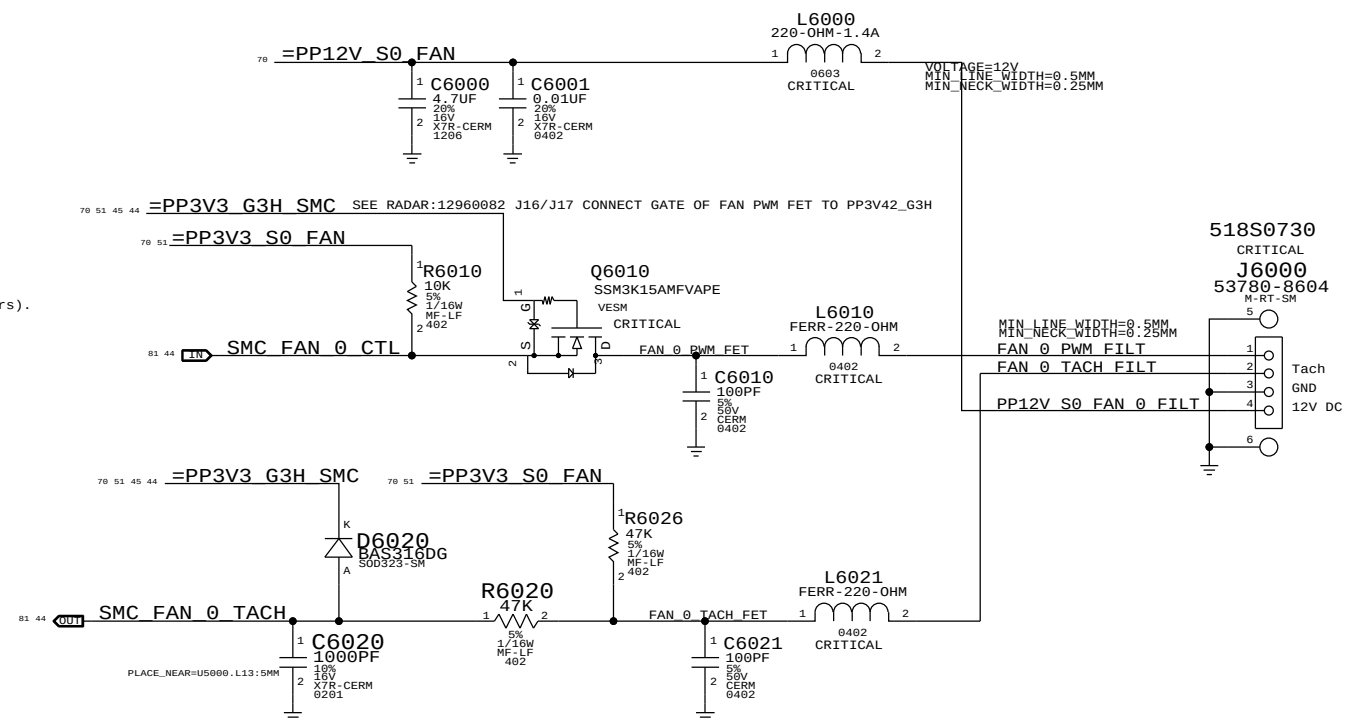
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PAGE TITLE		Temperature Sensors	
Apple Inc.		DRAWING NUMBER	051-0164
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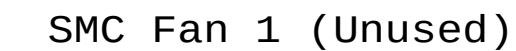
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

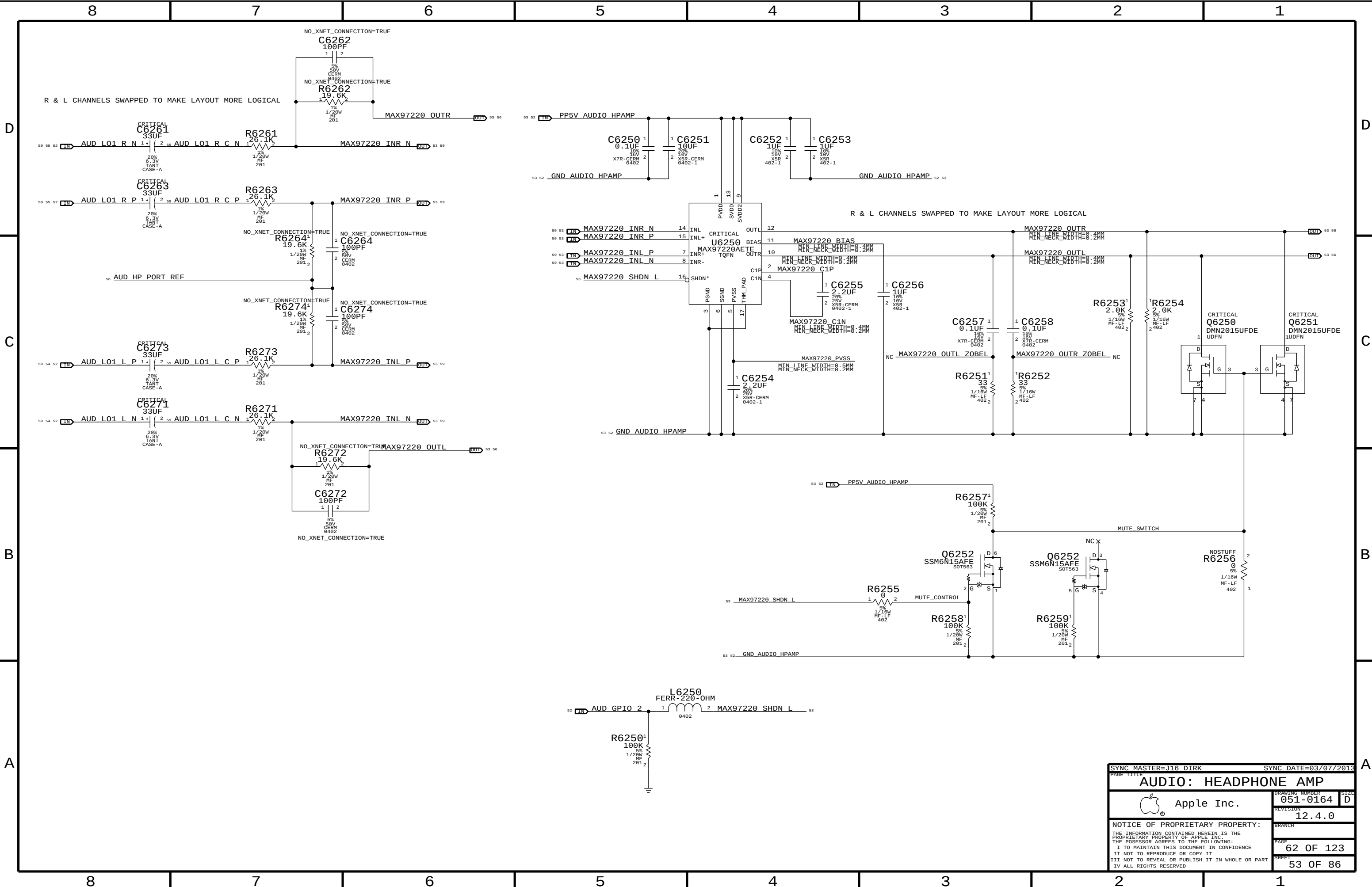
Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.



SYNC MASTER=J16 JERRY		SYNC DATE=01/07/2013	
PAGE TITLE			
System Fan			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-0164		D
	REVISION		
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BRANCH			
PAGE		60 OF 123	
SHEET		51 OF 86	



SYNC MASTER=J16 DIRK		SYNC DATE=03/07/2013	
PAGE TITLE		AUDIO: HEADPHONE AMP	
Apple Inc.		DRAWING NUMBER	051-0164
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LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6304
0 OHM
NOSTUFF

R6305
0 OHM
NOSTUFF

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6306
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF

R6307
0 OHM
47 KOHM
NOSTUFF
NOSTUFF
NOSTUFF

SYNC MASTER=J16 DIRK		SYNC DATE=03/07/2013	
PAGE TITLE		AUDIO: LEFT SPKR AMP	
Apple Inc.		DRAWING NUMBER	051-0164
		REVISION	12.4.0
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RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

ONLY WOOFERS ON UNDER WINDOWS

WOOFERS & TWEETERS ON UNDER MAC OS

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6404
0 OHM
NOSTUFF

R6405
NOSTUFF
0 OHM

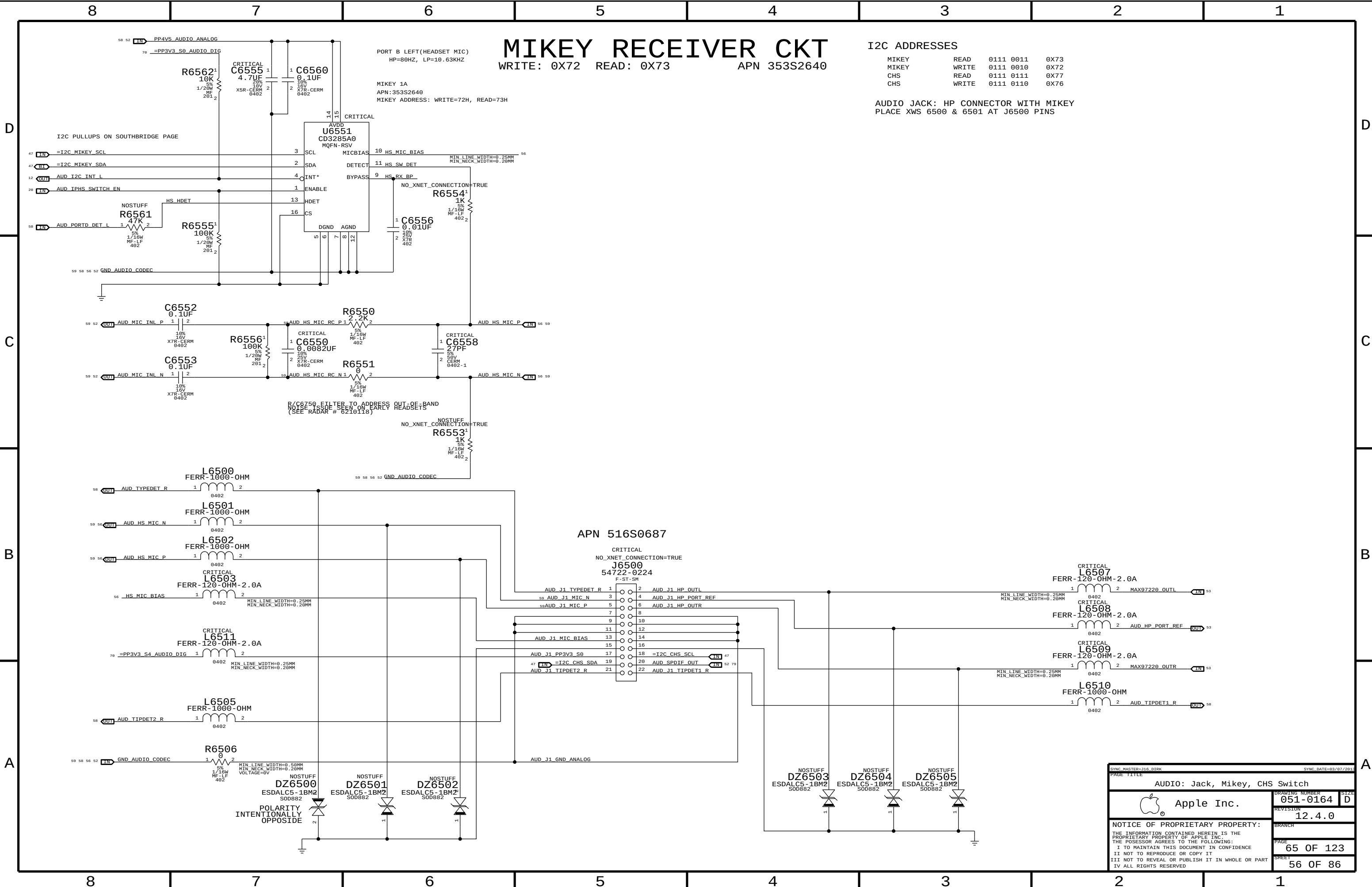
AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6406
NOSTUFF
NOSTUFF
47 KOHM
0 OHM

R6407
0 OHM
NOSTUFF
NOSTUFF
NOSTUFF

SYNC MASTER=J16 DIRK		SYNC DATE=03/07/2013	
PAGE TITLE		AUDIO: RIGHT SPKR AMP	
Apple Inc.		DRAWING NUMBER	051-0164
		REVISION	12.4.0
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2640

I2C ADDRESSES			
MIKEY	READ	0111 0011	0X73
MIKEY	WRITE	0111 0010	0X72
CHS	READ	0111 0111	0X77
CHS	WRITE	0111 0110	0X76

AUDIO JACK: HP CONNECTOR WITH MIKEY
PLACE XWS 6500 & 6501 AT J6500 PINS

APN 516S0687

Apple Inc.

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PAGE TITLE

AUDIO: Jack, Mikey, CHS Switch

DRAWING NUMBER

051-0164

REVISION

12.4.0

PAGE

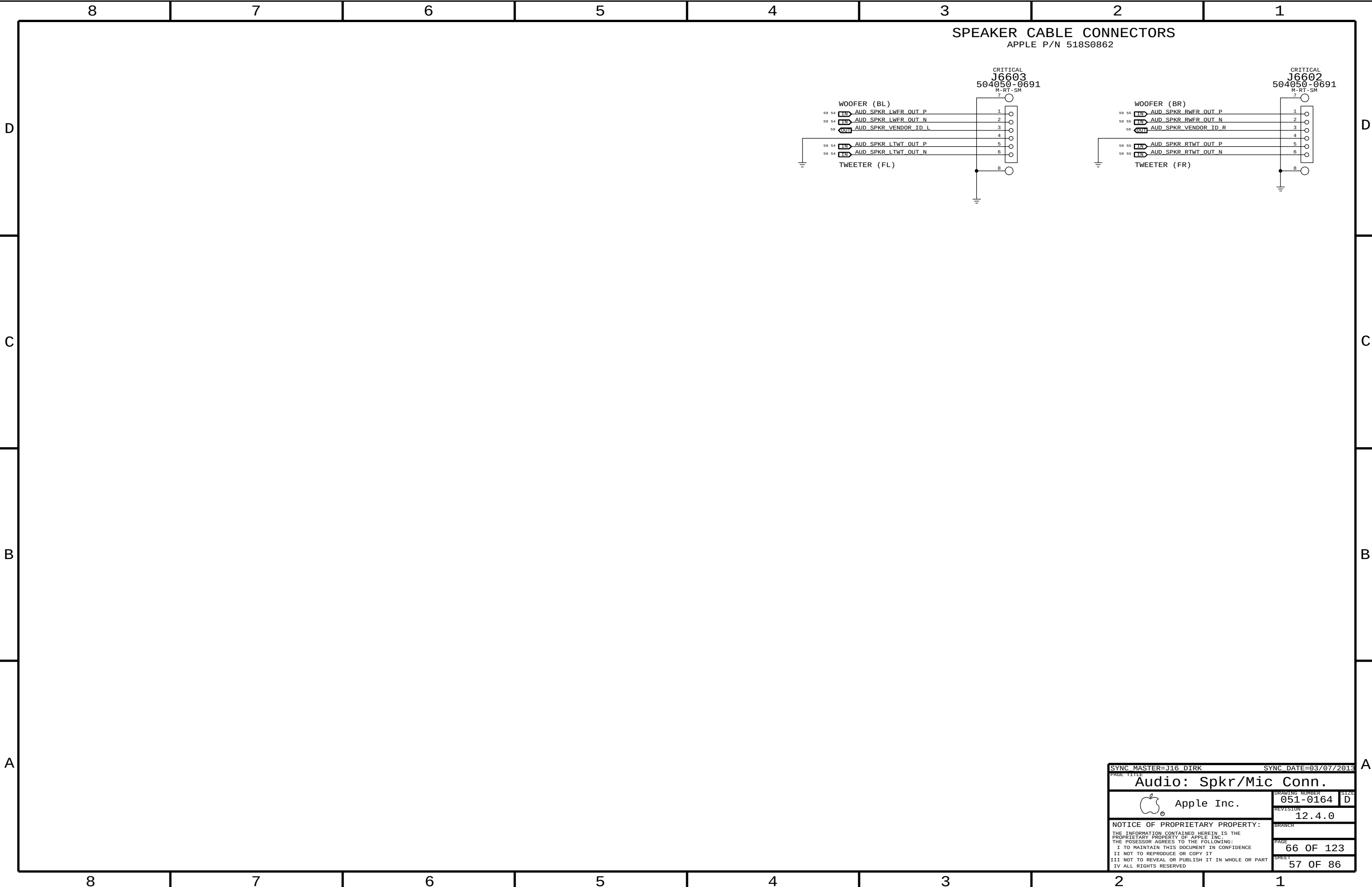
65 OF 123

SHEET

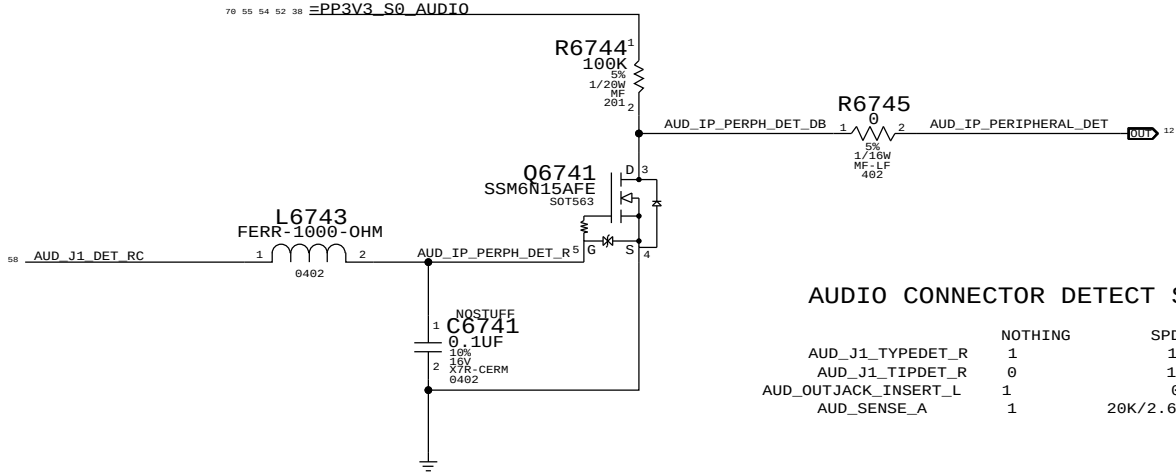
56 OF 86

SYNC MASTER=J10, DIRM

SYNC DATE=03/07/2011

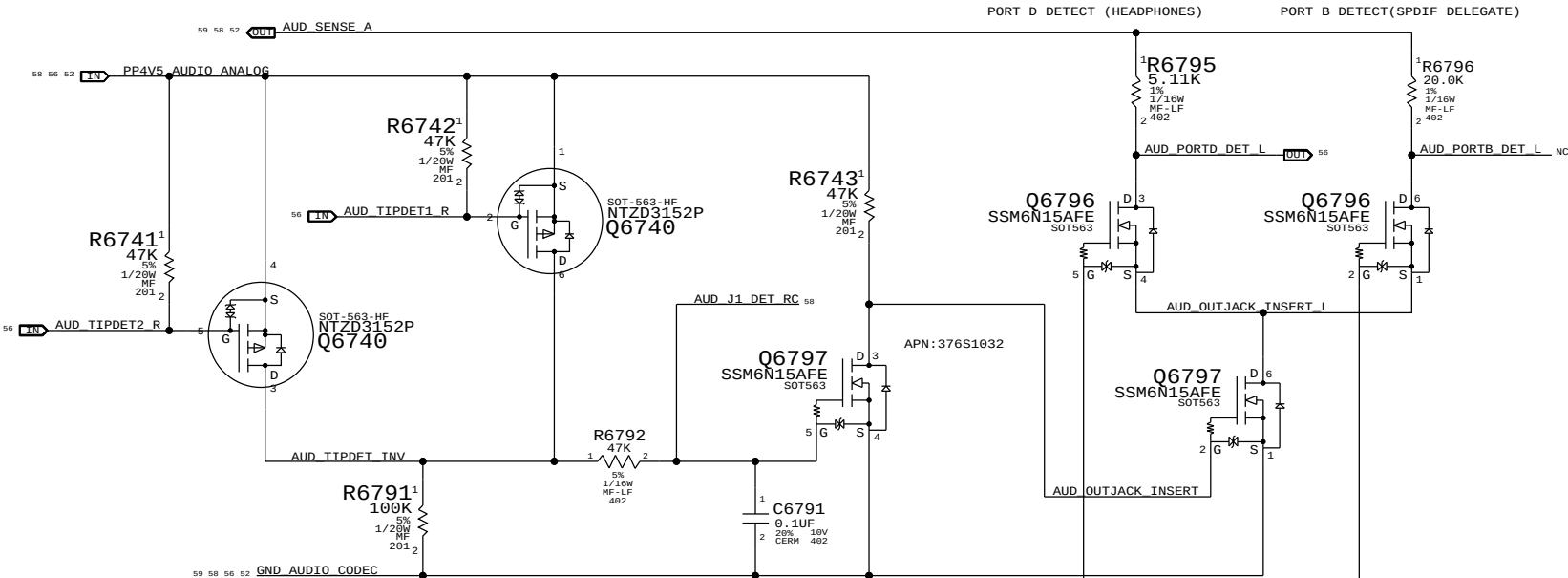
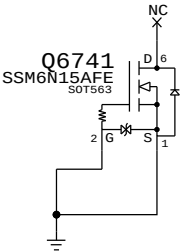


IPHS HS Detect Debounce CKT

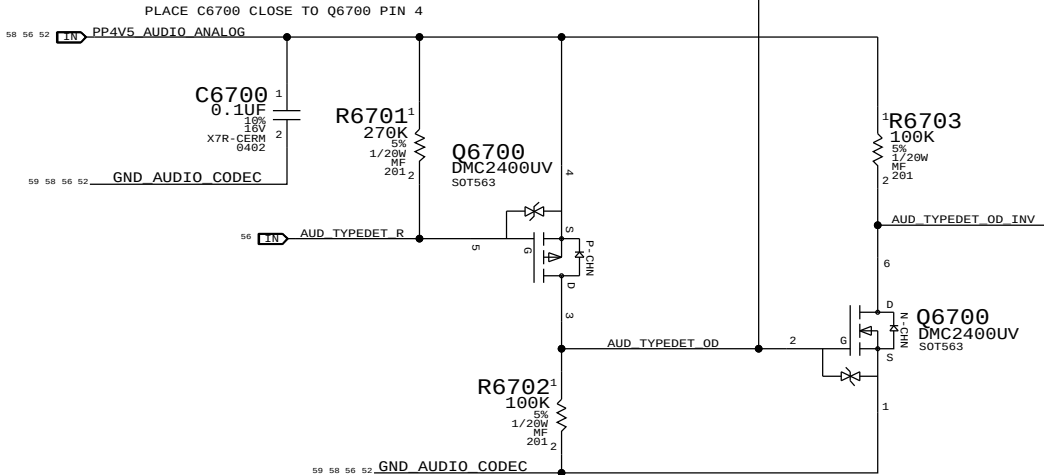
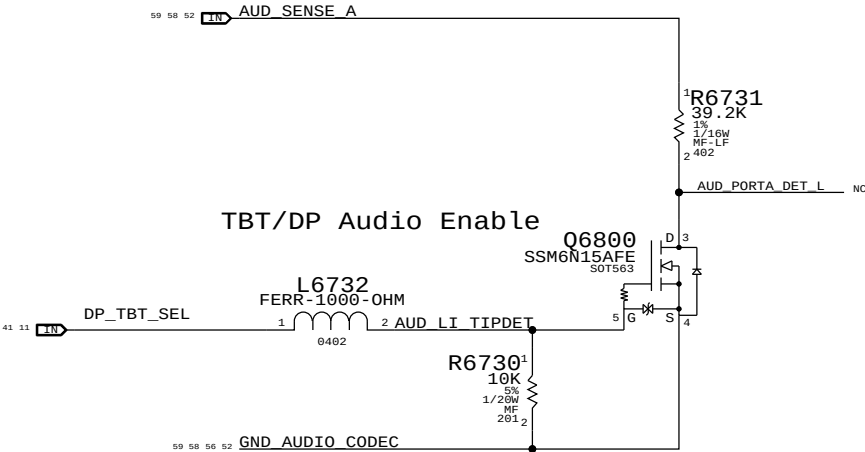


AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_J1_TYDEDET_R	1	1	0
AUD_J1_TIPDET_R	0	1	1
AUD_OUTJACK_INSERT_L	1	0	0
AUD_SENSE_A	1	20K/2.67K RDIV	39.2K/2.67K RDIV



Target Display Mode Detect

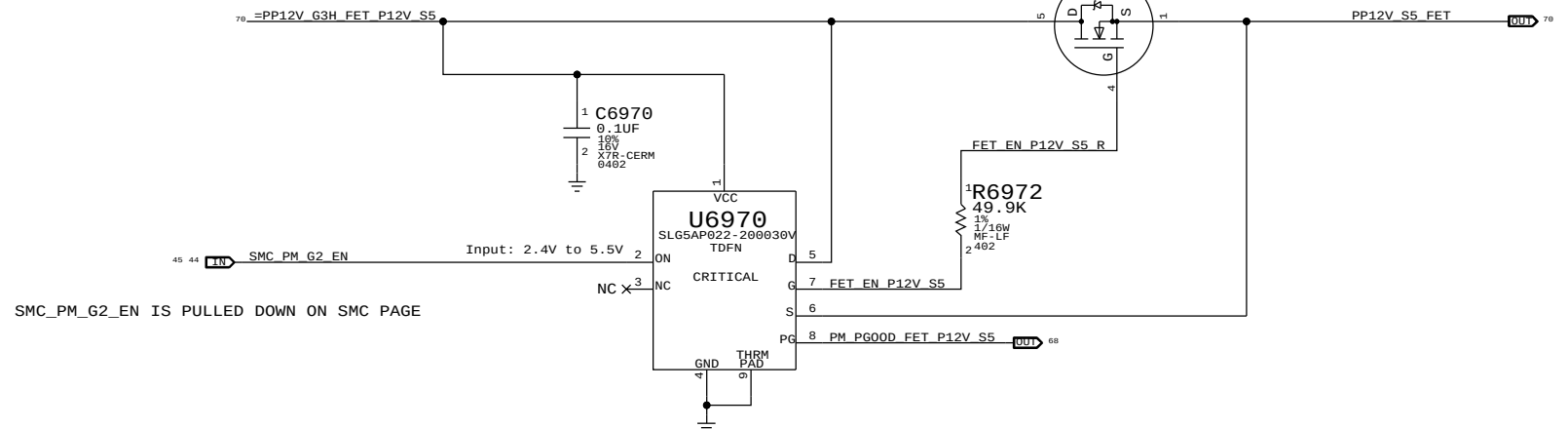
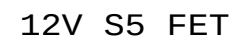
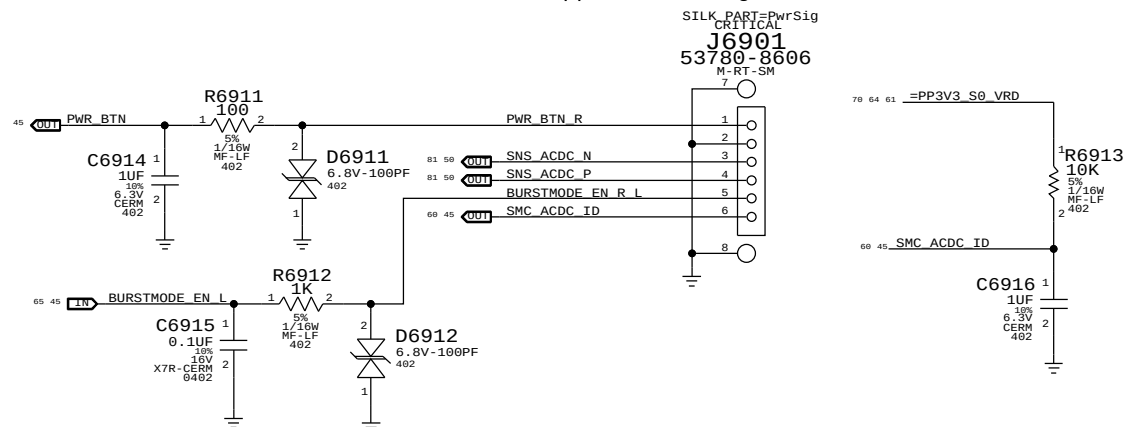
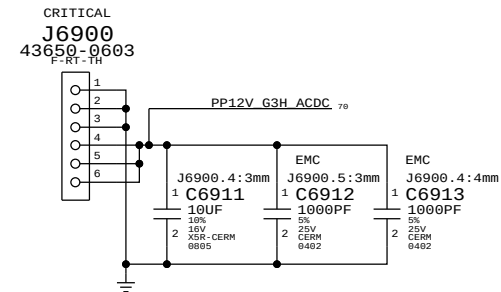


SYNC MASTER=J16 DIRK		SYNC DATE=03/07/2013	
PAGE TITLE		AUDIO: Detects/Grounding	
Apple Inc.		DRAWING NUMBER	051-0164
		REVISION	12.4.0
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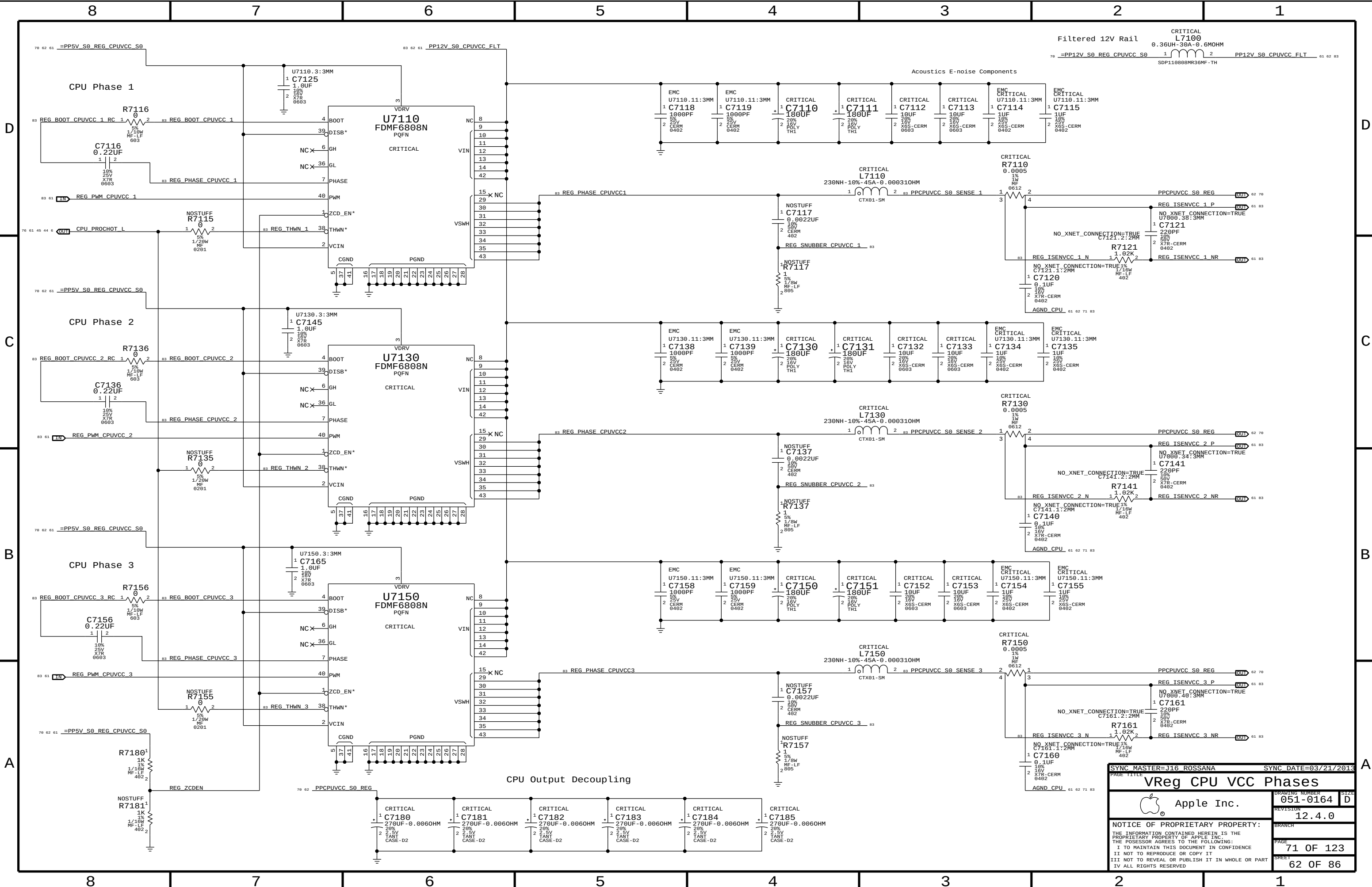
Switching freq: $409 \text{ kHz} = \frac{13.5}{L6901}$

Vout = 1.25V * (1

CRITICAL
06970
IRFH3702TRPBF
PQFN



SMC_PM_G2_EN IS PULLED DOWN ON SMC PAGE



SYNC MASTER=J16 ROSSANA		SYNC DATE=03/21/2013	
PAGE TITLE			
VReg CPU VCC Phases			
 Apple Inc.		DRAWING NUMBER	051-0164
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		PAGE	71 OF 123
		SHEET	62 OF 86

D

C

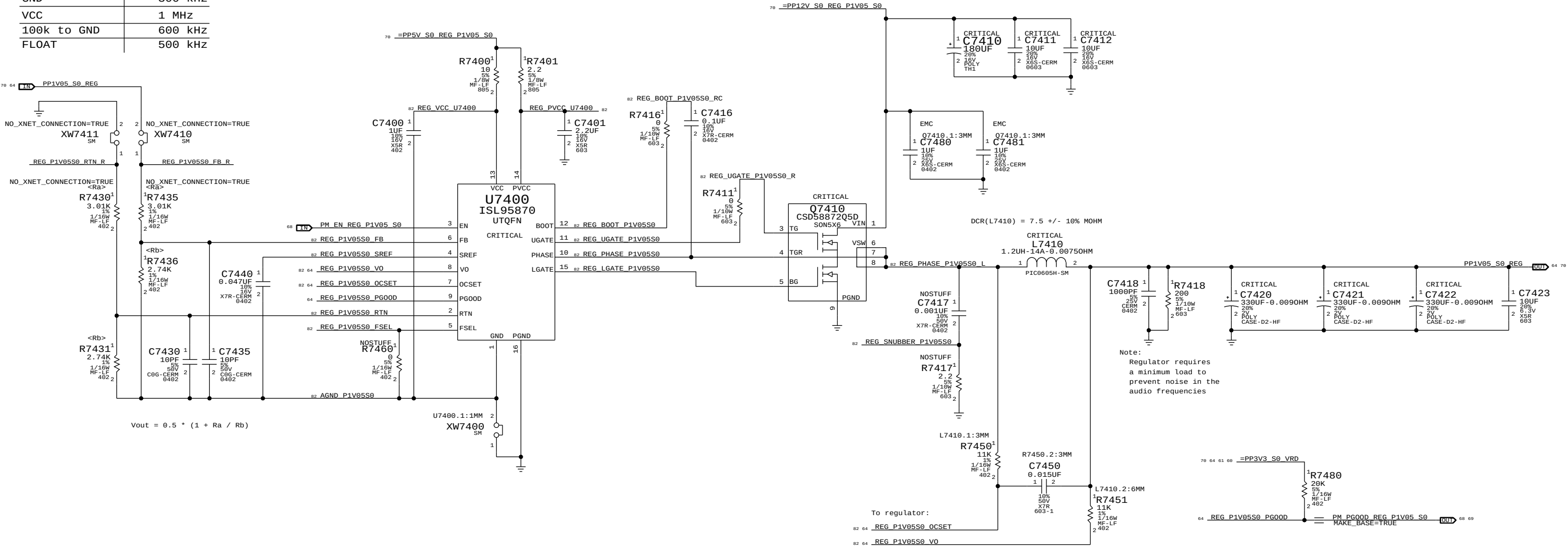


A

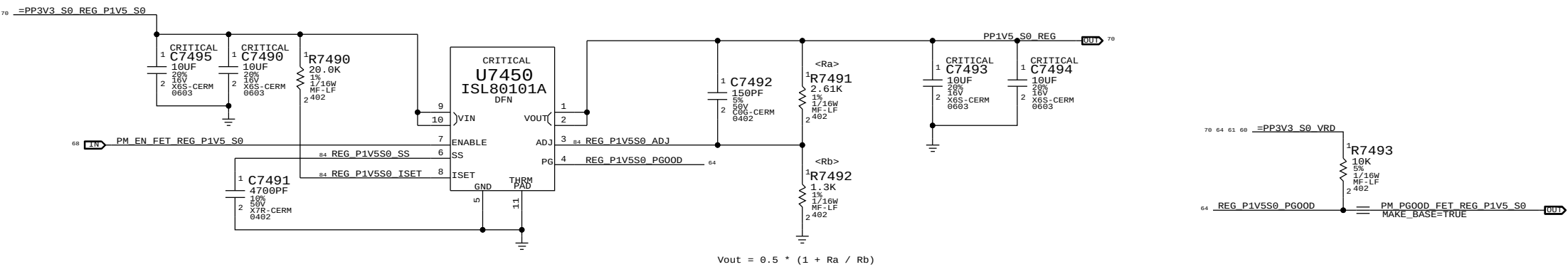
PCH/TBT (1.05V) S0 REGULATOR

Switching freq: 500 kHz OC trip point: 12.4 A = $\frac{R7450 \cdot 8.5 \cdot 10^{-6}}{DCR(L7410)}$

FSEL STRAP	SW FREQ
GND	300 kHz
VCC	1 MHz
100k to GND	600 kHz
FLOAT	500 kHz



1.5V S0 REGULATOR



SYNC MASTER=J16 ROSSANA		SYNC DATE=03/04/2013	
PAGE TITLE		VREG 1V05 S0 / 1V5 S0	
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		REVISION	12.4.0
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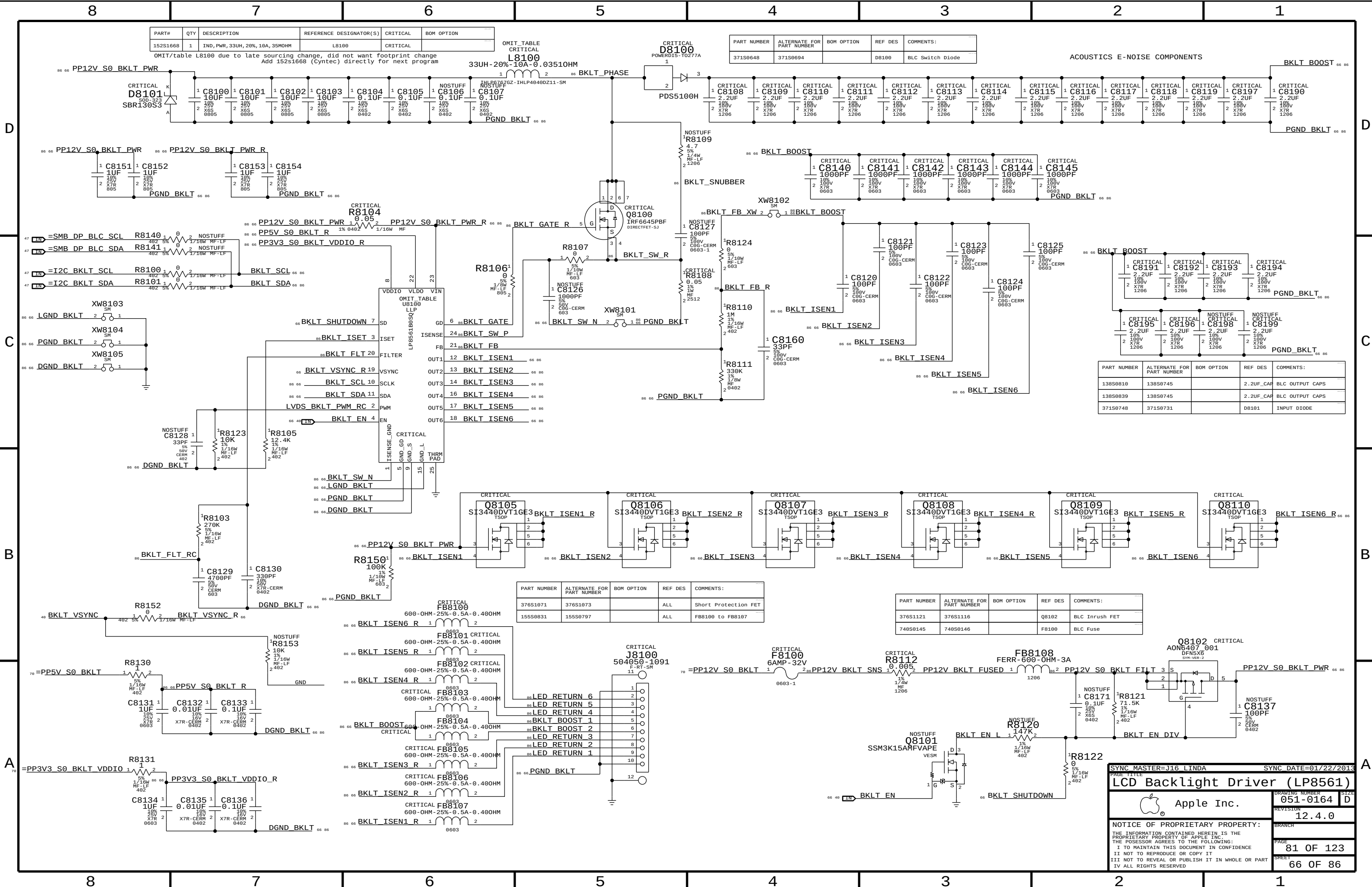
A

D

C



A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
-------	-----	-------------	-------------------------	----------	------------

OMIT/table L8100 due to late sourcing change, did not want footprint change
Add 152s1668 (Cyntec) directly for next program

OMIT TABLE
CRITICAL
L8100
33UH-20%-10A-0.03510HM

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
-------------	---------------------------	------------	---------	-----------

ACOUSTICS E-NOISE COMPONENTS

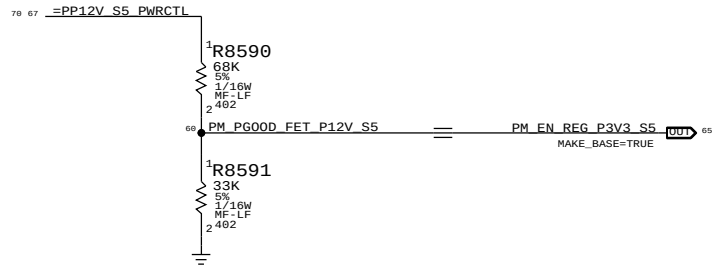
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0810	138S0745		2.2UF_CAP	BLC OUTPUT CAPS
138S0839	138S0745		2.2UF_CAP	BLC OUTPUT CAPS
371S0748	371S0731		D8101	INPUT DIODE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1071	376S1073		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8100 to FB8107

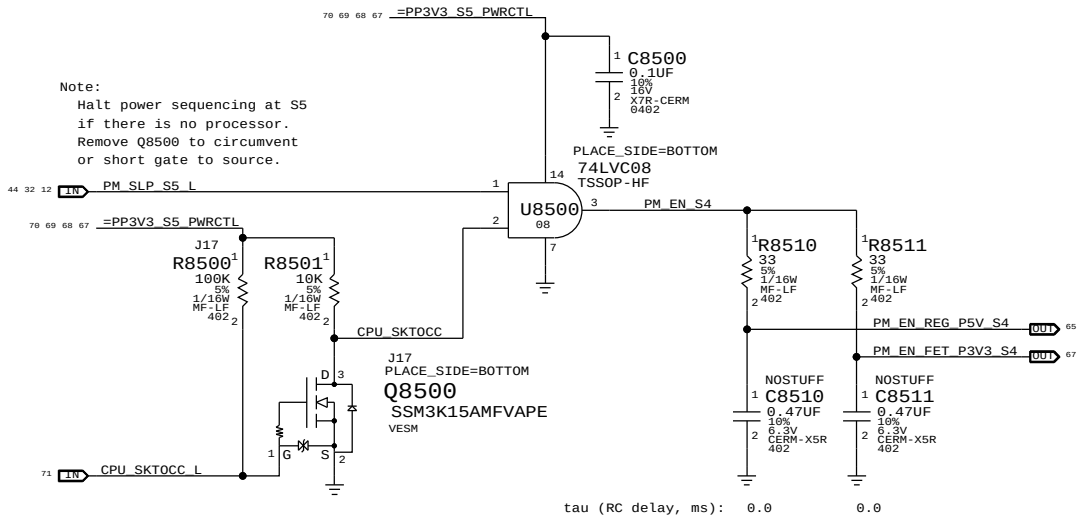
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1121	376S1116		Q8102	BLC Inrush FET
740S0145	740S0146		F8100	BLC Fuse

SYNC MASTER=J16 LINDA		SYNC DATE=01/22/2013	
PAGE TITLE		DRAWING NUMBER	
LCD Backlight Driver (LP8561)		051-0164	
Apple Inc.		REVISION	
		12.4.0	
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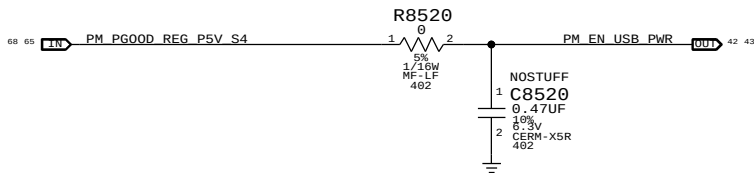
S5 Enable



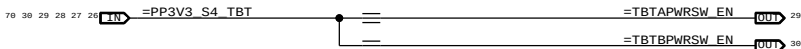
S4 Enables



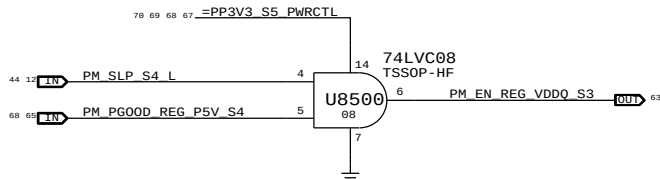
S4 USB Enable



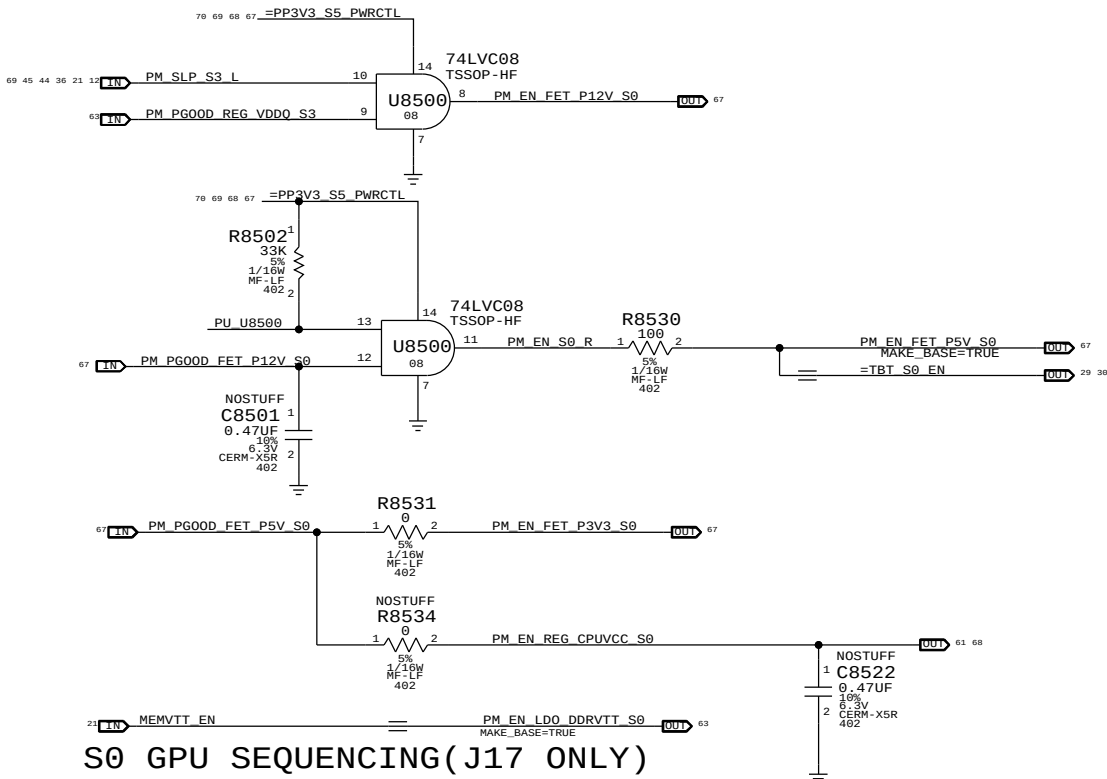
S4 TBT S4 Port Enable



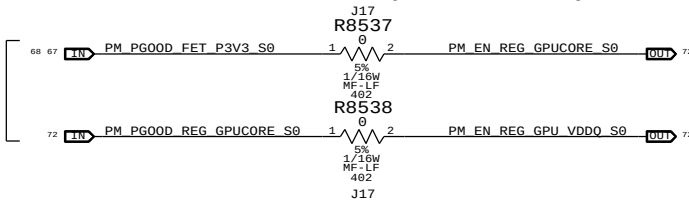
S3 VDDQ Enable



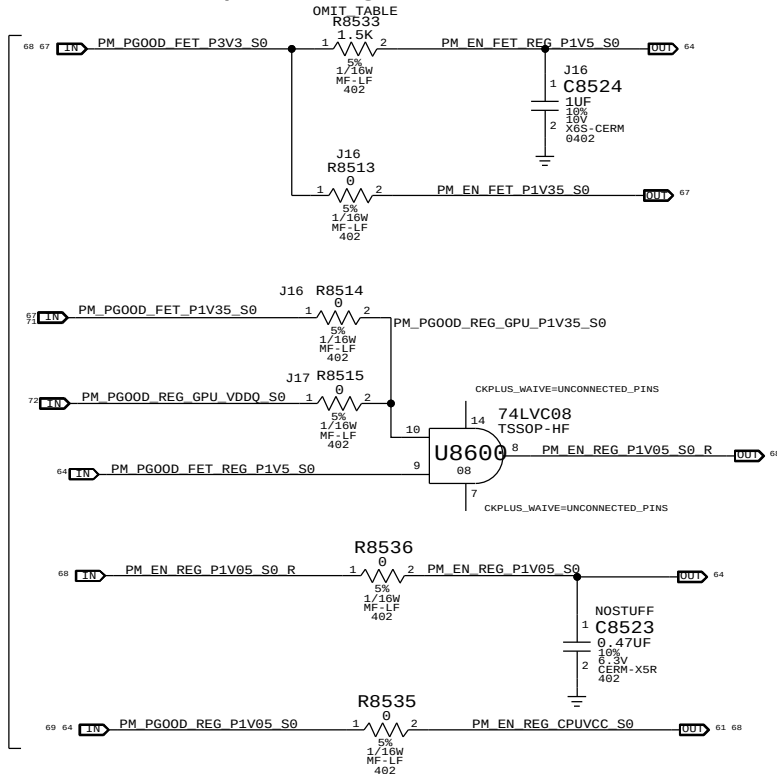
S0 Enables



S0 GPU SEQUENCING(J17 ONLY)



S0 PCH Sequencing



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0070	1	RES,1.5K,0402,5%	R8533	J16
116S0004	1	RES,00HM,0402,5%	R8533	J17

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.5 V, 1.05V for PCH/TBT/GPU)
Uncore: VDDQ

Notes on sequencing requirements

Intel:

1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

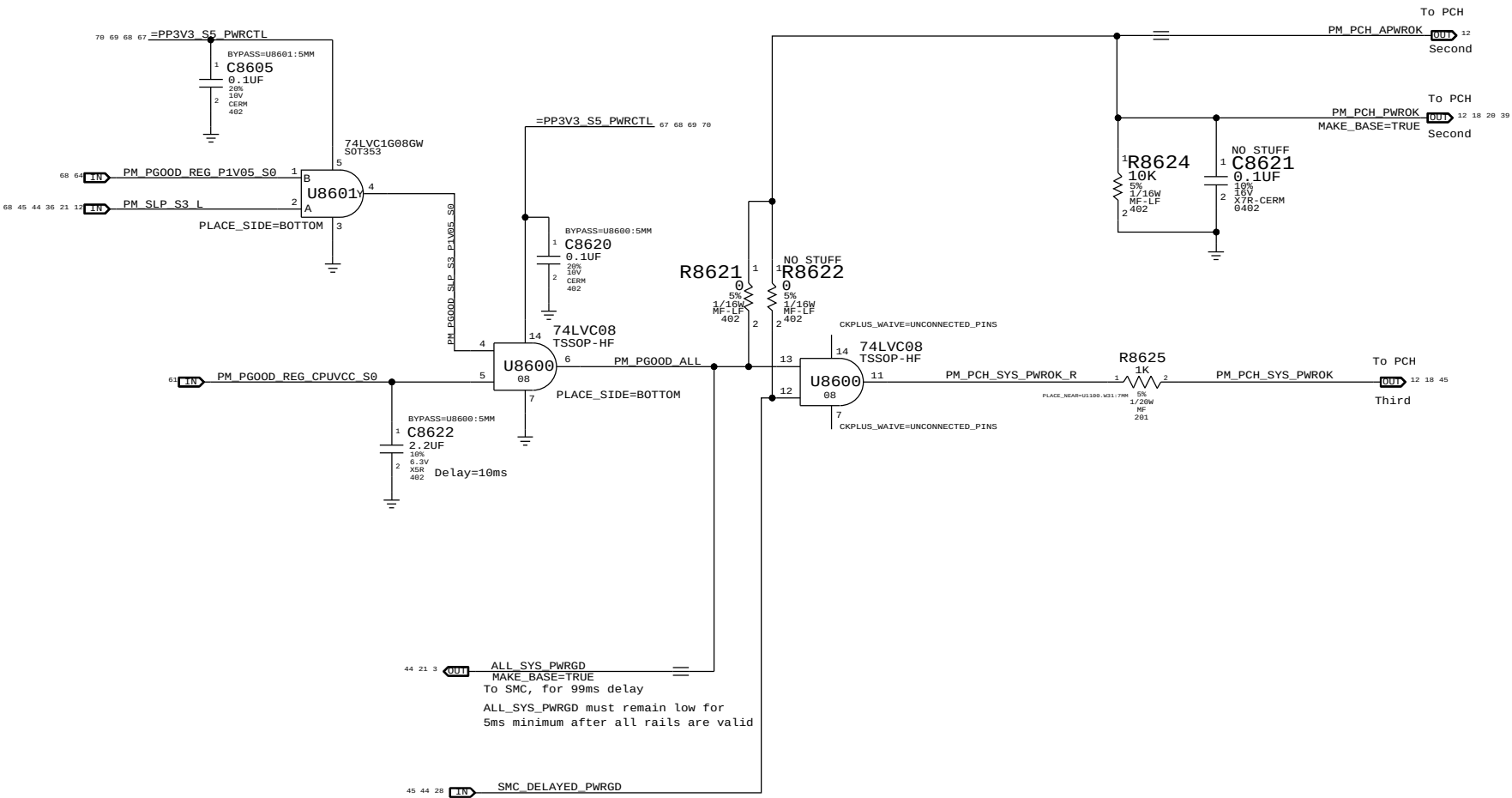
NVIDIA:

1. 3V3_S0 must ramp first
2. VDDQ MUST RAMP AFTER GPU_OP-HF
3. PEX_VDD with IFPC/D/E/F_IOVDD (1.05V) must ramp after VDDQ
4. All rails must reach their target voltages in more than 40 uS

SYNC MASTER=J16 AARON		SYNC DATE=02/21/2013	
PAGE TITLE		PM Regulator Enables	
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		REVISION	12.4.0
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ALL_SYS_PWRGD, PCH_PWROK & SYS_PWROK Generation

PCH Power Goods

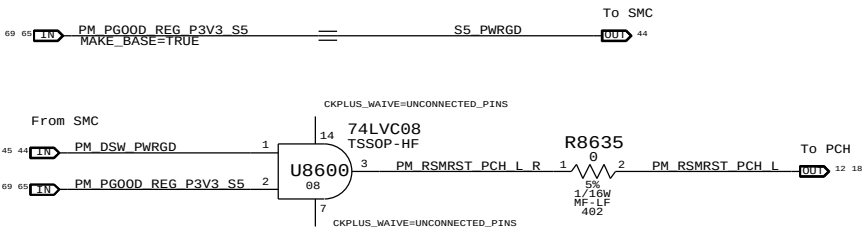


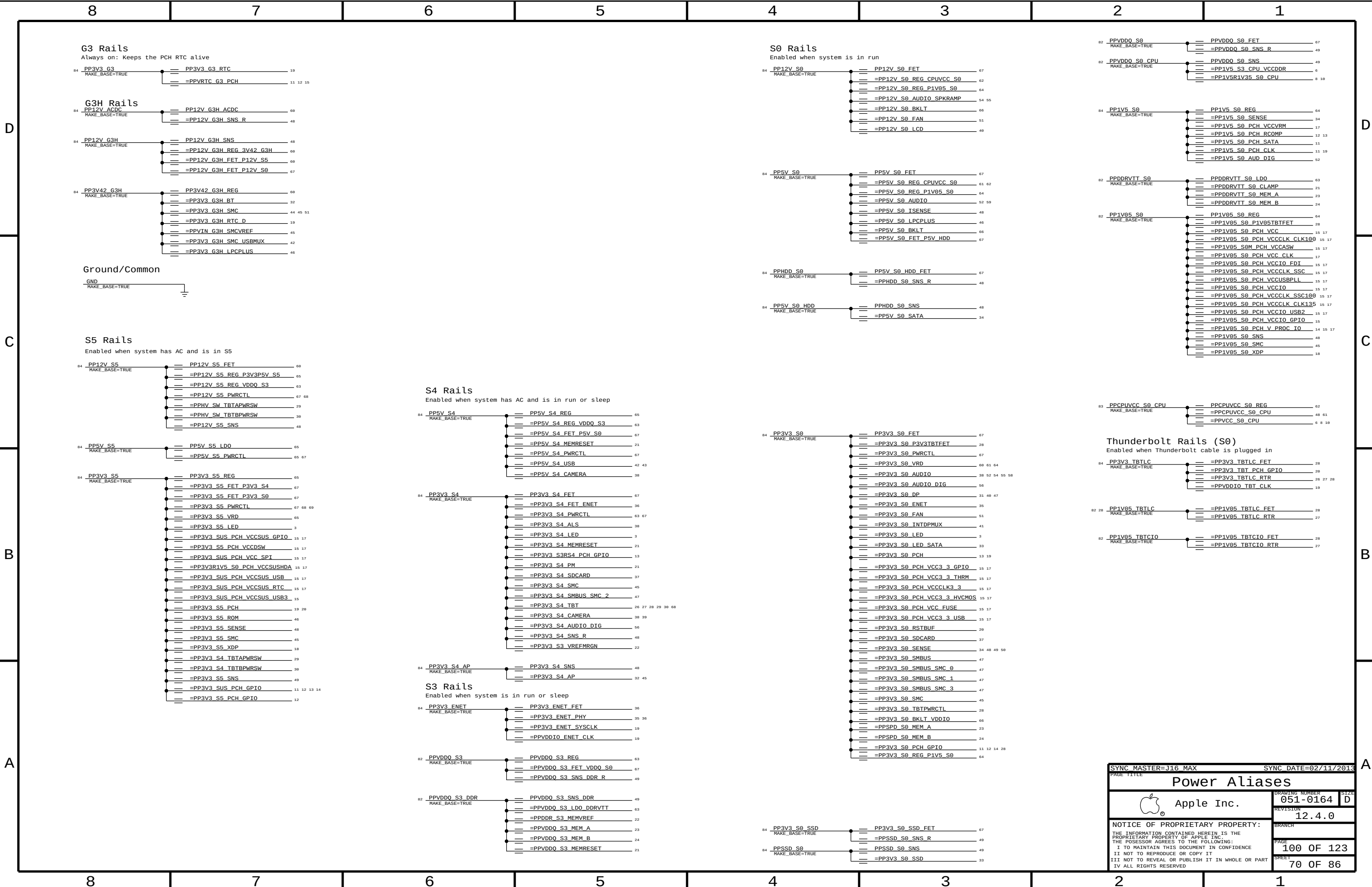
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8
Note:
The iMac J16/J17 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.





SYNC MASTER=J16 MAX

SYNC DATE=02/11/2013

PAGE TITLE

Power Aliases

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DRAWING NUMBER

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SHEET

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8	7	6	5	4	3	2	1
CPU Reserved		UNUSED GRAPHICS ALIASES					
78 18 6 CPU_CFG<15..12> == TP_CPU_CFG<15..12> MAKE_BASE=TRUE		20 TP_GPU_RESET_L == NC_TP_GPU_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE					
CPU Memory		UNUSED THUNDERBOLT ALIASES					
75 MEM_A_CLK_N<2..3> == NC_MEM_A_CLKN<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		26 TP_TBT_PCIE_RESET0_L == NC_TBT_PCIE_RESET0_L MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_A_CLK_P<2..3> == NC_MEM_A_CLKP<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		26 TP_TBT_PCIE_RESET1_L == NC_TBT_PCIE_RESET1_L MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_A_CS_L<2..3> == NC_MEM_A_CS_L<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		26 TP_TBT_PCIE_RESET2_L == NC_TBT_PCIE_RESET2_L MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_A_CKE<2..3> == NC_MEM_A_CKE<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		26 TP_TBT_PCIE_RESET3_L == NC_TBT_PCIE_RESET3_L MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_B_CLK_N<2..3> == NC_MEM_B_CLKN<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		26 TP_TBT_THERM_DP == NC_TBT_THERM_DP MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_B_CLK_P<2..3> == NC_MEM_B_CLKP<2..3> MAKE_BASE=TRUE NO_TEST=TRUE							
75 MEM_B_CS_L<2..3> == NC_MEM_B_CS_L<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		UNUSED VREG ALIASES					
75 MEM_B_CKE<2..3> == NC_MEM_B_CKE<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		61 REG_PWM_CPUVCC_4 == NC_REG_PWM_CPUVCC_4 MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_A_ODT<2..3> == NC_MEM_A_ODT<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		61 REG_ISENVCC_4_P == NC_REG_ISENVCC_4P MAKE_BASE=TRUE NO_TEST=TRUE					
75 MEM_B_ODT<2..3> == NC_MEM_B_ODT<2..3> MAKE_BASE=TRUE NO_TEST=TRUE							
PCH GPIO		UNUSED GPU ALIASES					
11 TP_PCH_GPIO64_CLKOUTFLEX0 == NC_PCH_GPIO64_CLKOUTFLEX0 MAKE_BASE=TRUE NO_TEST=TRUE		68 PM_EN_REG_GPUCORE_S0 == NC_PM_EN_REG_GPUCORE_S0 MAKE_BASE=TRUE NO_TEST=TRUE					
11 TP_PCH_GPIO65_CLKOUTFLEX1 == NC_PCH_GPIO65_CLKOUTFLEX1 MAKE_BASE=TRUE NO_TEST=TRUE		68 PM_PG00D_REG_GPUCORE_S0 == NC_PM_PG00D_REG_GPUCORE_S0 MAKE_BASE=TRUE NO_TEST=TRUE					
11 TP_PCH_GPIO66_CLKOUTFLEX2 == NC_PCH_GPIO66_CLKOUTFLEX2 MAKE_BASE=TRUE NO_TEST=TRUE		68 PM_EN_REG_GPU_VDDQ_S0 == NC_PM_EN_REG_GPU_VDDQ_S0 MAKE_BASE=TRUE NO_TEST=TRUE					
11 TP_PCH_GPIO67_CLKOUTFLEX3 == NC_PCH_GPIO67_CLKOUTFLEX3 MAKE_BASE=TRUE NO_TEST=TRUE		68 PM_PG00D_REG_GPU_VDDQ_S0 == NC_PM_PG00D_REG_GPU_VDDQ_S0 MAKE_BASE=TRUE NO_TEST=TRUE					
UNUSED IG DISPLAY		UNUSED PEG ALIASES					
5 TP_DP_IG_A_MLP<3..0> == NC_DP_IG_A_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		5 =PEG_D2R_P<0..15> == NC_PEG_D2R_P<0..15> MAKE_BASE=TRUE NO_TEST=TRUE					
5 TP_DP_IG_A_MLN<3..0> == NC_DP_IG_A_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		5 =PEG_D2R_N<0..15> == NC_PEG_D2R_N<0..15> MAKE_BASE=TRUE NO_TEST=TRUE					
5 TP_DP_IG_A_AUXCHP == NC_DP_IG_A_AUXCHP MAKE_BASE=TRUE NO_TEST=TRUE		5 =PEG_R2D_C_P<0..15> == NC_PEG_R2D_C_P<0..15> MAKE_BASE=TRUE NO_TEST=TRUE					
5 TP_DP_IG_A_AUXCHN == NC_DP_IG_A_AUXCHN MAKE_BASE=TRUE NO_TEST=TRUE		5 =PEG_R2D_C_N<0..15> == NC_PEG_R2D_C_N<0..15> MAKE_BASE=TRUE NO_TEST=TRUE					
PCH PCI							
13 TP_LPC_DREQ0_L == NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE							
PCH Miscellaneous							
11 TP_HDA_SDIN1 == NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE							
11 TP_HDA_SDIN2 == NC_HDA_SDIN2 MAKE_BASE=TRUE NO_TEST=TRUE							
11 TP_HDA_SDIN3 == NC_HDA_SDIN3 MAKE_BASE=TRUE NO_TEST=TRUE							
11 TP_PCI_CLK33M_OUT2 == NC_PCI_CLK33M_OUT2 MAKE_BASE=TRUE NO_TEST=TRUE							
11 TP_PCI_CLK33M_OUT3 == NC_PCI_CLK33M_OUT3 MAKE_BASE=TRUE NO_TEST=TRUE							
8	7	6	5	4	3	2	1

SYNC MASTER=J116 MAX

SYNC DATE=02/11/2013

PAGE TITLE

Unused Signal Aliases

 Apple Inc.

DRAWING NUMBER

051-0164

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SYNC MASTER=J16 MAX

SYNC DATE=02/11/2013

PAGE TITLE		Functional / ICT Test	
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J16 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, BOTTOM	NO_TYPE, BGA	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	12.7 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.215 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.215 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.170 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP, BOTTOM	Y	0.170 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP, BOTTOM	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.138 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.138 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.110 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.180 MM	0.085 MM	=STANDARD	0.140 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.180 MM	0.085 MM	=STANDARD	0.140 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.135 MM	0.085 MM	=STANDARD	0.160 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	0.160 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.089 MM	0.085 MM	=STANDARD	0.220 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.089 MM	0.085 MM	=STANDARD	0.220 MM	0.1 MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	*	0.076 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

Board Stack-up

Finished board thickness: 1.58 mm

Layer	Material	Thickness
Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 mm
2	Plane	1 oz
	Prepreg	0.076 mm
3	Signal	0.5 oz
	Prepreg	0.435 mm
4	Plane	1 oz
	Core	0.127 mm
5	Plane	1 oz
	Prepreg	0.435 mm
6	Signal	0.5 oz
	Prepreg	0.076 mm
2	Plane	1 oz
	Prepreg	0.071 mm
Btm	Signal	0.5 oz (Cu plated)

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
DDR_COMP	*	Y	0.395 MM	0.195 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_QDQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (Shark Bay PDG, Intel Doc# 486712)

Table	Trace	Design	Iso	Design	Comments
4-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
4-3	8	9.84	12	13.78	
4-4	6	7.87	12	13.78	
4-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

See Note (3)

See Note (1)

See Note (3)

See Note (1)

See Note (2)

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggested 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net

Note (3):

In order for the constraints $DDR_*_DQ_BYTE*$ to =SAME to win out over $DDR\{A,B\}_DQ_BYTE*$ to $DDR\{A,B\}_DQ_BYTE*$ so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL .

DDR3

Electrical Constraint Set	Physical	Spacing	
Channel A			
H470 DDR A CLK0	DDR CLK_PHY	DDR CLK	MEM A CLK P<1..0> 7 23
H471 DDR A CLK0	DDR CLK_PHY	DDR CLK	MEM A CLK N<1..0> 7 23
H472 DDR A CLK1	DDR CLK_PHY	DDR CLK	MEM A CLK P<3..2> 72
H480 DDR A CLK1	DDR CLK_PHY	DDR CLK	MEM A CLK N<3..2> 72
H480 DDR A CTRL0	DDR CTRL_PHY	DDR CTRL	MEM A CKE<1..0> 7 23
H481 DDR A CTRL0	DDR CTRL_PHY	DDR CTRL	MEM A CS L<1..0> 7 23
H482 DDR A CTRL0	DDR CTRL_PHY	DDR CTRL	MEM A ODT<1..0> 7 23
H483 DDR A CTRL1	DDR CTRL_PHY	DDR CTRL	MEM A CKE<3..2> 72
H484 DDR A CTRL1	DDR CTRL_PHY	DDR CTRL	MEM A CS L<3..2> 72
H485 DDR A CTRL1	DDR CTRL_PHY	DDR CTRL	MEM A ODT<3..2> 72
H486 DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A A<15..0> 7 23
H487 DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A BA<2..0> 7 23
H488 DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A RAS L 7 23
H489 DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A CAS L 7 23
H490 DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A WE L 7 23
H491 DDR A DQ_BYTE0	DDR DQ_PHY	DDR A DQ_BYTE0	MEM A DQ<7..0> 7 25
H492 DDR A DQ_BYTE1	DDR DQ_PHY	DDR A DQ_BYTE1	MEM A DQ<15..8> 7 25
H493 DDR A DQ_BYTE2	DDR DQ_PHY	DDR A DQ_BYTE2	MEM A DQ<23..16> 7 25
H494 DDR A DQ_BYTE3	DDR DQ_PHY	DDR A DQ_BYTE3	MEM A DQ<31..24> 7 25
H495 DDR A DQ_BYTE4	DDR DQ_PHY	DDR A DQ_BYTE4	MEM A DQ<39..32> 7 25
H496 DDR A DQ_BYTE5	DDR DQ_PHY	DDR A DQ_BYTE5	MEM A DQ<47..40> 7 25
H497 DDR A DQ_BYTE6	DDR DQ_PHY	DDR A DQ_BYTE6	MEM A DQ<55..48> 7 25
H498 DDR A DQ_BYTE7	DDR DQ_PHY	DDR A DQ_BYTE7	MEM A DQ<63..56> 7 25
H499 DDR A DQS0	DDR DQS_PHY	DDR A DQS0	MEM A DQS P<0> 7 25
H500 DDR A DQS0	DDR DQS_PHY	DDR A DQS0	MEM A DQS N<0> 7 25
H501 DDR A DQS1	DDR DQS_PHY	DDR A DQS1	MEM A DQS P<1> 7 25
H502 DDR A DQS1	DDR DQS_PHY	DDR A DQS1	MEM A DQS N<1> 7 25
H503 DDR A DQS2	DDR DQS_PHY	DDR A DQS2	MEM A DQS P<2> 7 25
H504 DDR A DQS2	DDR DQS_PHY	DDR A DQS2	MEM A DQS N<2> 7 25
H505 DDR A DQS3	DDR DQS_PHY	DDR A DQS3	MEM A DQS P<3> 7 25
H506 DDR A DQS3	DDR DQS_PHY	DDR A DQS3	MEM A DQS N<3> 7 25
H507 DDR A DQS4	DDR DQS_PHY	DDR A DQS4	MEM A DQS P<4> 7 25
H508 DDR A DQS4	DDR DQS_PHY	DDR A DQS4	MEM A DQS N<4> 7 25
H509 DDR A DQS5	DDR DQS_PHY	DDR A DQS5	MEM A DQS P<5> 7 25
H510 DDR A DQS5	DDR DQS_PHY	DDR A DQS5	MEM A DQS N<5> 7 25
H511 DDR A DQS6	DDR DQS_PHY	DDR A DQS6	MEM A DQS P<6> 7 25
H512 DDR A DQS6	DDR DQS_PHY	DDR A DQS6	MEM A DQS N<6> 7 25
H513 DDR A DQS7	DDR DQS_PHY	DDR A DQS7	MEM A DQS P<7> 7 25
H514 DDR A DQS7	DDR DQS_PHY	DDR A DQS7	MEM A DQS N<7> 7 25
Channel B			
H230 DDR B CLK0	DDR CLK_PHY	DDR CLK	MEM B CLK P<1..0> 7 24
H231 DDR B CLK0	DDR CLK_PHY	DDR CLK	MEM B CLK N<1..0> 7 24
H232 DDR B CLK1	DDR CLK_PHY	DDR CLK	MEM B CLK P<3..2> 72
H233 DDR B CLK1	DDR CLK_PHY	DDR CLK	MEM B CLK N<3..2> 72
H234 DDR B CTRL0	DDR CTRL_PHY	DDR CTRL	MEM B CKE<1..0> 7 24
H235 DDR B CTRL0	DDR CTRL_PHY	DDR CTRL	MEM B CS L<1..0> 7 24
H236 DDR B CTRL0	DDR CTRL_PHY	DDR CTRL	MEM B ODT<1..0> 7 24
H237 DDR B CTRL1	DDR CTRL_PHY	DDR CTRL	MEM B CKE<3..2> 72
H238 DDR B CTRL1	DDR CTRL_PHY	DDR CTRL	MEM B CS L<3..2> 72
H239 DDR B CTRL1	DDR CTRL_PHY	DDR CTRL	MEM B ODT<3..2> 72
H240 DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B A<15..0> 7 24
H241 DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B BA<2..0> 7 24
H242 DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B RAS L 7 24
H243 DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B CAS L 7 24
H244 DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B WE L 7 24
H245 DDR B DQ_BYTE0	DDR DQ_PHY	DDR B DQ_BYTE0	MEM B DQ<7..0> 7 25
H246 DDR B DQ_BYTE1	DDR DQ_PHY	DDR B DQ_BYTE1	MEM B DQ<15..8> 7 25
H247 DDR B DQ_BYTE2	DDR DQ_PHY	DDR B DQ_BYTE2	MEM B DQ<23..16> 7 25
H248 DDR B DQ_BYTE3	DDR DQ_PHY	DDR B DQ_BYTE3	MEM B DQ<31..24> 7 25
H249 DDR B DQ_BYTE4	DDR DQ_PHY	DDR B DQ_BYTE4	MEM B DQ<39..32> 7 25
H250 DDR B DQ_BYTE5	DDR DQ_PHY	DDR B DQ_BYTE5	MEM B DQ<47..40> 7 25
H251 DDR B DQ_BYTE6	DDR DQ_PHY	DDR B DQ_BYTE6	MEM B DQ<55..48> 7 25
H252 DDR B DQ_BYTE7	DDR DQ_PHY	DDR B DQ_BYTE7	MEM B DQ<63..56> 7 25
H253 DDR B DQS0	DDR DQS_PHY	DDR B DQS0	MEM B DQS P<0> 7 25
H254 DDR B DQS0	DDR DQS_PHY	DDR B DQS0	MEM B DQS N<0> 7 25
H255 DDR B DQS1	DDR DQS_PHY	DDR B DQS1	MEM B DQS P<1> 7 25
H256 DDR B DQS1	DDR DQS_PHY	DDR B DQS1	MEM B DQS N<1> 7 25
H257 DDR B DQS2	DDR DQS_PHY	DDR B DQS2	MEM B DQS P<2> 7 25
H258 DDR B DQS2	DDR DQS_PHY	DDR B DQS2	MEM B DQS N<2> 7 25
H259 DDR B DQS3	DDR DQS_PHY	DDR B DQS3	MEM B DQS P<3> 7 25
H260 DDR B DQS3	DDR DQS_PHY	DDR B DQS3	MEM B DQS N<3> 7 25
H261 DDR B DQS4	DDR DQS_PHY	DDR B DQS4	MEM B DQS P<4> 7 25

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PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.385 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP
CPU_ASYNC_PHY	*	CPU_50S

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?
CPU_ASYNC_ISO	*	=3:1_SPACING	?
CPU_MS_ISO	TOP, BOTTOM	=4.5:1_SPACING	?
CPU_MS_ISO	*	=3:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
CPU_ASYNC	*	*	CPU_ASYNC_ISO
CPU_ASYNC_MS	*	*	CPU_MS_ISO

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

CPU ASYNCHRONOUS

Electrical Constraint Set	Physical	Spacing	
E57D	CPU_ASYNC_PHY	CPU_ASYNC	CPU_PROCHOT_L 0 44 45 61 62
E57A	CPU_ASYNC_PHY	CPU_ASYNC	CPU_PROCHOT_R_L 0
E59D PECT	CPU_ASYNC_PHY	CPU_ASYNC_MS	CPU_PECT 0 14 44 45
E59D	CPU_ASYNC_PHY	CPU_ASYNC_MS	SMC_PECT_L 44 45
E57D	CPU_ASYNC_PHY	CPU_ASYNC	CPU_CATERR_L 0 45
E59D	CPU_ASYNC_PHY	CPU_ASYNC	CPU_PWRGD 0 14 18
E59D	CPU_ASYNC_PHY	CPU_ASYNC	PM_SYNC 0 12
E61D	CPU_ASYNC_PHY	CPU_ASYNC	PM_THRMTRIP_L 0 14 45
E61D	CPU_ASYNC_PHY	CPU_ASYNC	CPU_RESET_L 0 14 18
E61D XDP_BPM_L	CPU_ASYNC_PHY	CPU_ASYNC	XDP_BPM_L<1..0> 0 18

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing	
CPU PCIe Compensation	COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_RCOMP
CPU eDP Compensation	COMP_PCIE_PHY	COMP_PCIE	CPU_EDP_RCOMP
	COMP_PCIE_PHY	COMP_PCIE	CPU_CFG_RCOMP

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIE-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCie (PCH)

Electrical Constraint Set	Physical	Spacing
x4 Thunderbolt		
H18P0 PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
H18S0 PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
H18S0	PCIE_PHY	PCIE_TBT_R2D
H18S0	PCIE_PHY	PCIE_TBT_R2D
H18S0 PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
H18P0 PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
H18P0	PCIE_PHY	PCIE_TBT_D2R
H18S0	PCIE_PHY	PCIE_TBT_D2R
H18P0 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
H18P0 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 AirPort		
H18P0 PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
H18S0 PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
H18S0	PCIE_PHY	PCIE
H18P0	PCIE_PHY	PCIE
H18P0 PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
H18S0 PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
H18S0	PCIE_PHY	PCIE
H18P0 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
H18P0 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x1 Caesar IV		
H18S0 PCIE_GEN2_R2D	PCIE_PHY	PCIE
H18S0 PCIE_GEN2_R2D	PCIE_PHY	PCIE
H18S0	PCIE_PHY	PCIE
H18S0	PCIE_PHY	PCIE
H18P0 PCIE_GEN2_D2R	PCIE_PHY	PCIE
H18S0 PCIE_GEN2_D2R	PCIE_PHY	PCIE
H18S0	PCIE_PHY	PCIE
H18S0	PCIE_PHY	PCIE
H18P0 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
H18S0 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x2 SSD		
H18P0 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
H18P0 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCH PCIE Compensation		
H18W0	COMP_DMI_PHY	COMP_PCIE

CPU DP REF CLK

Electrical Constraint Set	Physical	Spacing
CPU DP REF CLK		
H18W0 CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
H18P0 CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
H18S0 CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
H18P0 CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE

DMI

Electrical Constraint Set	Physical	Spacing
DMI		
H18P0 DMT_N2S	PCIE_PHY	DMI_N2S
H18S0 DMT_N2S	PCIE_PHY	DMI_N2S
H18P0 DMT_S2N	PCIE_PHY	DMI_S2N
H18S0 DMT_S2N	PCIE_PHY	DMI_S2N
H18S0 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
H18W0 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
DMI Compensation		
H18P0	COMP_DMI_PHY	COMP_PCIE

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

E14

Electrical constraint Set	Physical	Spacing	
PCI Clock			
168B	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN 11 29
168C	CLK_PCT_55S	CLK_PCT	PCH_CLK33M_PCIOUT 11 19

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC 55S	LPC	LPC AD<3..0> 13 44 46
	LPC 55S	LPC	LPC AD R<3..0> 13
	LPC 55S	LPC	LPC FRAME_L 13 44 46
	LPC 55S	LPC	LPC FRAME_R_L 13
LPC Clocks			
	CLK LPC 55S	CLK LPC	LPC_CLK33M LPCPLUS 19 46
	CLK LPC 55S	CLK LPC	LPC_CLK33M LPCPLUS_R 11 19
	CLK LPC 55S	CLK LPC	LPC_CLK33M_SMC 19 44
	CLK LPC 55S	CLK LPC	LPC_CLK33M_SMC_R 11 19

PCH Clocks

Electrical Constraint Set	Physical	Spacing		
PCH Reference Clock				
H400	CLK_PCH_55S	CLK_PCH	SYSLCK_CLK25M_SB	11 19
H400	CLK_PCH_55S	CLK_PCH	SYSLCK_CLK25M_SB_R	11
PCH RTC 32K				
H400	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	11 19
H400	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	11 19
H400	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	19
SMC 32K				
H400	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R	12 45
H400	CLK_PCH_55S	CLK_PCH	SMC_CLK32K	44 45

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
H37P	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1 19
H80B	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2 19
H81P	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R 19
25M Reference Clocks			
H82P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET 19 35
H83P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R 19
H86B	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT 19 26
H87P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R 26

HDA

E1

Electrical Constraint Set	Physical	Spacing	
HDA			
HDB	HDA_55S	HDA	HDA BIT_CLK 11 52
HDB	HDA_55S	HDA	HDA BIT_CLK_R 11
HDB	HDA_55S	HDA	HDA_RST_L 11 52
HDB	HDA_55S	HDA	HDA_RST_R_L 11
HDB	HDA_55S	HDA	HDA_SDOUT 11 52
HDB	HDA_55S	HDA	HDA_SDOUT_R 11 19
HDB	HDA_55S	HDA	HDA_SYNC 11 52
HDB	HDA_55S	HDA	HDA_SYNC_R 11
HDB	HDA_55S	HDA	HDA_SDIN0 11 52
HDB	HDA_55S	HDA	AUD_SDI_R 52
SPDIF			
SPD		HDA	AUD_SPDIF_CHIP 52
SPD		HDA	AUD_SPDIF_OUT 52 56

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H00P	SPI 50S	SPI	SPI CLK R 13 46
H00Q	SPI 50S	SPI	SPI CLK 46
H00R	SPI 50S	SPI	SPI ALT CLK 46
H00S	SPI 50S	SPI	SPI SMC CLK 44 46
H40Y	SPI 50S	SPI	SPI MLB CLK 46
H00T	SPI 50S	SPI	SPI CS0 R L 13 46
H00U	SPI 50S	SPI	SPI CS0 L 46
H00V	SPI 50S	SPI	SPI ALT CS L 46
H40U	SPI 50S	SPI	SPI SMC CS L 44 46
H00W	SPI 50S	SPI	SPI MLB CS L 46
H00D	SPI 50S	SPI	SPI MOSI R 13 46
H40U	SPI 50S	SPI	SPI MOSI 46
H00A	SPI 50S	SPI	SPI ALT MOSI 46
H00I	SPI 50S	SPI	SPI SMC MOSI 44 46
H40A	SPI 50S	SPI	SPI MLB MOSI 46
H40B	SPI 50S	SPI	SPI MISO 13 46
H00B	SPI 50S	SPI	SPI ALT MISO 46
H40B	SPI 50S	SPI	SPI SMC MISO 44 46
H40B	SPI 50S	SPI	SPI MLB MISO 46
H00B	SPI 50S	SPI	SPIROM USE MLB 14 46

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=1.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PwR_P2MM
SENSE	GND	*	GND_P2MM

SMC Generic Control Line Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMC_ISO	*	=1:1_SPACING	?

Constraints

NET_SPACING_TYPE1		NET_SPACING_TYPE2		AREA_TYPE	SPACING_RULE_SET
SMC_CTRL	*	*			SMC_ISO

SMC Generic Control Line Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMC_GEN	*	SMC_50S

Current/Voltage Sense

Electrical Constraint Set		Physical	Spacing
Common			
REG		SENSE	GND_SMC_AVSS
12V S5 (System Total)			
PG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
PG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
PG		SENSE	SNS_P12VG3H_P
PG		SENSE	SNS_P12VG3H_N
PG		SENSE	ISNS_P12VG3H_R
PG		SENSE	ISNS_P12VG3H
PG		SENSE	VSNS_P12VG3H
HDD			
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG		SENSE	SNS_HDD_P
REG		SENSE	SNS_HDD_N
REG		SENSE	ISNS_HDD50_R
REG		SENSE	ISNS_HDD50
REG		SENSE	VSNS_HDD50
SSD			
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG		SENSE	SNS_SSD_P
REG		SENSE	SNS_SSD_N
REG		SENSE	ISNS_SSD50_R
REG		SENSE	ISNS_SSD50
REG		SENSE	VSNS_P3V3S5
VDDQ S3 (DDR)			
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG		SENSE	SNS_VDDQ53_DDR_P
REG		SENSE	SNS_VDDQ53_DDR_N
REG		SENSE	ISNS_VDDQ53_DDR_R
REG		SENSE	ISNS_VDDQ53_DDR
REG		SENSE	VSNS_VDDQ53_DDR
CPU Core			
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG		SENSE	REG_CPUVCC_IMON_R
REG		SENSE	ISNS_CPUVCC_FB_R
REG		SENSE	ISNS_CPUVCC_FB
REG		SENSE	ISNS_CPUVCC
REG		SENSE	VSNS_CPUVCC
PP1V05_S0_PCH			
REG		SENSE	VSNS_P1V05S0_PCH
PP1V5_S0			
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG		SENSE	SNS_PVDDQ50_P
REG		SENSE	SNS_PVDDQ50_N
REG		SENSE	ISNS_PVDDQ50_R
REG		SENSE	ISNS_PVDDQ50
REG		SENSE	VSNS_PVDDQ50
Airport			
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG	SNS_CURRENT	SNS_DTEFF_PHY	SENSE
REG		SENSE	SNS_P3V3S4_AP_P
REG		SENSE	SNS_P3V3S4_AP_N
REG		SENSE	ISNS_P3V3S4_AP_R
REG		SENSE	ISNS_P3V3S4_AP

SMC

Electrical Constraint Set	Physical	Spacing	
SMC			
E1	CLK_XTAL	XTAL	SMC XTAL 44.4
E2	CLK_XTAL	XTAL	SMC EXTAL 44.4
E31	SMC_GEN	SMC_CTLR	SMC LRESET L 26.4
E32	SMC_GEN	SMC_CTLR	SMC RUNTIME SCI L 14.4
E33	SMC_GEN	SMC_CTLR	SMC WAKE SCI L 14.4
E34	SMC_GEN	SMC_CTLR	SMC FAN_0 CTL 44.5
E35	SMC_GEN	SMC_CTLR	SMC FAN_0 TACH 44.5

SMBus

Electrical Constraint Set	Physical	Spacing	
SMC			
	SMB_PHY	SMB	SMBUS SMC 0 S0_SCL 44 41
	SMB_PHY	SMB	SMBUS SMC 0 S0_SDA 44 41
	SMB_PHY	SMB	SMBUS SMC 1 S0_SCL 44 41
	SMB_PHY	SMB	SMBUS SMC 1 S0_SDA 44 41
	SMB_PHY	SMB	SMBUS SMC 2 S4_SCL 44 41
	SMB_PHY	SMB	SMBUS SMC 2 S4_SDA 44 41
	SMB_PHY	SMB	SMBUS SMC 3_SCL 44 41
	SMB_PHY	SMB	SMBUS SMC 3_SDA 44 41
	SMB_PHY	SMB	SMBUS SMC 5 G3H_SCL 44 45
	SMB_PHY	SMB	SMBUS SMC 5 G3H_SDA 44 45
PCH			
	TBT_T2C_55S	TBT_T2C	SMBUS PCH_CLK 13 41
	TBT_T2C_55S	TBT_T2C	SMBUS PCH_DATA 13 41
	SMB_PHY	SMB	SML_PCH_0_CLK 13 41
	SMB_PHY	SMB	SML_PCH_0_DATA 13 41
Display Tcon			
	SMB_PHY	SMB	SMB_DP_TCON_SCL 40 41
	SMB_PHY	SMB	SMB_DP_TCON_SDA 40 41

Temperature Sense

Electrical Contrast Set	Physical	Spacing
EMC1414-1 (Production)		
E67 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T1 1 P
E68 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T1 1 N
E69 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T1 3 P
E69 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T1 3 N
E65 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS ACDC_P
E66 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS ACDC_N
E920 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T1 2 P
E920 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T1 2 N
TMP423 (Development)		
E67 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T2 1 P
E68 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T2 1 N
E69 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T2 2 P
E70 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T2 2 N
E71 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T2 3 P
E72 SNS_TEMP	SNS_DTEF_PHY	SENSE SNS T2 3 N
HDD Out-of-Band		
E26		SENSE HDD_00B1_D2R_L
E27		SENSE HDD_00B1_D2R_F_L
E28		SENSE HDD_00B1_D2R_R_L
E33		SENSE SMC_00B1_D2R_L
E29		SENSE SMC_00B1_R2D_L
E310		SENSE SMC_00B1_R2D_R_L
SSD Out-of-Band		
E77		SENSE SMC_00B2_R2D_L
E78		SENSE SMC_00B2_D2R_L

DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	D1D2	NO_TEST	
Input Bus						
E1	POWER	POWER	5V			REG VCC U7400 64
E2	POWER	POWER	5V			REG PVCC U7400 64
Local Ground						
E3	GND	GND	0V			AGND P1V05S0 64
1.05V S0						
E5	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0 64
E6	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0_L 64
E7	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG BOOT P1V05S0 64
E8	VR_DTDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT P1V05S0_RC 64
E9	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0 64
E10	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0_R 64
E11	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P1V05S0 64
E12	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P1V05S0 64
E13	VR_CTL_PHY	VR_CTL				REG P1V05S0_OCSSET 64
E14	VR_CTL_PHY	VR_CTL				REG P1V05S0_V0 64
E19		SENSE				REG P1V05S0_FB 64
E18		SENSE				REG P1V05S0_RTN 64
E21	VR_CTL_PHY	VR_CTL				REG P1V05S0_SREF 64
E20	VR_CTL_PHY	VR_CTL				REG P1V05S0_FSEL 64
Output Bus						
E22	POWER	POWER	1.05V			PP1V05_S0 70
FET Switched						
E23	POWER	POWER	1.05V			PP1V05_TBTLIC 28 70
E25	POWER	POWER	1.05V			PP1V05_TBTCIO 70

VDDQ S3 (1.35V)/VTT S0

Physical	Spacing	Voltage	IDIT	NO_TEST	
Input Bus					
E43 POWER	POWER	5V			REG_V5IN_U7390
Local Ground					
E44 GND	GND	0V			AGND_VDDQS3
VDDQ S3					
E45 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3
E47 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L
E46 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3
E48 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC
E49 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3
E50 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R
E49 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VDDQS3
E51 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3
E53 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS
E55 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF
E56 VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN
E57 VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE
E58 VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP
E59 VR_CTL_PHY	VR_CTL				LDO_DDRVTT0_SNS
E61 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VTTREF
Output Bus					
E60 POWER	POWER	1.35V			PPVDDQ_S3
E62 POWER_DDR	POWER_DDR	0.675V			PPDDRVTI_S0
FET Switched					
E63 POWER	POWER	1.35V			PPVDDQ_S0
Sensed					
E64 POWER	POWER	1.35V			PPVDDQ_S3_DDR
E67 POWER	POWER	1.35V			PPVDDQ_S0_CPU

SYNC MASTER=316 NICK		SYNC DATE=01/10/2013	
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VReg Constraints			
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<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th></tr><tr><td colspan="6">Input Bus</td></tr><tr><td>103D</td><td>POWER</td><td>POWER</td><td>12V</td><td></td><td></td><td>PP12V_S0_CPUVCC FLT</td><td>01 02</td></tr><tr><td>103B</td><td>POWER</td><td>POWER</td><td>5V</td><td></td><td></td><td>REG VCC U7000</td><td>01</td></tr><tr><td colspan="6">Local Ground</td></tr><tr><td>102D</td><td>GND</td><td>GND</td><td>0V</td><td></td><td></td><td>AGND_CPU</td><td>01 02 71</td></tr><tr><td colspan="6">Phase 1</td></tr><tr><td>108D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG THWN 1</td><td>02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG PWM CPUVCC 1</td><td>01 02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG PWM CPUVCC 1 R</td><td>01</td></tr><tr><td>101B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUVCC 1</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUVCC1</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG BOOT CPUVCC 1</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG BOOT CPUVCC 1 RC</td><td>02</td></tr><tr><td>108B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG SNUBBER CPUVCC 1</td><td>02</td></tr><tr><td>103B</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td><td>PPCPUVCC_S0 SENSE 1</td><td>02</td></tr><tr><td>102B</td><td>TSNS_CPU CORE</td><td>SNS_DIEF_PHY</td><td></td><td></td><td></td><td>REG ISENVCC 1 P</td><td>01 02</td></tr><tr><td>102B</td><td>TSNS_CPU CORE</td><td>SNS_DIEF_PHY</td><td></td><td></td><td></td><td>REG ISENVCC 1 N</td><td>02</td></tr><tr><td>102B</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENVCC 1 NR</td><td>01 02</td></tr><tr><td colspan="6">Phase 2</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG THWN 2</td><td>02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG PWM CPUVCC 2</td><td>01 02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG PWM CPUVCC 2 R</td><td>01</td></tr><tr><td>108B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUVCC 2</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUVCC2</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG BOOT CPUVCC 2</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG BOOT CPUVCC 2 RC</td><td>02</td></tr><tr><td>108B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG SNUBBER CPUVCC 2</td><td>02</td></tr><tr><td>103B</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td><td>PPCPUVCC_S0 SENSE 2</td><td>02</td></tr><tr><td>102B</td><td>TSNS_CPU CORE</td><td>SNS_DIEF_PHY</td><td></td><td></td><td></td><td>REG ISENVCC 2 P</td><td>01 02</td></tr><tr><td>102B</td><td>TSNS_CPU CORE</td><td>SNS_DIEF_PHY</td><td></td><td></td><td></td><td>REG ISENVCC 2 N</td><td>02</td></tr><tr><td>102B</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG ISENVCC 2 NR</td><td>01 02</td></tr><tr><td colspan="6">Phase 3</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG THWN 3</td><td>02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG PWM CPUVCC 3</td><td>01 02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG PWM CPUVCC 3 R</td><td>01</td></tr><tr><td>108B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUVCC 3</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG PHASE CPUVCC3</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG BOOT CPUVCC 3</td><td>02</td></tr><tr><td>102B</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG BOOT CPUVCC 3 RC</td><td>02</td></tr><tr><td>108B</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG SNUBBER CPUVCC 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CPUVCC1	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG BOOT CPUVCC 1	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT CPUVCC 1 RC	02	108B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER CPUVCC 1	02	103B	POWER	POWER	1.8V			PPCPUVCC_S0 SENSE 1	02	102B	TSNS_CPU CORE	SNS_DIEF_PHY				REG ISENVCC 1 P	01 02	102B	TSNS_CPU CORE	SNS_DIEF_PHY				REG ISENVCC 1 N	02	102B		SENSE				REG ISENVCC 1 NR	01 02	Phase 2						108B	VR_CTL_PHY	VR_CTL				REG THWN 2	02	108B	VR_CTL_PHY	VR_CTL				REG PWM CPUVCC 2	01 02	108B	VR_CTL_PHY	VR_CTL				REG PWM CPUVCC 2 R	01	108B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUVCC 2	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUVCC2	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG BOOT CPUVCC 2	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT CPUVCC 2 RC	02	108B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER CPUVCC 2	02	103B	POWER	POWER	1.8V			PPCPUVCC_S0 SENSE 2	02	102B	TSNS_CPU CORE	SNS_DIEF_PHY				REG ISENVCC 2 P	01 02	102B	TSNS_CPU CORE	SNS_DIEF_PHY				REG ISENVCC 2 N	02	102B		SENSE				REG ISENVCC 2 NR	01 02	Phase 3						108B	VR_CTL_PHY	VR_CTL				REG THWN 3	02	108B	VR_CTL_PHY	VR_CTL				REG PWM CPUVCC 3	01 02	108B	VR_CTL_PHY	VR_CTL				REG PWM CPUVCC 3 R	01	108B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUVCC 3	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE CPUVCC3	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG BOOT CPUVCC 3	02	102B	VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT CPUVCC 3 RC	02	108B	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER CPUVCC 3	02	103B	POWER	POWER	1.8V			PPCPUVCC_S0 SENSE 3	02	102B	TSNS_CPU CORE	SNS_DIEF_PHY				REG ISENVCC 3 P	01 02	102B	TSNS_CPU CORE	SNS_DIEF_PHY				REG ISENVCC 3 N	02	102B		SENSE				REG ISENVCC 3 NR	01 02	<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th></tr><tr><td colspan="6">ISL6372</td></tr><tr><td>106D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG CPUVCC_DVC</td><td>01</td></tr><tr><td>106D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUVCC_DVC RC</td><td>01</td></tr><tr><td>106D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUVCC_FB_RC_2</td><td>01</td></tr><tr><td>106D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG CPUVCC_COMP</td><td>01</td></tr><tr><td>106D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUVCC_COMP_RC</td><td>01</td></tr><tr><td>106D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG 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Set	Physical	Spacing	Voltage	DIDT	NO_TEST	ISL6372						106D	VR_CTL_PHY	VR_CTL				REG CPUVCC_DVC	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_DVC RC	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC_2	01	106D	VR_CTL_PHY	VR_CTL				REG CPUVCC_COMP	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_COMP_RC	01	106D	VR_CTL_PHY	VR_CTL				REG CPUVCC_FB	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_1	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_2	01	106D	VR_CTL_PHY	VR_CTL				CPUVCC_PSICOMP_RC	01	106D	VR_CTL_PHY	VR_CTL				REG CPUVCC_PSICOMP	01	106D	VR_CTL_PHY	VR_CTL				REG CPUVCC_HFCOMP	01	106B	VSNS_CPU CORE	SNS_DIEF_PHY				CPU_VCCSENSE_P	0 01	106B	VSNS_CPU CORE	SNS_DIEF_PHY				CPU_VCCSENSE_N	0 01	106B		SNS_DIEF_PHY				CPU_VCCSENSE_R_P	01	106B		SNS_DIEF_PHY				CPU_VCCSENSE_R_N	01	106B		SNS_DIEF_PHY				SNS_VCC_XW_P	01	106B		SNS_DIEF_PHY				SNS_VCC_XW_N	01	106B		SENSE				REG CPUVCC_VSEN	01	106B		SENSE				REG CPUVCC_RGND	01	106B		SENSE				REG CPUVCC_VIN	01	106B	VR_CTL_PHY	VR_CTL				REG 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Apple Inc.

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DRAWING NUMBER
051-0164

REVISION
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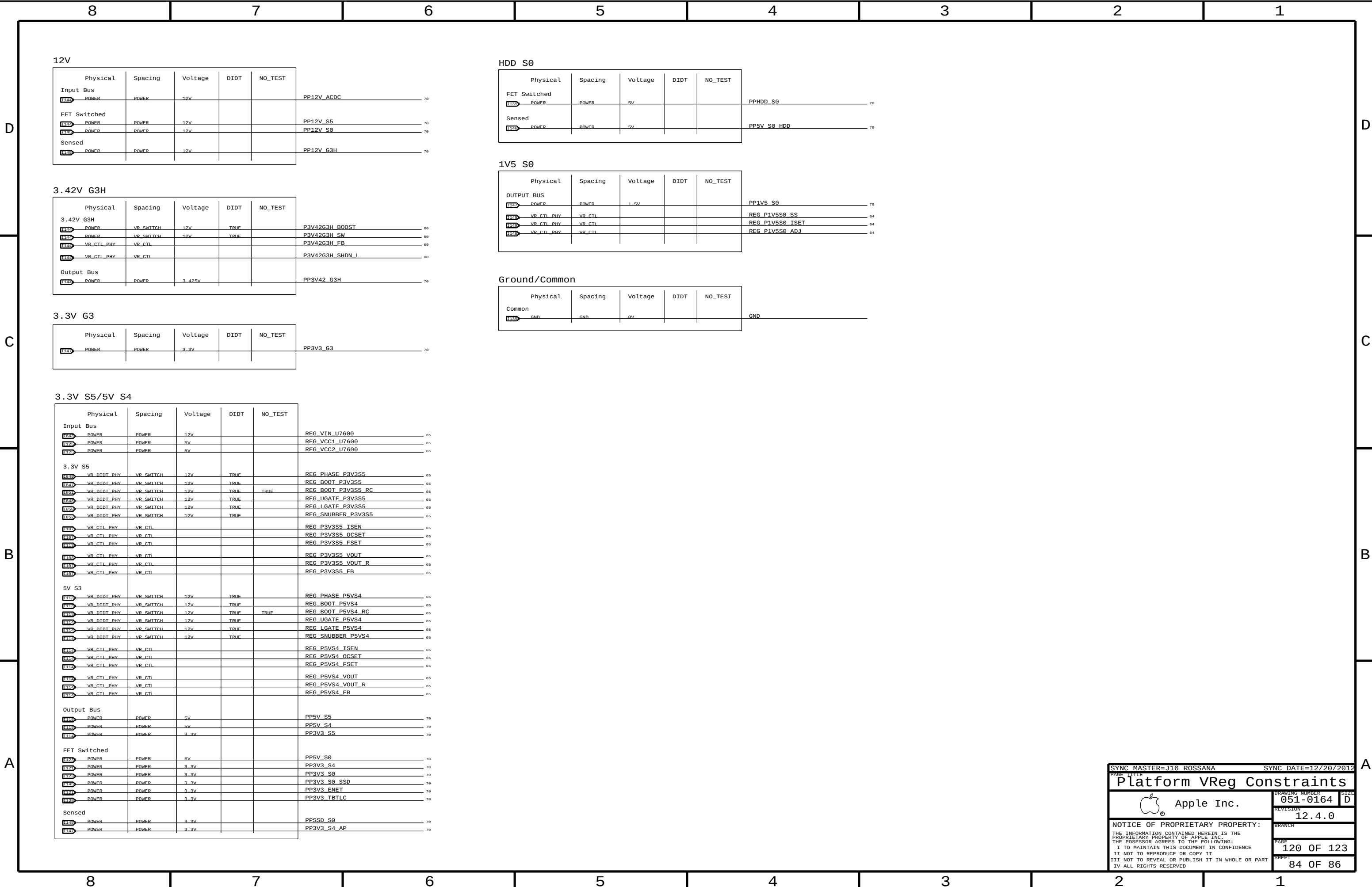
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SYNC DATE=12/14/2012

CPU VReg Constraints

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Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTD_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE+TO+LINE SPACING	WEIGHT
TBT_I2C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP	*	=5x_DIELECTRIC	?
TBTDP	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

Pairs should be within 100 mils of clock length.

Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.

DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

Electrical Constraint Set	Physical	Spacing		
H000	DP_850	DTSP1AYPORT	DP_TBTSNK0_ML_C_P<3...0>	26 71
H002	DP_850	DTSP1AYPORT	DP_TBTSNK0_ML_C_N<3...0>	26 71
H003 DP_TBTSNK0_ML	DP_850	DTSP1AYPORT	DP_TBTSNK0_ML_P<3...0>	26
H004 DP_TBTSNK0_ML	DP_850	DTSP1AYPORT	DP_TBTSNK0_ML_N<3...0>	26
H010	DP_850	DTSP1AYPORT	DP_TBTSNK0_AUXCH_C_P	26 71
H015	DP_850	DTSP1AYPORT	DP_TBTSNK0_AUXCH_C_N	26 71
H019 DP_TBTSNK0_AUX	DP_850	DTSP1AYPORT	DP_TBTSNK0_AUXCH_P	26
H021 DP_TBTSNK0_AUX	DP_850	DTSP1AYPORT	DP_TBTSNK0_AUXCH_N	26
H035	DP_850	DTSP1AYPORT	DP_TBTSNK1_ML_C_P<3...0>	26 71
H040	DP_850	DTSP1AYPORT	DP_TBTSNK1_ML_C_N<3...0>	26 71
H042 DP_TBTSNK1_ML	DP_850	DTSP1AYPORT	DP_TBTSNK1_ML_P<3...0>	26 71
H043 DP_TBTSNK1_ML	DP_850	DTSP1AYPORT	DP_TBTSNK1_ML_N<3...0>	26
H049	DP_850	DTSP1AYPORT	DP_TBTSNK1_AUXCH_C_P	26 71
H050	DP_850	DTSP1AYPORT	DP_TBTSNK1_AUXCH_C_N	26 71
H052 DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	DP_TBTSNK1_AUXCH_P	26
H053 DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	DP_TBTSNK1_AUXCH_N	26
H060 DP_INTPN1_TBT_ML_MUX	DP_850	DTSP1AYPORT	DP_TBTSRC_ML_P<3...0>	41
H061 DP_INTPN1_TBT_ML_MUX	DP_850	DTSP1AYPORT	DP_TBTSRC_ML_N<3...0>	41
H062 DP_INTPN1_TBT_ML_MUX	DP_850	DTSP1AYPORT	DP_TBTSRC_ML_C_P<3...0>	41
H063 DP_INTPN1_TBT_ML_MUX	DP_850	DTSP1AYPORT	DP_TBTSRC_ML_C_N<3...0>	41
H069 DP_INTPN1_TBT_AUX_MUX	DP_850	DTSP1AYPORT	DP_TBTSRC_AUXCH_P	41
H070 DP_INTPN1_TBT_AUX_MUX	DP_850	DTSP1AYPORT	DP_TBTSRC_AUXCH_N	41
H071	DP_850	DTSP1AYPORT	DP_TBTSRC_AUX_C_P	41
H072	DP_850	DTSP1AYPORT	DP_TBTSRC_AUX_C_N	41
H080	TBT_T2C_55S	TBT_T2C	I2C_TBTRTR_SCL	26
H081	TBT_T2C_55S	TBT_T2C	I2C_TBTRTR_SDA	26
H090 TRT_SPT_CLK	TRT_SPT_55S	TRT_SPT	TBT_SPT_CLK	26
H091 TRT_SPT_MOSI	TRT_SPT_55S	TRT_SPT	TBT_SPT_MOSI	26
H092 TRT_SPT_MISO	TRT_SPT_55S	TRT_SPT	TBT_SPT_MISO	26
H093 TRT_SPT_CS_1	TRT_SPT_55S	TRT_SPT	TBT_SPT_CS_1	26

*: Only used on hosts supporting T29 video-in

DisplayPort

Electrical Constraint Set	Physical	Spacing	
Graphics Source			
FE059 DP INTPNL EG MI_MUX	DP 850	DISPLAYPORT	DP INT_ML P<1..0> 41 71
FE060 DP INTPNL EG MI_MUX	DP 850	DISPLAYPORT	DP INT_ML N<1..0> 41 71
FE061 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT_AUX P 41 71
FE062 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT_AUX N 41 71
FE063 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT_AUX C_P 41 71
FE064 DP INTPNL EG AUX_MUX	DP 850	DISPLAYPORT	DP INT_AUX C_N 41
Internal Panel			
FE065	DP 850	DISPLAYPORT	DP INTPNL_ML C_P<3..0> 41
FE066	DP 850	DISPLAYPORT	DP INTPNL_ML C_N<3..0> 41
FE067 DP INTPNL_ML_CONN	DP 850	DISPLAYPORT	DP INTPNL_ML P<3..0> 40 41
FE068 DP INTPNL_ML_CONN	DP 850	DISPLAYPORT	DP INTPNL_ML N<3..0> 40 41
FE069 DP INTPNL_AUX_CONN	DP 850	DISPLAYPORT	DP INTPNL_AUX P 40 41
FE070 DP INTPNL_AUX_CONN	DP 850	DISPLAYPORT	DP INTPNL_AUX N 40 41
Internal DP SPDIF			
FE071		HDA	DP_INT_SPDIF_AUDIO 40 52

TBT/DP Net Properties

Electrical Constraint Set	Physical	Spacing	
Port A			
E621 TBT_A_R2D1	TBTDP_960	TBTDP	TBT A R2D C P<1>
E622 TBT_A_R2D1	TBTDP_960	TBTDP	TBT A R2D C N<1>
E623 TBT_A_R2D8	TBTDP_960	TBTDP	TBT A R2D C P<0>
E624 TBT_A_R2D8	TBTDP_960	TBTDP	TBT A R2D C N<0>
E625	TBTDP_960	TBTDP	TBT A R2D P<1..0>
E626	TBTDP_960	TBTDP	TBT A R2D N<1..0>
E627 DP_TBTPA_ML1	DP_850	DISPLAYPORT	DP TBTPA ML C P<1>
E628 DP_TBTPA_ML1	DP_850	DISPLAYPORT	DP TBTPA ML C N<1>
E629 DP_TBTPA_ML3	DP_850	DISPLAYPORT	DP TBTPA ML C P<3>
E630 DP_TBTPA_ML3	DP_850	DISPLAYPORT	DP TBTPA ML C N<3>
E631	DP_850	DISPLAYPORT	DP TBTPA ML P<1>
E632	DP_850	DISPLAYPORT	DP TBTPA ML N<1>
E633	DP_850	DISPLAYPORT	DP TBTPA ML P<3>
E634	DP_850	DISPLAYPORT	DP TBTPA ML N<3>
E635 DP_A_LSX	DP_850	DISPLAYPORT	DP A LSX ML P<1>
E636 DP_A_LSX	DP_850	DISPLAYPORT	DP A LSX ML N<1>
E637	TBTDP_960	TBTDP	TBT A D2R C P<1..0>
E638	TBTDP_960	TBTDP	TBT A D2R C N<1..0>
E639 TBT_A_D2R1	TBTDP_960	TBTDP	TBT A D2R P<1>
E640 TBT_A_D2R1	TBTDP_960	TBTDP	TBT A D2R N<1>
E641 TBT_A_D2R8	TBTDP_960	TBTDP	TBT A D2R P<0>
E642 TBT_A_D2R8	TBTDP_960	TBTDP	TBT A D2R N<0>
E643 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP TBTPA AUXCH C P
E644 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP TBTPA AUXCH C N
E645	DP_850	DISPLAYPORT	DP TBTPA AUXCH P
E646	DP_850	DISPLAYPORT	DP TBTPA AUXCH N
E647 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP A AUXCH DDC P
E648 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP A AUXCH DDC N
E649	TBTDP_960	TBTDP	TBT A D2R1 AUXDDC P
E650	TBTDP_960	TBTDP	TBT A D2R1 AUXDDC N
Port B			
E651 TBT_B_R2D1	TBTDP_960	TBTDP	TBT B R2D C P<1>
E652 TBT_B_R2D1	TBTDP_960	TBTDP	TBT B R2D C N<1>
E653 TBT_B_R2D8	TBTDP_960	TBTDP	TBT B R2D C P<0>
E654 TBT_B_R2D8	TBTDP_960	TBTDP	TBT B R2D C N<0>
E655	TBTDP_960	TBTDP	TBT B R2D P<1..0>
E656	TBTDP_960	TBTDP	TBT B R2D N<1..0>
E657 DP_TBTPB_ML1	DP_850	DISPLAYPORT	DP TBTPB ML C P<1>
E658 DP_TBTPB_ML1	DP_850	DISPLAYPORT	DP TBTPB ML C N<1>
E659 DP_TBTPB_ML3	DP_850	DISPLAYPORT	DP TBTPB ML C P<3>
E660 DP_TBTPB_ML3	DP_850	DISPLAYPORT	DP TBTPB ML C N<3>
E661	DP_850	DISPLAYPORT	DP TBTPB ML P<1>
E662	DP_850	DISPLAYPORT	DP TBTPB ML N<1>
E663	DP_850	DISPLAYPORT	DP TBTPB ML P<3>
E664	DP_850	DISPLAYPORT	DP TBTPB ML N<3>
E665 DP_B_LSX	DP_850	DISPLAYPORT	DP B LSX ML P<1>
E666 DP_B_LSX	DP_850	DISPLAYPORT	DP B LSX ML N<1>
E667	TBTDP_960	TBTDP	TBT B D2R C P<1..0>
E668	TBTDP_960	TBTDP	TBT B D2R C N<1..0>
E669 TBT_B_D2R1	TBTDP_960	TBTDP	TBT B D2R P<1>
E670 TBT_B_D2R1	TBTDP_960	TBTDP	TBT B D2R N<1>
E671 TBT_B_D2R8	TBTDP_960	TBTDP	TBT B D2R P<0>
E672 TBT_B_D2R8	TBTDP_960	TBTDP	TBT B D2R N<0>
E673 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP TBTPB AUXCH C P
E674 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP TBTPB AUXCH C N
E675	DP_850	DISPLAYPORT	DP TBTPB AUXCH P
E676	DP_850	DISPLAYPORT	DP TBTPB AUXCH N
E677 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP B AUXCH DDC P
E678 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP B AUXCH DDC N
E679	TBTDP_960	TBTDP	TBT B D2R1 AUXDDC P
E680	TBTDP_960	TBTDP	TBT B D2R1 AUXDDC N

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